Radio Frequency Technology : („*Hochfrequenztechnik 3*“)

Lecture : weekly Tuesday, 9:45 – 11:15, V 47.04

→ Prof. Dr. Jan Hesselbarth, ETI 1, 3.215

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Exercise : weekly Friday, 9:45 – 11:15, V 47.04

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WS 2026-2027

Website :

→ **ILIAS & CAMPUS**

possibly Feb 24, 2027

Exam : written, 120 minutes, all materials allowed, ~ end of Feb 2027

no programmable devices
no communication devices

Note that in the future (e.g., summer 2027), the exam *may* take place as oral exam !

Radio Frequency Technology — plan WS 2026-2027

Lecture:	Exercise:
Di, 13.10.26, 09:45-11:15, V47.04 … recap	Fr, 16.10.26, 09:45-11:15, V47.04 – Ü1 … recap
Di, 20.10.26, 09:45-11:15, V47.04 … recap	Fr, 23.10.26, 09:45-11:15, V47.04 – Ü2 … recap
~~Di, 27.10.26, 09:45-~~ Lecture cancelled	Fr, 30.10.26, 09:45-11:15, V47.04 – Ü3 … recap
Di, 03.11.26, 09:45-11:15, V47.04	Fr, 06.11.26, 09:45-11:15, V47.04 – ?
Di, 10.11.26, 09:45-11:15, V47.04	Fr, 13.11.26, 09:45-11:15, V47.04 – Ü4
Di, 17.11.26, 09:45-11:15, V47.04	Fr, 20.11.26, 09:45-11:15, V47.04 – Ü5
Di, 24.11.26, 09:45-11:15, V47.04	Fr, 27.11.26, 09:45-11:15, V47.04 – Ü6
Di, 01.12.26, 09:45-11:15, V47.04	Fr, 04.12.26, 09:45-11:15, V47.04 – Ü7
Di, 08.12.26, 09:45-11:15, V47.04	Fr, 11.12.26, 09:45-11:15, V47.04 – Ü8
Di, 15.12.26, 09:45-11:15, V47.04	Fr, 18.12.26, 09:45-11:15, V47.04 – Ü9
~~Di, 22.12.26, 09:45-~~ Lecture cancelled	~~Fr, 08.01.27, 09:45-~~ Exercise cancelled
Di, 12.01.27, 09:45-11:15, V47.04	Fr, 15.01.27, 09:45-11:15, V47.04 – Ü10
Di, 19.01.27, 09:45-11:15, V47.04	Fr, 22.01.27, 09:45-11:15, V47.04 – Ü11
Di, 26.01.27, 09:45-11:15, V47.04	Fr, 29.01.27, 09:45-11:15, V47.04 – Ü12
Di, 02.02.27, 09:45-11:15, V47.04	Fr, 05.02.27, 09:45-11:15, V47.04 – Ü13
Di, 09.02.27, 09:45-11:15, V47.04	Fr, 12.02.27, 09:45-11:15, V47.04 – … exam prep Ü?

On short notice (→ email via ILIAS), changes are possible!

Curriculum

@ ILIAS: complete English script available !!!

Infotech:

1st sem.: „R.F.T. – Introduction“

2nd sem.: „Basics of R.F.T.“

BSc. ETIT:

3rd sem.: „Nachrichtentechnik 1“

5th sem.: „Hochfrequenztechnik 1“

Msc. Infotech / EENG / ETIT:

winter sem.:
- **„Radio frequency technology“**
- „Antennas“

summer sem.:
- „Microwave System Design and Technologies“
- „Microwave Acoustic Devices and RF MEMS“ (new in 2027)

also (WS+SS):
- laboratory course / Fachpraktikum „Microwave Components“,
- research project (Forschungsarbeit), MSc thesis

Radio Frequency Technology : („Hochfrequenztechnik 3“)

Prerequisites :
- some knowledge of electrodynamics („Maxwell‘s equations“)
- RF engineering basics (Transistor, transmission lines, Smith chart)
- slides of „Hochfrequenztechnik 1“ ***in english*** are available on ILIAS

… prerequisite knowledge covered by, e.g., „Hochfrequenztechnik 1“ (partly equivalent to Infotech: „Basics of Radio Frequency Technology“) ——— note that only “Nachrichtentechnik 1” is NOT sufficient !

Note that „Radio Frequency Technology“ is NOT an introductory course!
The basics of microwave engineering are considered prerequisites!

Literature :

engl.:
- H.L. Hartnagel, eds., Fundamentals of RF and Microwave Techniques and Technologies
- R.E. Collin, Foundations for Microwave Engineering, IEEE
- D.M. Pozar, Microwave Engineering, Wiley
- R.E. Collin, Field theory of guided waves, IEEE
- R.F. Harrington, Time-harmonic electromagnetic fields, McGraw Hill
- IEEE Transactions on Microwave Theory and Techniques

dt.:
- P. Leuchtmann, Einführung in die elektromagnetische Feldtheorie, Pearson
- Zinke / Brunswig, Lehrbuch der Hochfrequenztechnik, Springer

AOB: time effort: it is suggested to attend the exercises **and** to contemplate at home every week

research project: it is suggested to make a good exam in this module and/or to attend the lab course

Motivation

microwave engineering – aka *radio frequency technology*

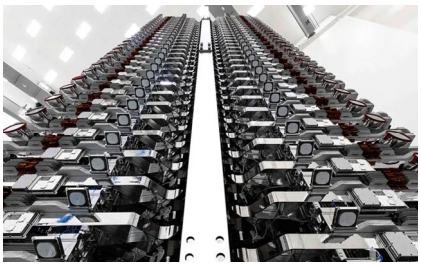
Microwave Engineering today covers electronic devices, circuits and concepts operating at frequencies between, roughly, 100 MHz and 100 GHz. In a wider context, some concepts also apply to higher frequencies up to (and including) optics (infrared and visible-light).

One main application of microwave engineering is communication: the physical infrastructure of wireless and fiber-optic communication. Other applications of today are in radar and sensing, localization, medicine and in industrial applications such as heating. Thus, microwave engineering applications are ubiquituous, and the related commercial impact is gigantic and growing.

If I have seen further than others, it is by standing upon the shoulders of giants.

Isaac Newton

Motivation



2023

Parts of electronics circuits:

- passives (RLC),
- (active) semiconductors,
- wires (short pieces of metal connections)

Parts of microwave circuits:

- same as above
  - … but RLC must consider parasitics
  - … but semiconductors must consider parasitics and nonlinearities
- transmission lines
  - … „long“ pieces of connections, not necessarily metallic
- transmission line resonators
  - … incl. cavity resonators
- non-reciprocal (and passive) components
- antennas — *separate lecture (56460) !*

This lecture:

…

…

…

+ measurement techniques

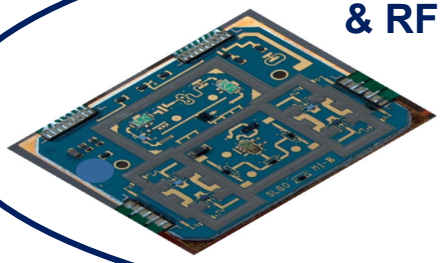
Motivation:

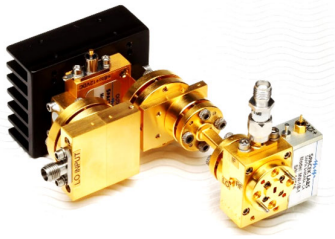
Prerequisite:

„microwaves 101“

This lecture:

passive components
& RF measurement techniques





IHF:

Microwave packaging and simulation techniques

ILH / INT:

active circuits & optics

IHF:

antennas

SAW, BAW, RF-MEMS

Radio Frequency Technology — main topics :

Repetition:

- line equivalent circuit
- impedance transformation
- planar transmission lines

A: introduction & motivation

B: transmission lines & waveguides

- rectangular waveguide
- slab waveguide
- circular waveguide, dielectric rod & fiber

C: resonators

- resonators

D: non-reciprocal components

- non-reciprocity
- isolator & circulator

E: RF measurement techniques

- power measurement
- about S-parameters
- network analyzer
- noise measurement
- signal analysis

Transmission Lines & Waveguides

- general comments
- transmission line equivalent circuit
- impedance transformation
- S parameters and few example circuits

brief repetition

- coaxial waveguide
- planar lines
- rectangular waveguide & circular waveguide
- slab waveguide & dielectric waveguides

Transmission Lines & Waveguides

have been met before:

- two-conductor („two-wire“) line : TEM mode
  … in electrical and electronics eng. (low freq & DC)
- micro-strip line : quasi-TEM mode
  … the „typical“ circuit board transmission line (low freq & DC)
- rectangular waveguide : fundamental TE_{10} mode
  … feeding the horn antenna
- optical fiber : monomode / multimode
  … high-speed communication medium

transmission lines & waveguides: infinite homogeneous structures

- exception: periodic structures
  … where extension of one period is 0.1 … 1 x wavelength
  … example : heliflex cable, flexible waveguide, corrugated horn

- feature wave propagation in (infinite number of) discrete modes
  … most useful : propagation modes
  … others : evanescent modes (i.e., $f_{operation}$ below cutoff frequency)

- practical structures combine waveguides and „discontinuities“
  … e.g., a waveguide excitation structure,
  … e.g., a waveguide bend
  … e.g., a transition to another waveguide (or another mode)
  … e.g., the end of a waveguide (e.g., short, match, open end)

- waveguide modes are analyzed as a 2D (cross-sectional) problem
  … 2D resonance → Eigenvalue problem

typical transmission line & waveguide cross sections

- two separate PEC structures in infinite vacuum
  … e.g., two-wire line
  … e.g., co-axial line

- vacuum-filled, completely enclosing PEC pipe
  … e.g., rectangular waveguide
  … e.g., circular, elliptical, ridge waveguide

- dielectric rod in infinite vacuum
  … e.g., simplification of optical fiber

- combination of PEC structures, dielectrics and vacuum
  … e.g., micro-strip line
  … e.g., optical fiber
  … e.g., surface-wave waveguides (slab, Goubau-line)

- consideration of lossy metals and lossy dielectrics

the effect of lossy metal — the only boundary is at infinity

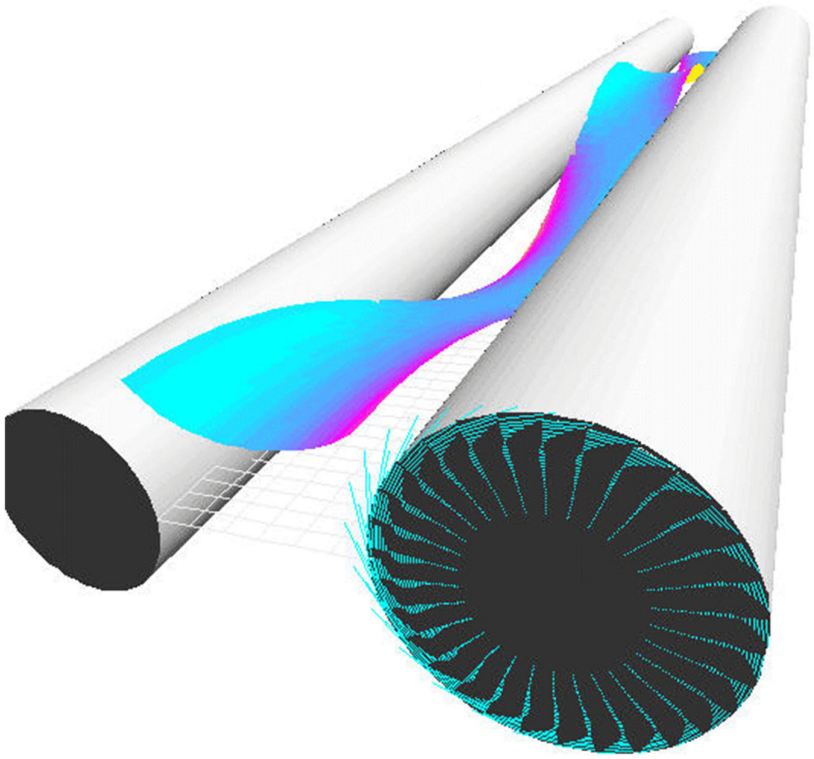
- magnetic field on a two wire line:

Picture shows magnitude of H field between conductors in the symmetry plane (grid)

Direction of H is perpendicular to this symmetry plane

Inside conductor (cross section): Magnitude of magnetic field and vectorial direction over radius

Note the skin effect and the large difference in the propagation speed along the line (f = 100 MHz, λ = 3m) and perpendicular into the wire ($\varnothing \approx$ 1m, σ = 10 S/m (!!!))



[P. Leuchtmann]

reducing the size of the cross section — boundaries

- most conductors are „good conductors“
  … PEC first to solve for the fields and consider finite σ thereafter
  … small discrepancy for high conductivity σ and „low“ frequency *f* :

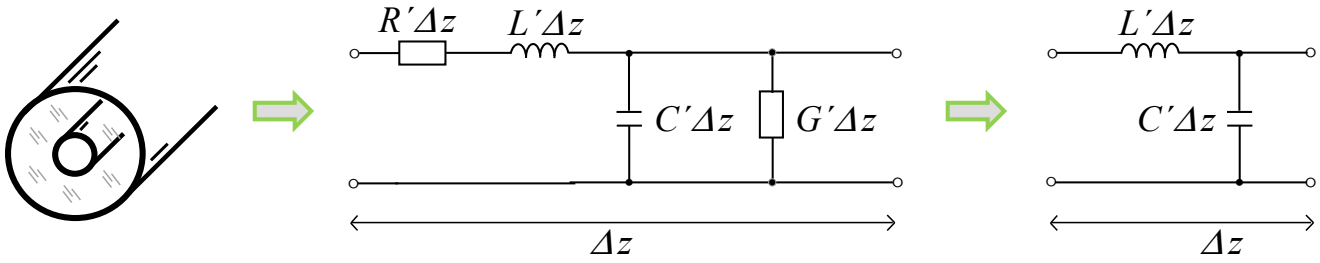
$$\delta \approx \sqrt{\frac{1}{\pi f \mu \sigma}}$$

… for a plane wave (normally) incidden on a conducting halfspace of conductivity σ , the amplitude of the wave is reduced to 1/e after a distance of δ

- often dielectrics are „low-loss dielectrics“
  … loss-less first to solve for the fields and consider tanδ thereafter
  … large loss will drastically change the fields (e.g., human body)

- helpful ist the use of symmetries
  … PEC symmetries (Dirichlet boudaries)
  … PMC symmetries (Neumann boundaries)
  … others, e.g., rotational symmetries (→ 1D problem)

transmission line equivalent circuit and wave propagation



- loss-less case:

$$U(z+\Delta z,t)-U(z,t)=-L'\,\Delta z\,\frac{\partial I(z,t)}{\partial t}$$

$$I(z+\Delta z,t)-I(z,t)=-C'\,\Delta z\,\frac{\partial U(z+\Delta z,t)}{\partial t}$$

$$I(z+\Delta z,t)-I(z,t)=-C'\,\Delta z\,\frac{\partial U(z,t)}{\partial t}+L'C'\,(\Delta z)^2\,\frac{\partial^2 I(z,t)}{\partial t^2}$$

$$\frac{\partial U(z,t)}{\partial z}=-L'\,\frac{\partial I(z,t)}{\partial t}\quad,\quad\frac{\partial I(z,t)}{\partial z}=-C'\,\frac{\partial U(z,t)}{\partial t}$$

...

transmission line equivalent circuit and wave propagation

$$\frac{\partial^2 U(z,t)}{\partial z^2}=L'C'\,\frac{\partial^2 U(z,t)}{\partial t^2}\quad,\quad\frac{\partial^2 I(z,t)}{\partial z^2}=L'C'\,\frac{\partial^2 I(z,t)}{\partial t^2}$$

„time harmonic“ (single frequency ω)

$$\frac{\partial^2 \underline{U}(z)}{\partial z^2}=-\omega^2\,L'C'\,\underline{U}(z)\quad,\quad\frac{\partial^2 \underline{I}(z)}{\partial z^2}=-\omega^2\,L'C'\,\underline{I}(z)$$

$$\underline{U}(z)=\underline{U}_0^+\,e^{-j\beta z}+\underline{U}_0^-\,e^{+j\beta z}\quad,\quad\underline{I}(z)=\frac{\underline{U}_0^+}{Z_0}\,e^{-j\beta z}-\frac{\underline{U}_0^-}{Z_0}\,e^{+j\beta z}$$

$$\beta=\omega\sqrt{L'C'}\quad,\quad v=\frac{\omega}{\beta}=\frac{1}{\sqrt{L'C'}}\quad,\quad Z_0=\sqrt{\frac{L'}{C'}}$$

- lossy case:

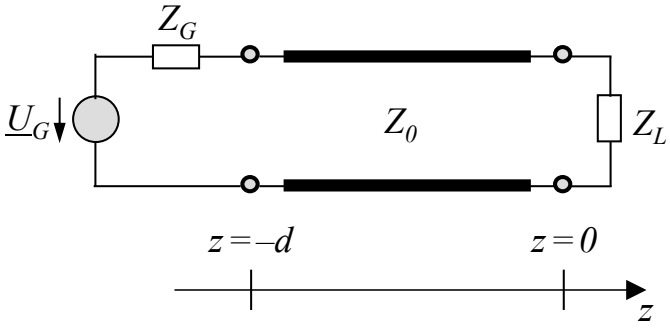
$$\underline{U}(z)=\underline{U}_0^+\,e^{-\alpha z}\,e^{-j\beta z}+\underline{U}_0^-\,e^{+\alpha z}\,e^{+j\beta z}$$

$$\underline{I}(z)=\frac{\underline{U}_0^+}{\underline{Z}_0}\,e^{-\alpha z}\,e^{-j\beta z}-\frac{\underline{U}_0^-}{\underline{Z}_0}\,e^{+\alpha z}\,e^{+j\beta z}$$

$$\gamma^2=(R'+j\omega L')(G'+j\omega C')=(\alpha+j\beta)^2\quad,\quad\underline{Z}_0=\sqrt{\frac{R'+j\omega L'}{G'+j\omega C'}}$$

reflection coefficient and impedance transformation

- reflection occurs at impedance mismatch between loss-less line Z_0 (real) ending at z=0 and load $\underline{Z}_L$ (complex):



$$\Gamma(z=0) = \Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0}$$

$$\Gamma(z=-d) = \Gamma_L\, e^{-2j\beta d}$$

$$Z(z=-d) = Z_0\,\frac{e^{j\beta d} + \Gamma_L\, e^{-j\beta d}}{e^{j\beta d} - \Gamma_L\, e^{-j\beta d}} = Z_0\,\frac{Z_L + jZ_0 \tan\beta d}{Z_0 + jZ_L \tan\beta d}$$

- special case: quarter-wave transformer:

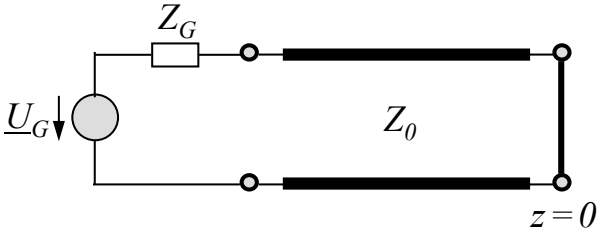
$\mathrm{d} = \lambda/4$
$\beta\mathrm{d} = \pi/2$
$\tan\beta\mathrm{d} \to \infty$

$$Z(z=-d) = \frac{Z_0^2}{Z_L}$$

reflection coefficient and impedance transformation

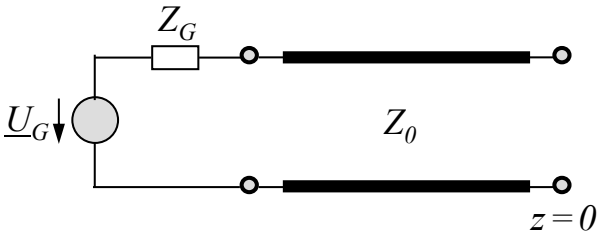
- input impedance of a short-circuited line (loss-less): (inductive if shorter than λ/4)

$$Z_{sc}(z=-d) = jZ_0 \tan\frac{2\pi d}{\lambda}$$

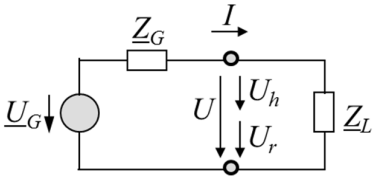


- input impedance of an open-circuited line (loss-less): (capacitive if shorter than λ/4)

$$Z_{oc}(z=-d) = \frac{-j\,Z_0}{\tan\frac{2\pi d}{\lambda}}$$



- generalized reflection coefficient: (complex source impedance)



$$\Gamma_p = \frac{\underline{Z}_L - \underline{Z}_G^*}{\underline{Z}_L + \underline{Z}_G}$$

reflection coefficient and impedance transformation

- Smith chart — a conformal mapping $\boldsymbol{r \leftrightarrow \zeta}$:

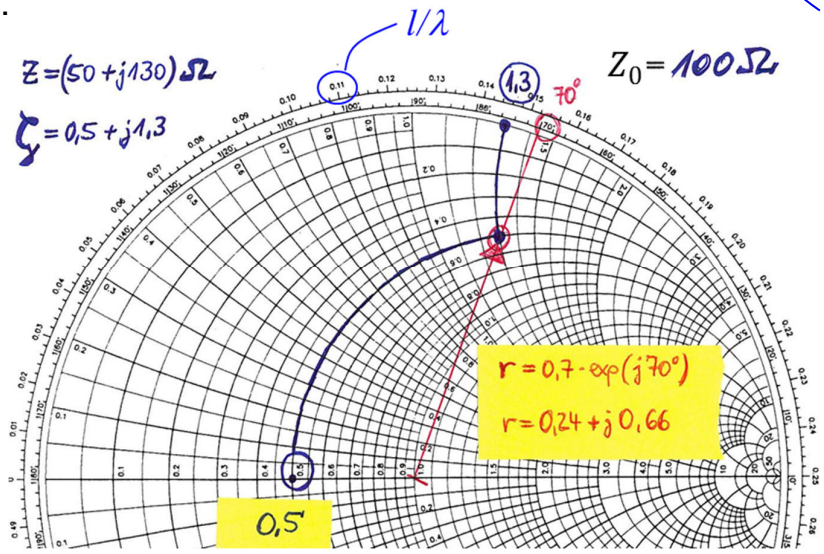
$$(\Gamma =)\, r = \frac{\frac{Z_L}{Z_0} - 1}{\frac{Z_L}{Z_0} + 1} = \frac{\zeta - 1}{\zeta + 1}$$

$$\zeta = \frac{Z_L}{Z_0}$$

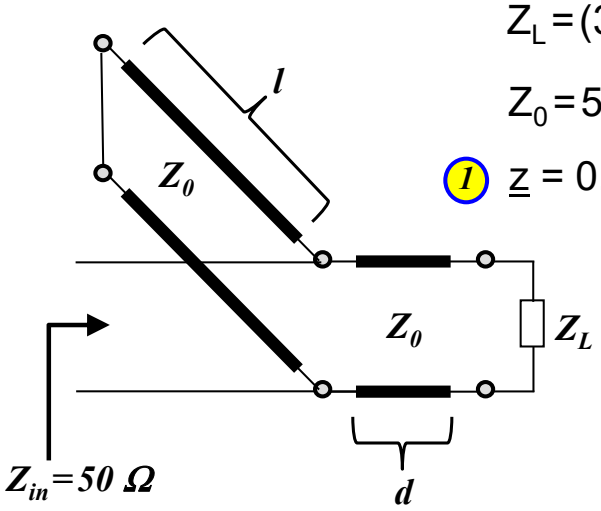
normalized impedance

Two coordinate systems overlapped:
→ cartesian coordinate system (not depicted) for $\underline{\Gamma}$
→ „circles coordinate system“ – shown – for $\boldsymbol{\zeta}$

- example:



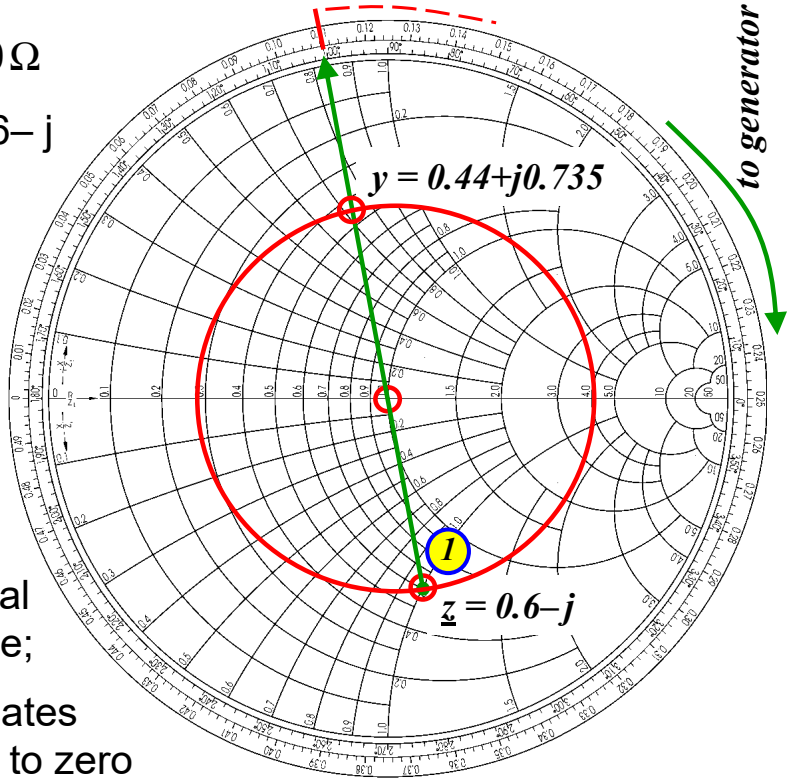
example: matching $\underline{Z}_L$ using line and shorted stub using Smith chart



$Z_L = (30 - j50)\,\Omega$

$Z_0 = 50\,\Omega$

① $\underline{z} = 0.6 - j$

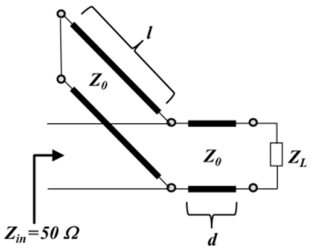


procedure:

(i) line of length *d* transforms real part of load admittance to one;

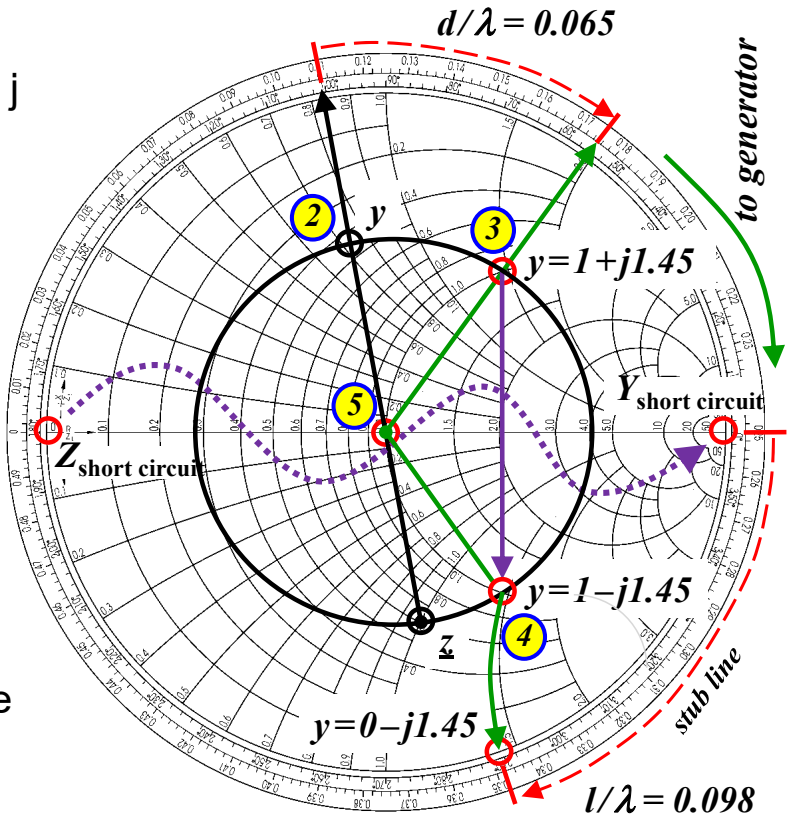
(ii) stub line of length *l* compensates imaginary part of admittance to zero

… matching $\underline{Z}_L$ using line and shorted stub using Smith chart (2/2)



$Z_0 = 50\,\Omega$

$\underline{z} = 0.6 - j$



(2) inversion $\underline{Z} \rightarrow \underline{Y}$

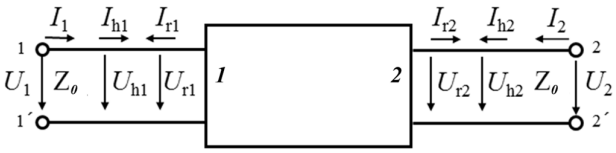
(3) rotate (CW) until real part (of transformed y) is one

(4) the admittance at the node becomes real, once the input admittance of the shorted stub (which is purely imaginary) cancels the

(5) sum of the two suseptances

S parameters

- consider a two-port (= four-pole) network, each port is connected to a line with impedance Z_0
- on the attached lines, current- and voltage-waves travel in both directions

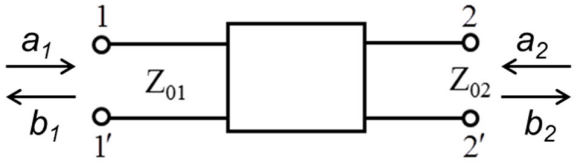


$$U_{r1} = S_{11} U_{h1} + S_{12} U_{h2}$$

$$U_{r2} = S_{21} U_{h1} + S_{22} U_{h2}$$

$$\begin{pmatrix} \underline{U}_{r1} \\ \underline{U}_{r2} \end{pmatrix} = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} \begin{pmatrix} \underline{U}_{h1} \\ \underline{U}_{h2} \end{pmatrix} = [\mathrm{S}] \begin{pmatrix} \underline{U}_{h1} \\ \underline{U}_{h2} \end{pmatrix}$$

- generalization: different reference-Z per port and “power waves” a and b (no U nor I needed)



$$a_i = \frac{U_{hi}}{\sqrt{Z_{0i}}} = I_{hi}\sqrt{Z_{0i}} \qquad b_i = \frac{U_{ri}}{\sqrt{Z_{0i}}} = I_{ri}\sqrt{Z_{0i}} \qquad [\sqrt{\mathrm{W}}]$$

$$u_{i,normiert} = \frac{U_{hi} + U_{ri}}{\sqrt{Z_{0i}}} = a_i + b_i \qquad [\sqrt{\mathrm{W}}]$$

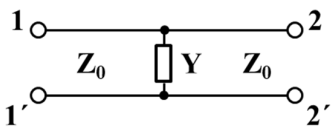
$$i_{i,normiert} = (I_{hi} - I_{ri})\sqrt{Z_{0i}} = a_i - b_i \qquad [\sqrt{\mathrm{W}}]$$

$$\begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = [\mathrm{S}] \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$$

S parameters: Example microwave circuits: shunt, power splitters

- shunt admittance:

$$[S] = \begin{bmatrix} -\frac{YZ_0}{2+YZ_0} & \frac{2}{2+YZ_0} \\ \frac{2}{2+YZ_0} & -\frac{YZ_0}{2+YZ_0} \end{bmatrix}$$



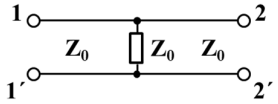
- splitter, loss-less, not matched:

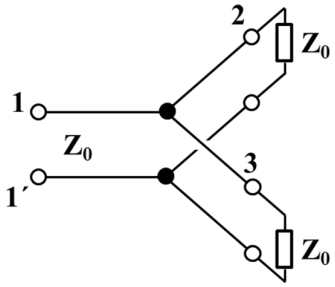
$$[S] = \frac{1}{3}\begin{bmatrix} -1 & 2 & 2 \\ 2 & -1 & 2 \\ 2 & 2 & -1 \end{bmatrix}$$

$$1 = |S_{11}|^2 + |S_{21}|^2 + |S_{31}|^2$$

$$S_{11} = \frac{Z_0 || Z_0 - Z_0}{Z_0 || Z_0 + Z_0} = \frac{-\frac{1}{2}Z_0}{\frac{3}{2}Z_0} = -\frac{1}{3}$$

$$S_{21} = \frac{2}{3} \quad :$$





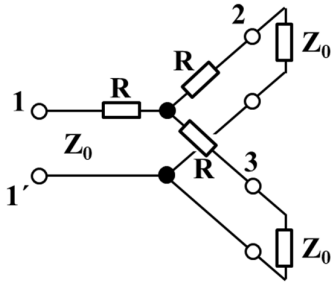
- splitter, lossy, matched:

$$Z_0 = R + \frac{1}{2}(R + Z_0) \quad \rightarrow \quad R = \frac{Z_0}{3}$$

$$[S] = \frac{1}{2}\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

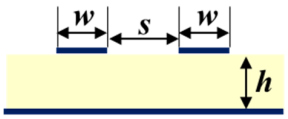
Insertion loss

6.5 dB, DC to 18 GHz
7 dB, 18 GHz to 26.5 GHz
8 dB, 26.5 GHz to 40 GHz
8.5 dB, 40 GHz to 50 GHz

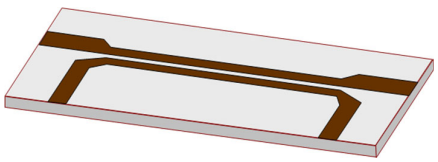


S parameters: Example microwave circuits: tandem coupler

- quarter-wave coupled-line directional coupler:



coupled lines

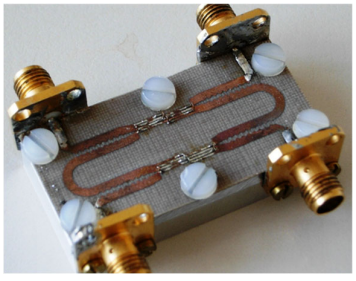


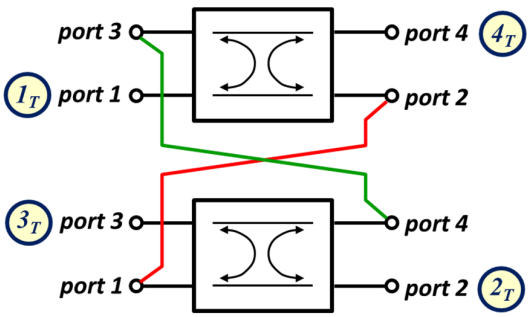
example microstrip layout

$$[S] = \begin{bmatrix} 0 & S_{21} & S_{31} & 0 \\ S_{21} & 0 & 0 & S_{31} \\ S_{31} & 0 & 0 & S_{21} \\ 0 & S_{31} & S_{21} & 0 \end{bmatrix}$$

coupler S matrix

- increased effective coupling coefficient ($k_T > k$) — tandem coupler:





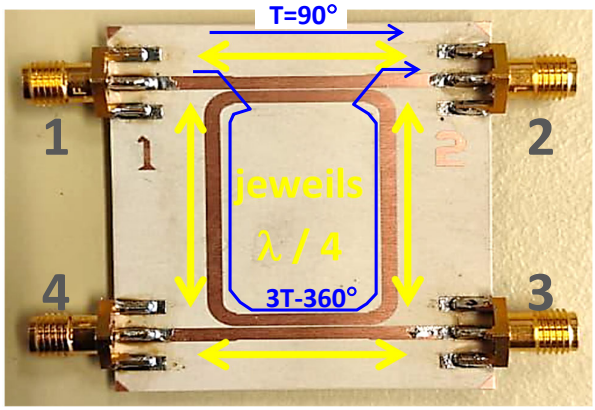
$$S_{21,T} = S_{21}^2 + S_{31}^2$$

$$S_{31,T} = 2\, S_{21}\, S_{31}$$

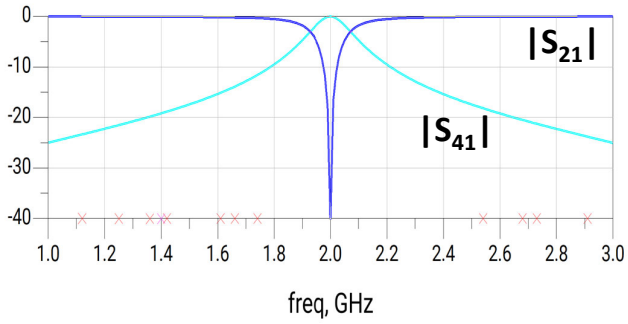
$$|S_{31,T}| = 2\,|S_{31}|\sqrt{1-|S_{31}|^2}$$

$$k_T = 2\,k\sqrt{1-k^2}$$

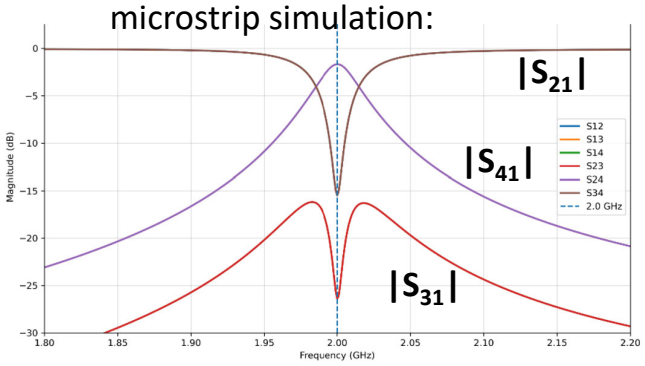
S parameters: Example microwave circuits: ring resonator filter



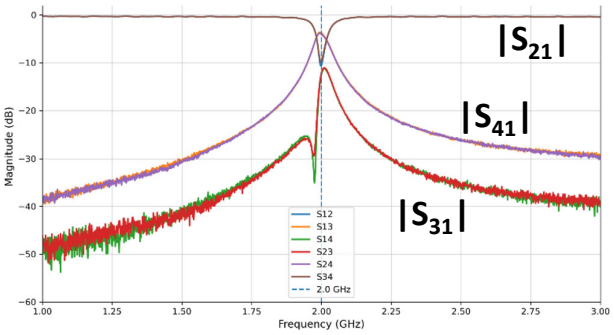
ideal simulation:



microstrip simulation:



measurement:



Transmission Lines & Waveguides

… see also script/slides „Hochfrequenztechnik 1“ (in English) → ILIAS

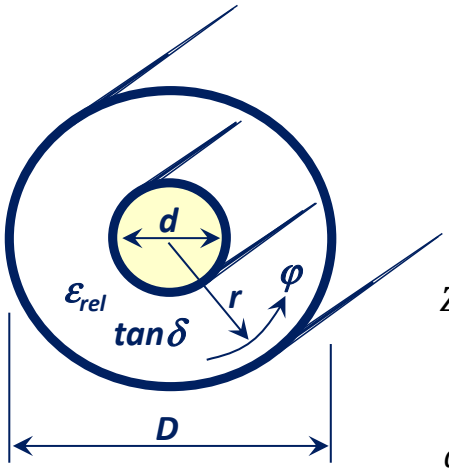
- general comments
- transmission line equivalent circuit
- impedance transformation
- S parameters and few example circuits

brief repetition

- coaxial waveguide
- planar lines
- rectangular waveguide & circular waveguide
- slab waveguide & dielectric waveguides

coaxial waveguide

… in its cross section, the coaxial waveguide possesses two isolated conductors, therefore, it can carry a TEM mode with zero cutoff



$$H_\varphi = \frac{I}{2\pi r}$$

$$E_r = \frac{U}{r \ln {}^{D}/_{d}}$$

$$Z_{coax} = \frac{60\ \Omega}{\sqrt{\varepsilon_{\mathrm{rel}}}} \ln {}^{D}/_{d}$$

$$\alpha_{cond} = \sqrt{\frac{f\mu}{\pi\sigma}} \frac{\left(1 + \frac{D}{d}\right)}{2DZ_L}$$

$$\alpha_{diel} = \frac{\pi f}{c_0} \sqrt{\varepsilon_{\mathrm{rel}}} \tan\delta$$

… for a given outer diameter D, the lowest conductor loss is when:

$$Z_{coax,lowloss} = \frac{77\ \Omega}{\sqrt{\varepsilon_{\mathrm{rel}}}}$$

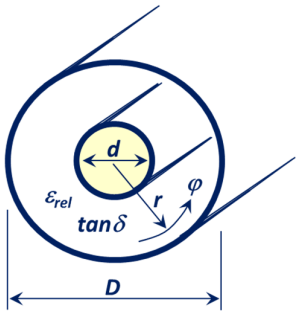
… about 50 Ω for „normal“ $\varepsilon_{\mathrm{rel}}$

… *very low* frequency behaviour:

$$\underline{Z}_0 = \sqrt{\frac{R' + j\omega L'}{G' + j\omega C'}}$$

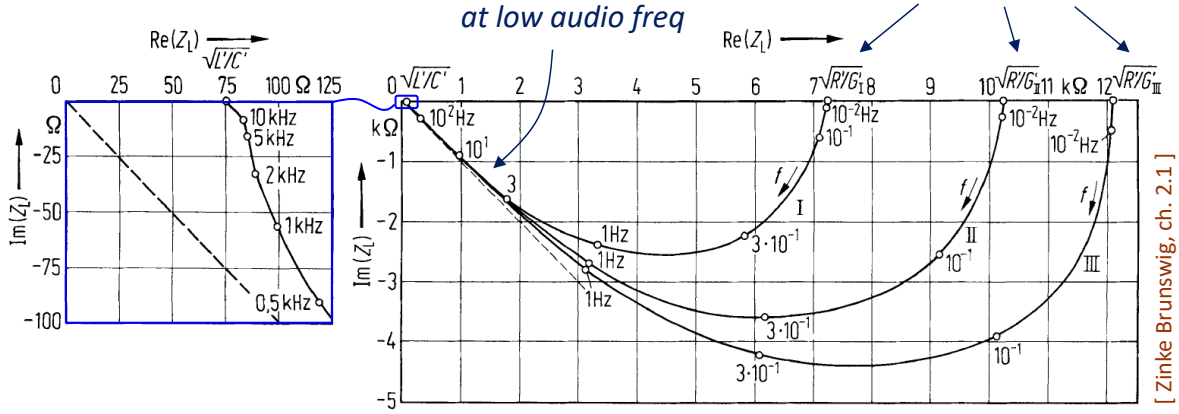
low freq $\quad \underline{Z}_0 = \sqrt{\frac{R'}{G'}}$

high freq $\quad \underline{Z}_0 = \sqrt{\frac{L'}{C'}}$



phase($\underline{Z}_0$) ≈ -45 ° at low audio freq

different cable geometries / loss contributions

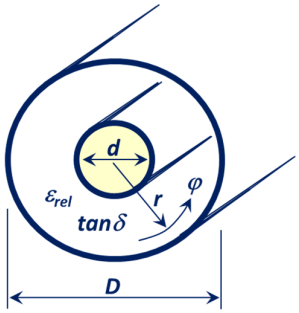


[Zinke Brunswig, ch. 2.1]

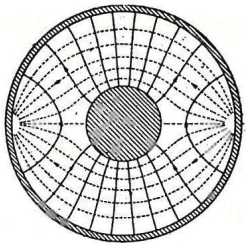
… the larger the cross section, the lower the conductive loss at microwaves. But …

… at some high frequency, other modes will appear

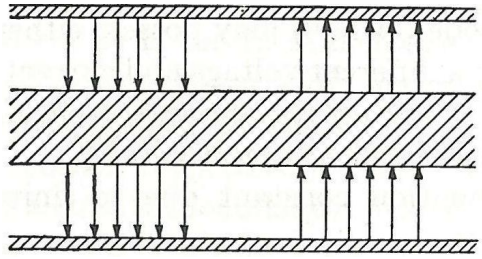
… of practical importance is the cutoff of the first non-TEM mode, the TE_{11} mode:



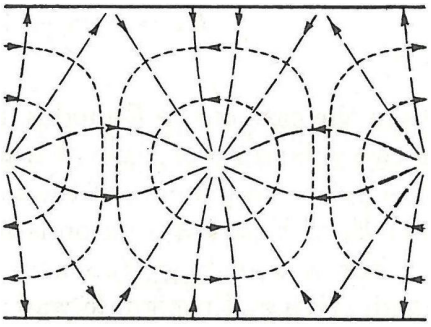
$$f_{\text{cutoff, TE11}} \approx \frac{2c_0}{\pi\sqrt{\varepsilon_{\text{rel}}}\,(D+d)}$$



cross section



vertical cut, E-field



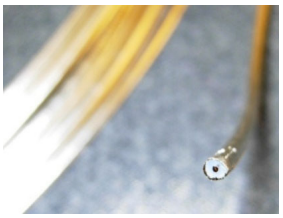
inner surface, H-field and currents

coaxial cables

… large cross section and low permittivity needed for low loss

… at high frequency, cross section is small / restricted due to the appearance of higher-order modes

… example: semi-flexible / semi-rigid cables:



(typical datasheet values) (PTFE/FEP dielectric, $e_{rel} \approx 2.0$) $f_{\text{cutoff, TE11}} \approx \frac{2c_0}{\pi\sqrt{\varepsilon_{\text{rel}}}\,(D+d)}$

cable type:	d_{inner} [mm]	d_{outer} [mm]	$\varnothing_{outer}$ [mm]	$\varnothing_{outer}$ ["]	$f_{c,TE11}$ [GHz]	loss/$_{1GHz}$ [dB/m]	loss/$_{18GHz}$ [dB/m]	loss/$_{40GHz}$ [dB/m]	loss/$_{65GHz}$ [dB/m]
23	0.142	0.432	0.584	0.023	235			13.8	17.6
34	0.20	0.66	0.864	0.034	157			10.9	14.3
47	0.31	0.94	1.19	0.047	108	1.17	5.52	8.75	
86	0.53	1.65	2.18	0.086	62	0.67	3.39	5.57	
141	0.95	2.95	3.58	0.141	34.6	0.39	2.23		
250	1.67	5.24	6.35	0.250	19.5	0.24	1.45		

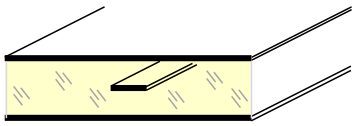
… alternative:

… „U.FL“ coax connector on 0.40 mm dielectric diameter (outer shield diameter 0.81 mm) cable: 6.5 dB/m @ 3 GHz, 9.5 dB/m @ 6 GHz !

planar lines

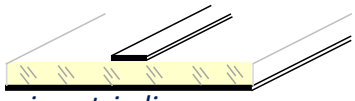
… to be realized in (multi-layer-) planar technology
(printed circuit boards / PCB,
low-temperature co-fired ceramic / LTCC,
on-chip back-end-of-line / BEOL).

TEM

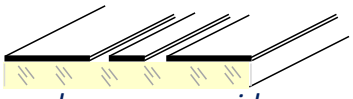


stripline

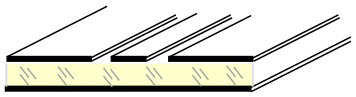
quasi-TEM



microstrip line

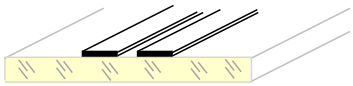


co-planar waveguide

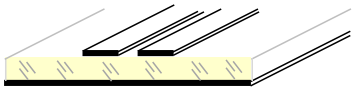


grounded co-planar waveguide

balanced

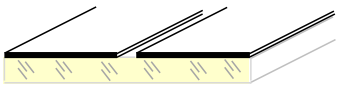


co-planar strips



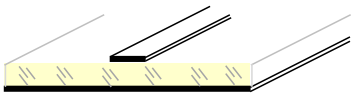
grounded co-planar strips

non-TEM

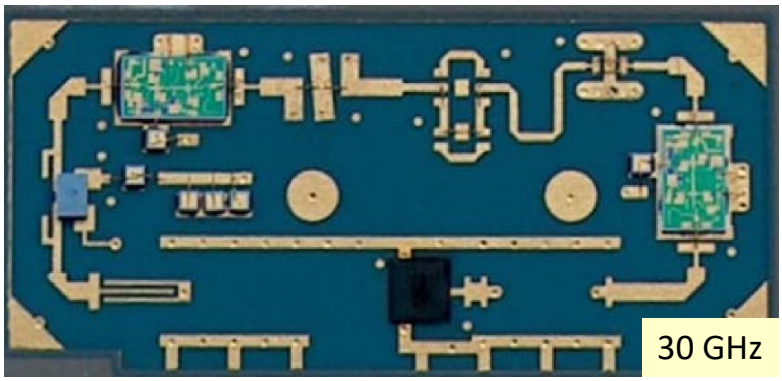


slotline

microstrip line

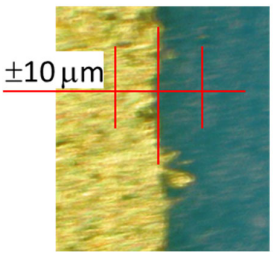


- most commonly used
  planar transmission line at microwaves
- zero cutoff, low dispersion, „quasi-TEM“
- thin, high-ε substrate concentrates fields below strip,
  causing higher loss.
- current concentrates at strip edges



microstrip circuit

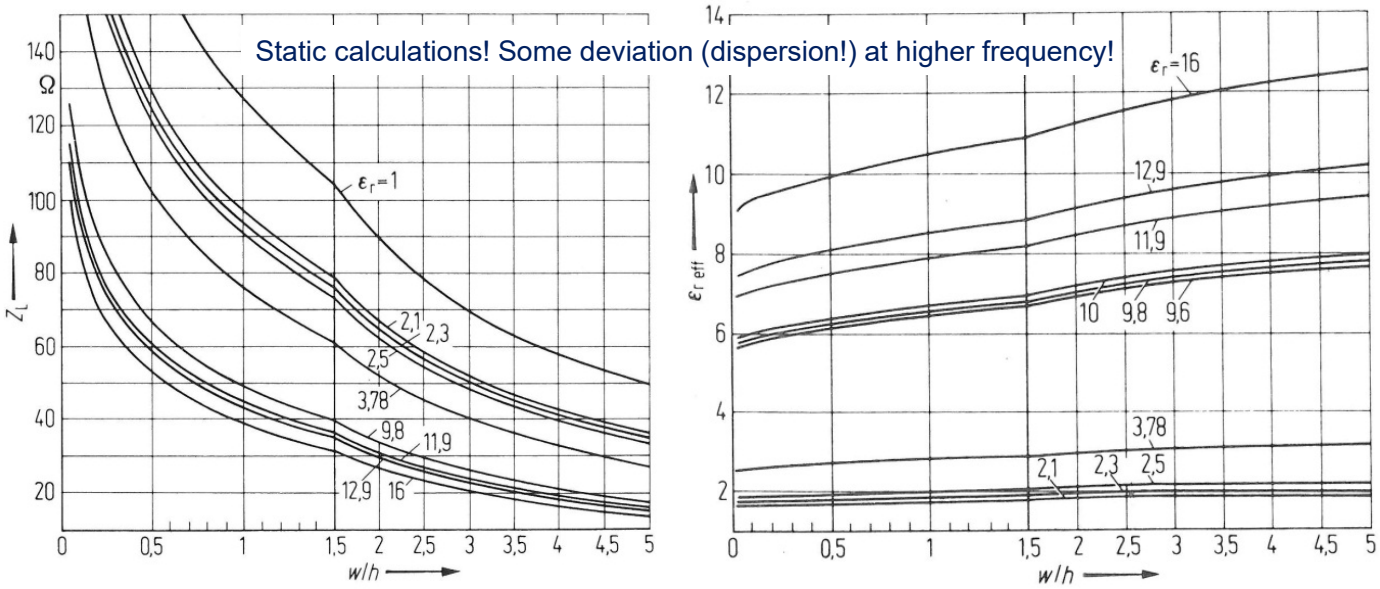
metal trace edge
(thickfilm on LTCC):



critical strip edge

microstrip line

- cross-sectional geometry defines impedance and effective permittivity:

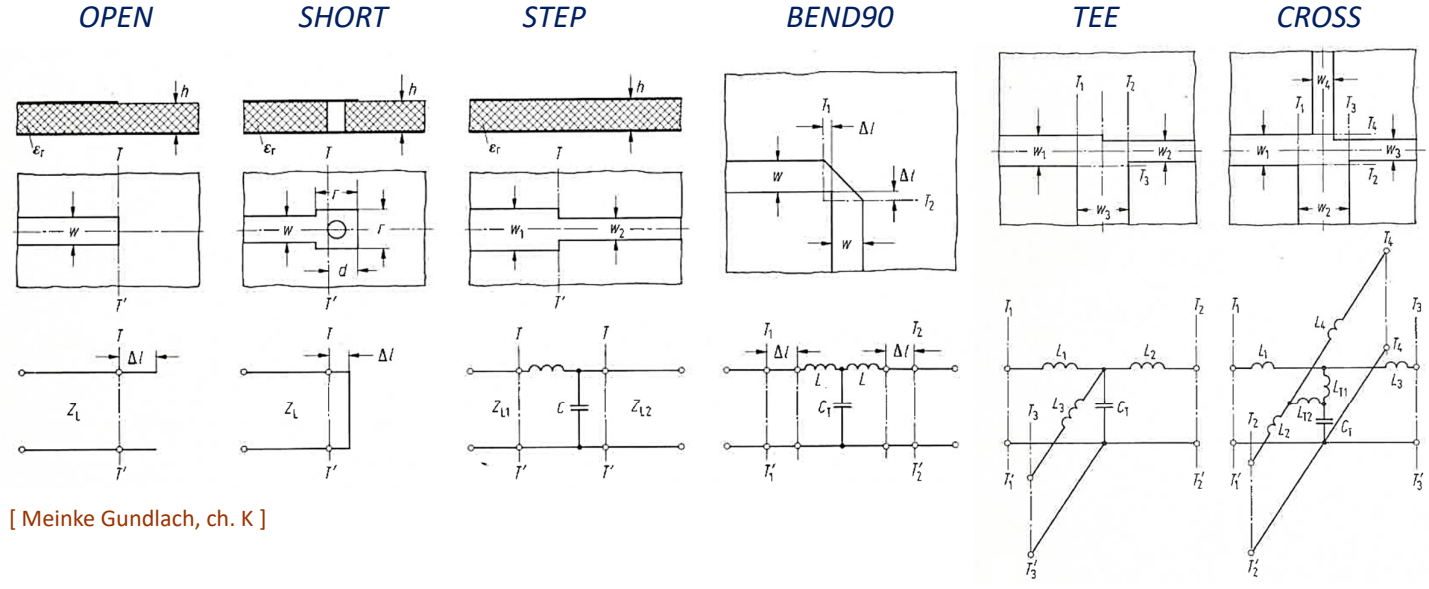


[Zinke Brunswig] , [Wheeler 1977]

- impedance range limited to approx. ratio 2…2.5 : 1

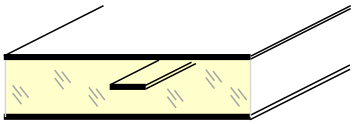
microstrip line

- many well-developed models of discontinuities available:

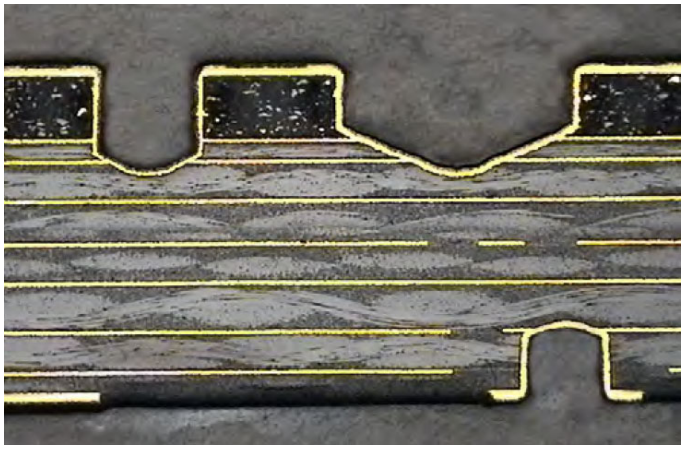


[Meinke Gundlach, ch. K]

stripline



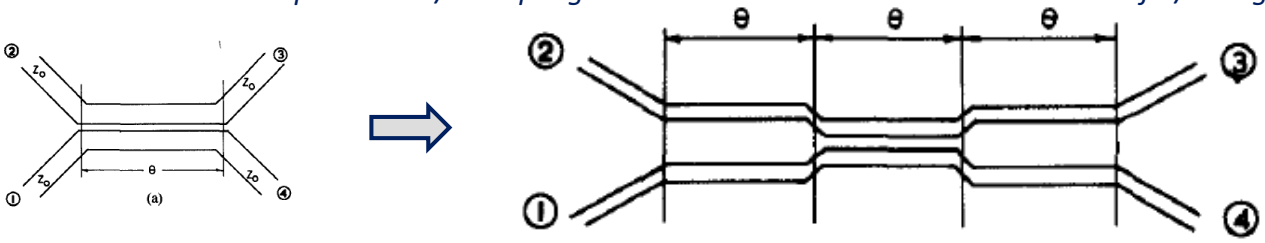
- TEM line with close-to-zero dispersion
- standard in multilayer boards
- special application in directional couplers:
  strong coupling by broadside coupling



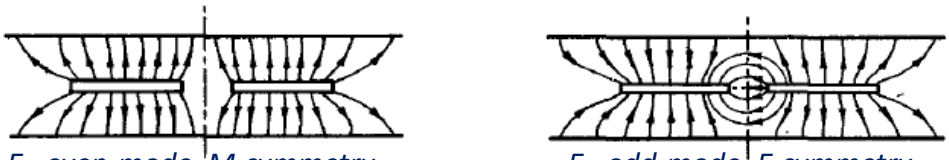
cut through multi-layer circuit board

- directional couplers:
  coupled striplines and offset-broadside-coupled striplines

coupled-line directional coupler with λ/4 coupling → broadband version with 3 sections of λ/4 length

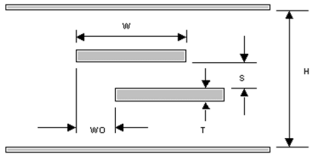


coupling described by even-mode-odd-mode impedances (and even/odd propagation constants):



E , even-mode, M-symmetry *E , odd-mode, E-symmetry*

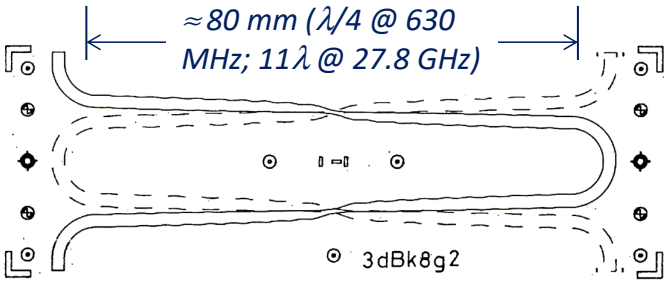
strong coupling achieved by broadside coupling → offset-broadside-coupled stripline:

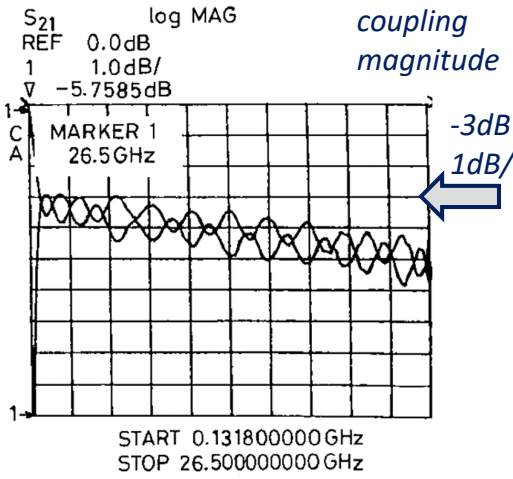


- directional coupler example:
  a 5-octave 3dB directional coupler in offset-broadside-coupled stripline

750 MHz to 26.5 GHz

tandem coupler topology (two couplers of -8.34 dB realize one coupler of -3dB)



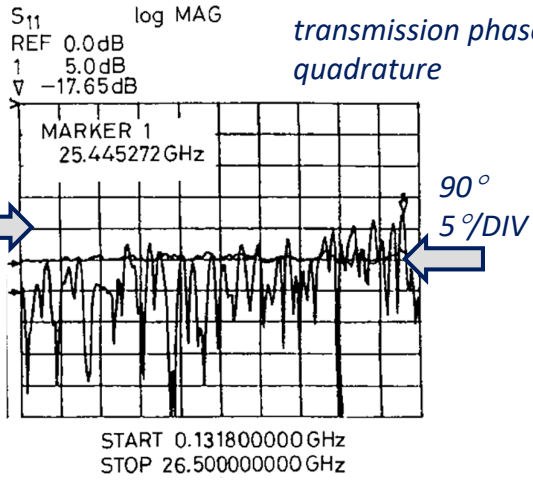


coupling magnitude

-3dB
1dB/DIV

input reflection

-20dB
5dB/DIV

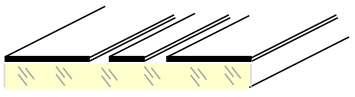


transmission phase quadrature

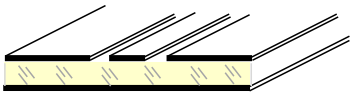
90°
5°/DIV

[Potter Hjipieris, IEE-Proc-H, 1992]

co-planar waveguide CPW and grounded co-planar waveguide GCPW



CPW

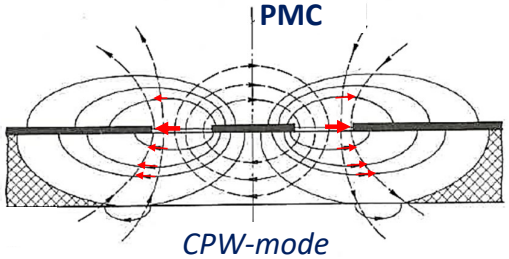


GCPW

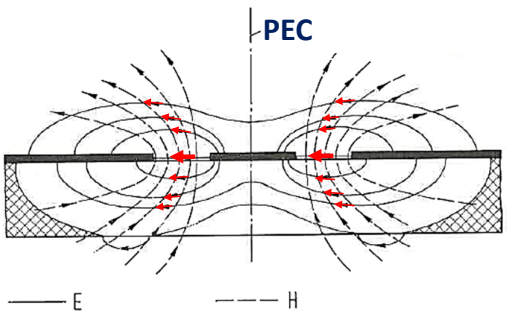
- CPW preferred on-chip line:
  - + does not require via-through for short
  - + lower loss and higher impedance than microstrip on very thin dielectric

- GCPW preferred on-board (top-layer) line:
  - + less coupling to close-by neighboring lines

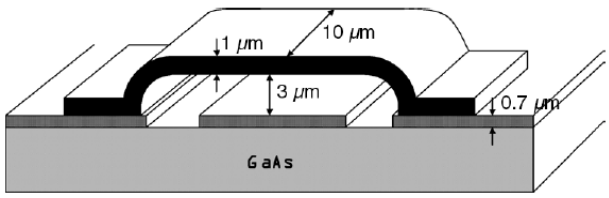
- CPW is a 3-conductor, multi-mode structure:

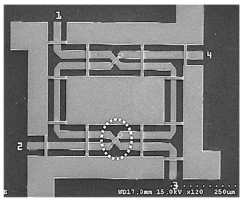


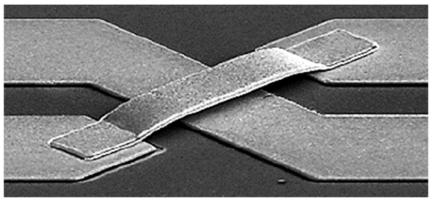
CPW-mode G-S-G



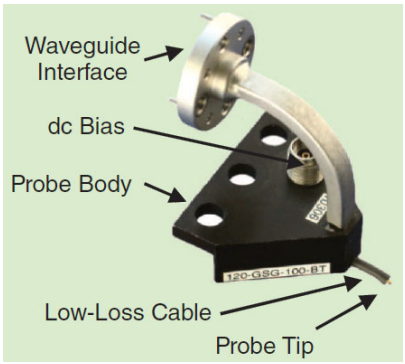
- CPW needs air bridges for ground balance at asymmetric discontinuities:





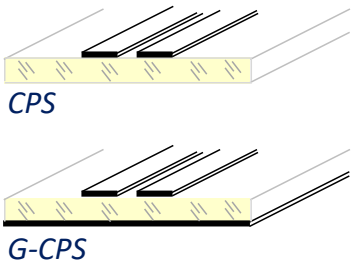


- CPW is the best choice for probe tips for (calibrated) on-chip RF testing:

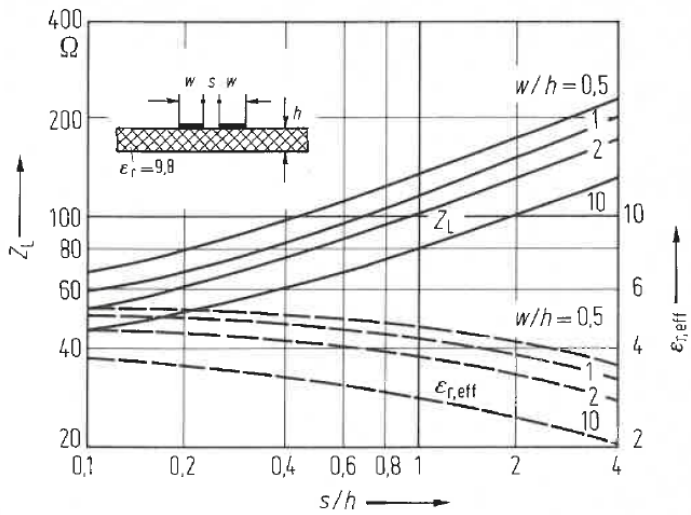


[Rumiantsev, Doerner, Microwave Magazine, 2013]

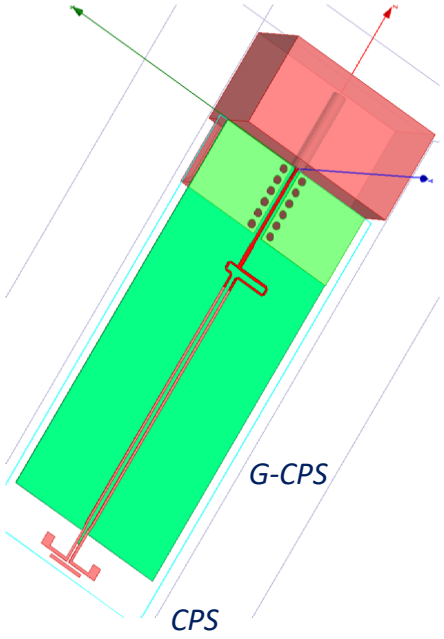
co-planar strips CPS and grounded co-planar strips

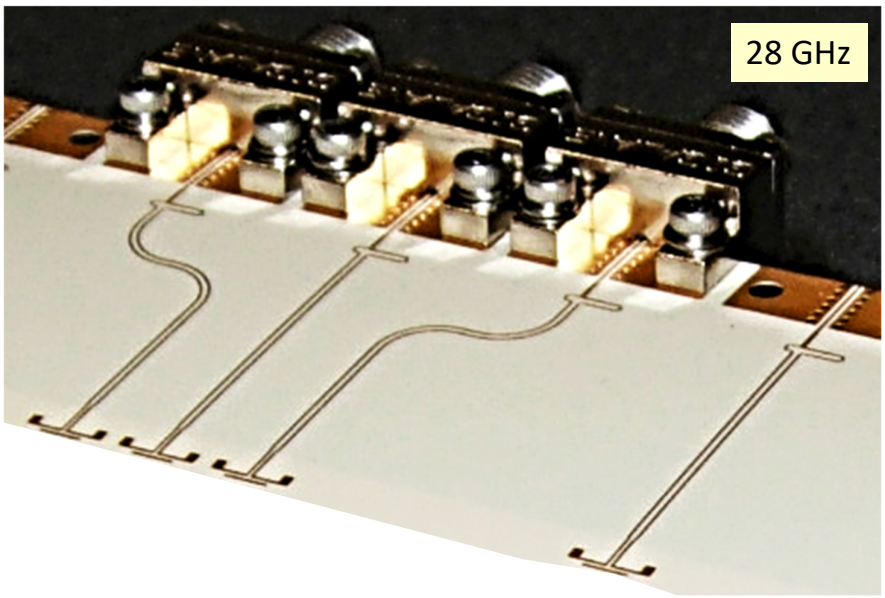


- zero cutoff and rather low dispersion
- balanced line
  but unbalanced mode possible (for G-CPS)
- applications: antenna feed, mixers and low-noise interconnects

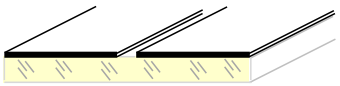


- example: dipole antenna feed

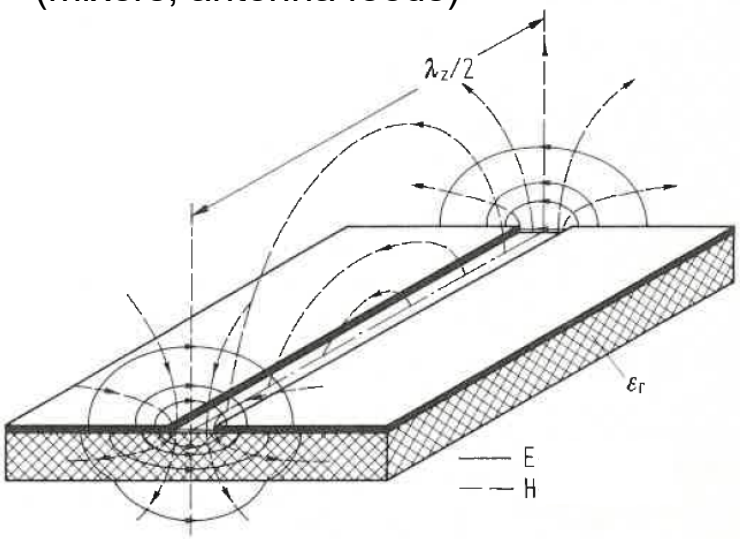


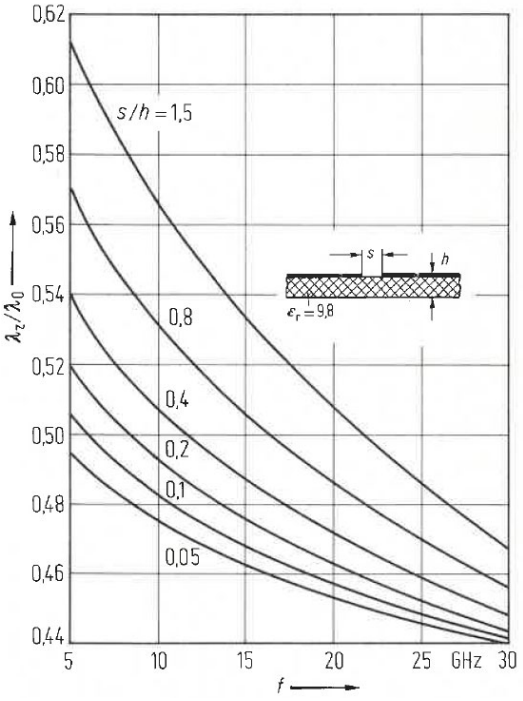


slotline

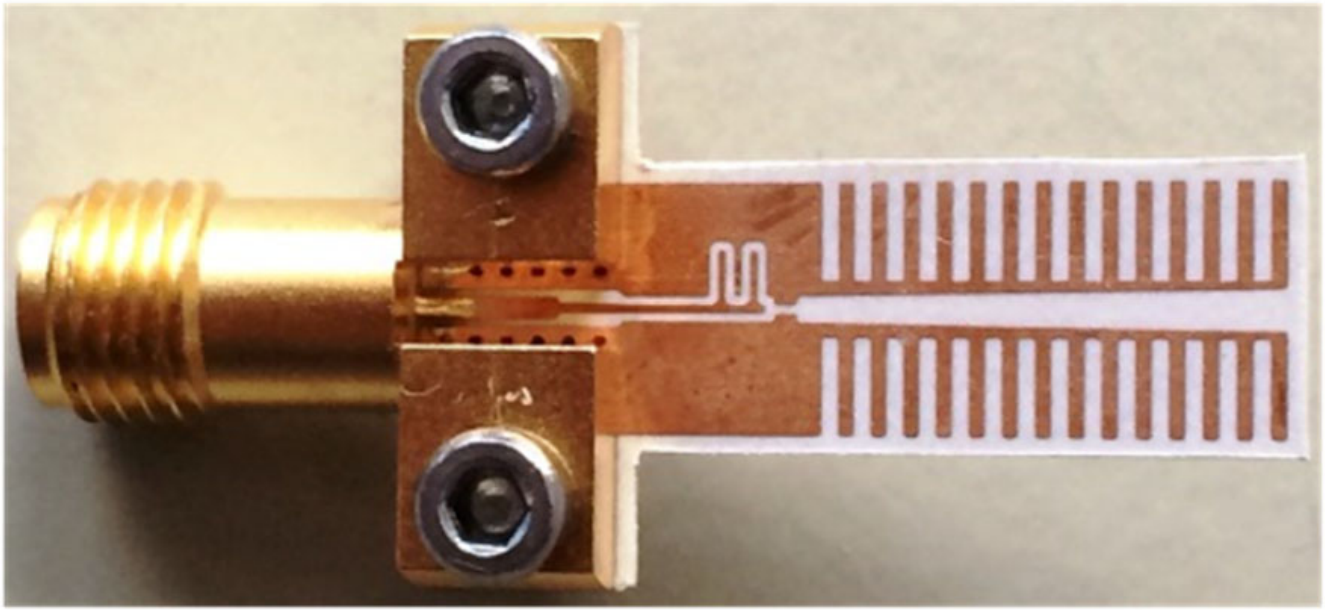


- zero cutoff but significant dispersion
- field extends significantly in transverse plane
- TE (significantly non-TEM)
- applications: special cases (mixers, antenna feeds)





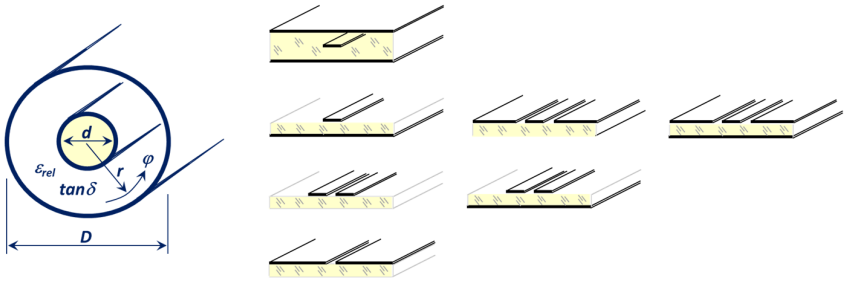
transition coax-to-slotline example (30 GHz)



coax *grounded co-planar waveguide* *co-planar waveguide* *slotline* *antenna*

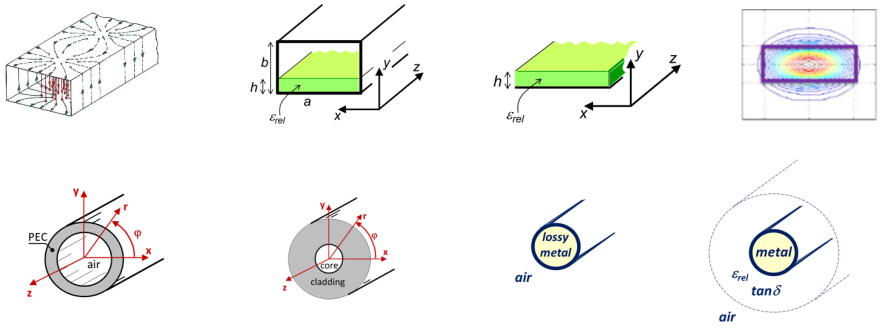
so far:

„transmission lines“



what is next:

„waveguides“



Note that „transmission lines“ are in fact „waveguides“ and *vice versa*

what is a waveguide mode

- math.: Eigen-solution of the wave equation in the cross section
  (i.e., a resonance in the transverse plane)

- engineering: the „way“ a single-frequency signal
  is allowed to travel along the waveguide

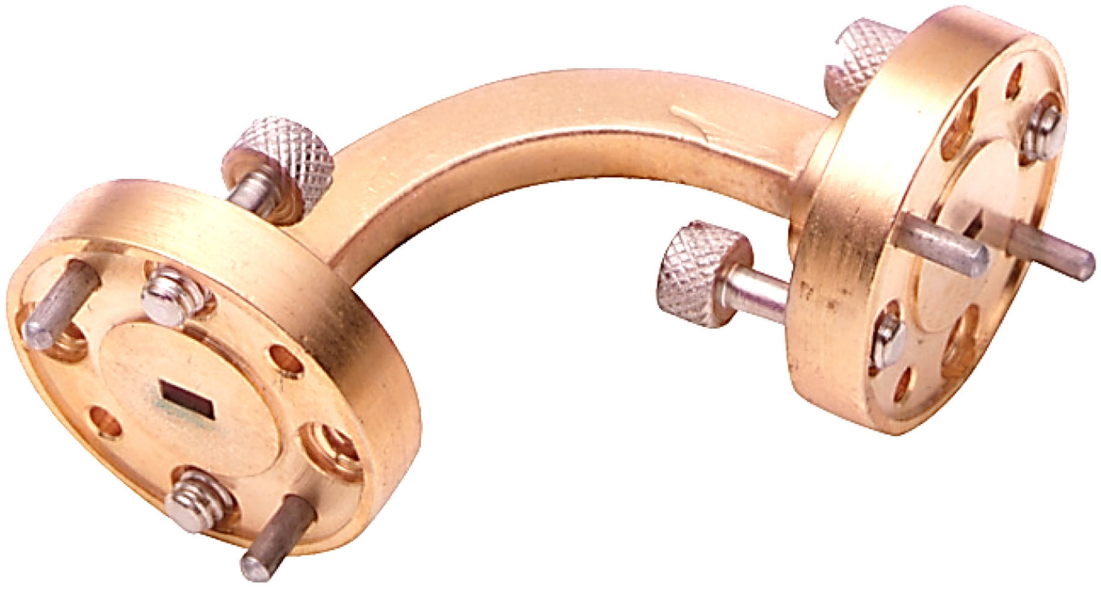
  as described by electromagnetic fields, guided wavelength, propagation constant etc

  several possible „ways“ mean several different modes

  that is, only some („some“ = discrete number, can be one or zero) field distributions / ways of signal propagation are allowed in/along a given waveguide at a given frequency

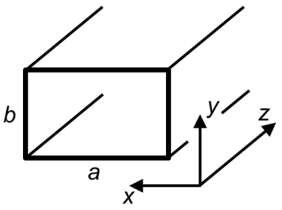
To characterize waveguides in the following, the wave equation will be solved in the transverse plane for the given waveguide cross-section

rectangular waveguide

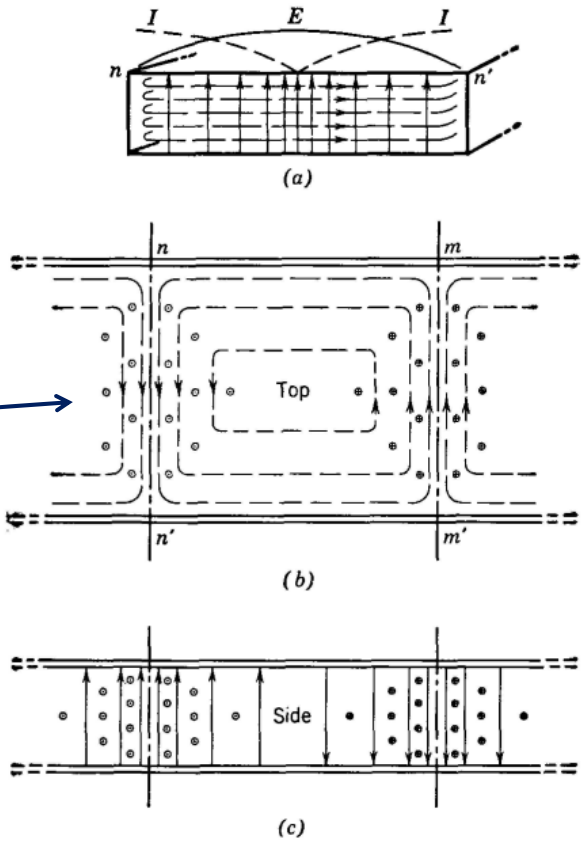


The fields and wave propagation characteristics of the fundamental mode (TE_{10}) in the (straight) rectangular waveguide can be derived from plane-wave considerations (which is supposed to be known from „microwaves 101“ and repeated in the following)

the TE_{10} mode in the rectangular waveguide



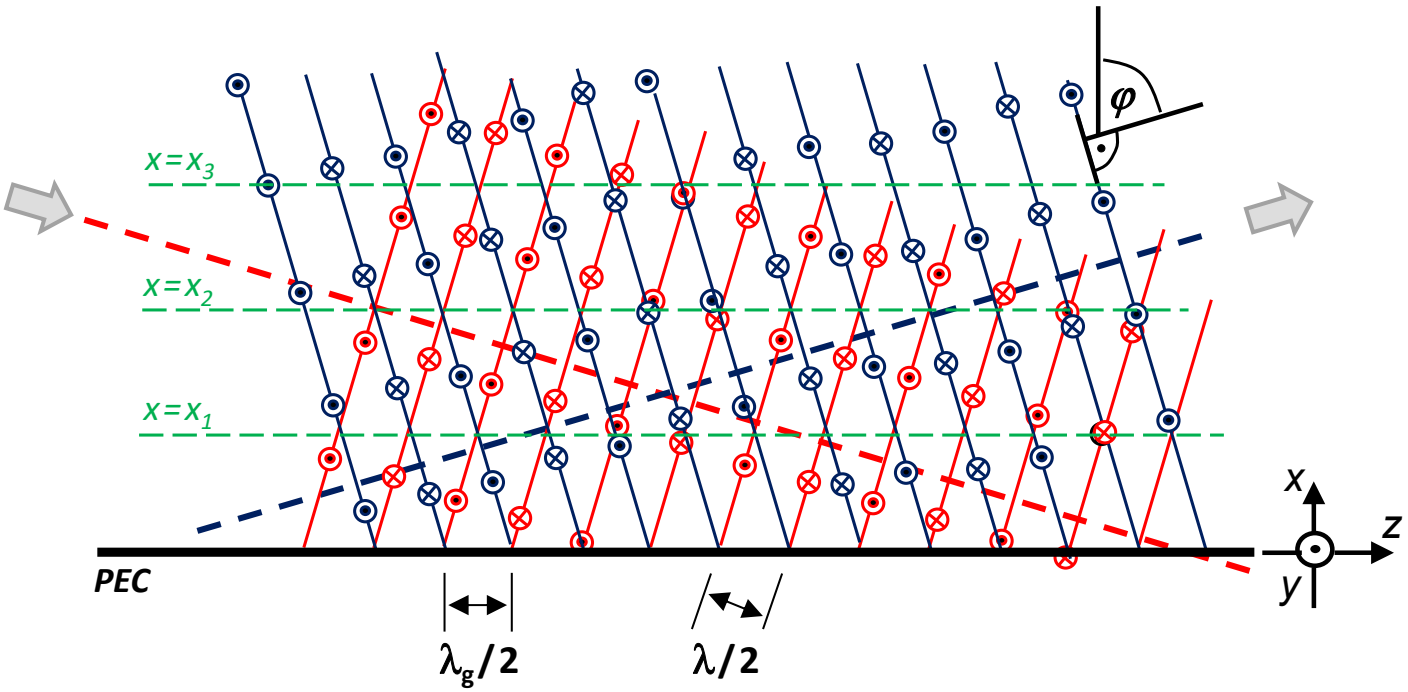
- dashed : H
- solid : E



[A.L. Albert, Electrical Communication, 1964]

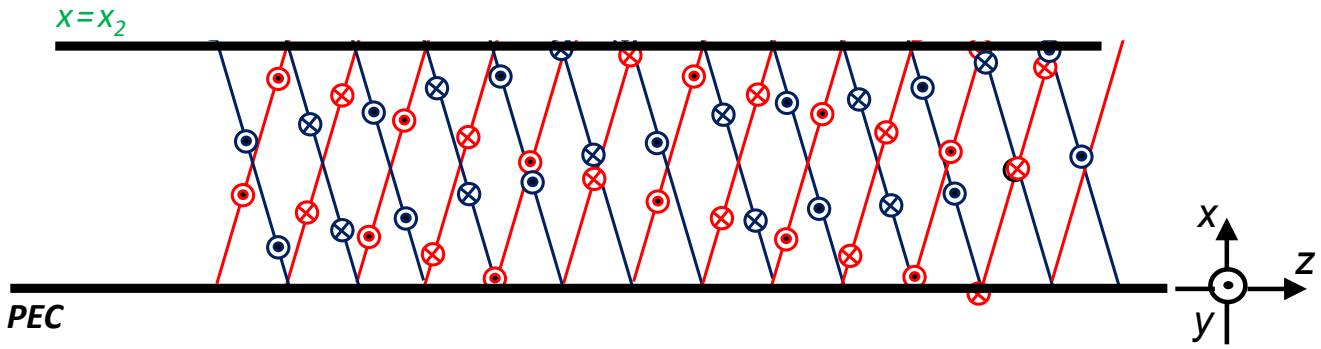
Ey and Hx are maximum at the same z locations

Reflection of a plane wave at a PEC surface (normal polarized E field):



- extinction of ***E*** at $x=0$ (boundary condition PEC) and at $x=x_2$
- doubling of ***E*** at $x=x_1$ and at $x=x_3$

Introduction of a second PEC surface at $x = x_2$ without disturbing fields (gives a parallel-plate waveguide):



Two additional PEC surfaces in xz-plane could be added (without disturbing the fields!) thereby forming rectangular waveguide TE_{10} mode

the modes in the rectangular waveguide derived from Maxwell

- vacuum filled rectangular PEC pipe, dimensions : width *a*, height *b*

Ampère's law

$$\operatorname{curl} \vec{H} = \sigma \vec{E} + \varepsilon \frac{\partial \vec{E}}{\partial t}$$

Faraday's law

$$\operatorname{curl} \vec{E} = -\mu \frac{\partial \vec{H}}{\partial t}$$

$$\operatorname{curl}\operatorname{curl} \vec{H} = -\sigma\mu \frac{\partial \vec{H}}{\partial t} - \varepsilon\mu \frac{\partial^2 \vec{H}}{\partial t^2} \quad , \quad \operatorname{curl}\operatorname{curl} \vec{E} = -\sigma\mu \frac{\partial \vec{E}}{\partial t} - \varepsilon\mu \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\operatorname{curl}\operatorname{curl} \underline{\vec{A}} = \operatorname{grad}\operatorname{div} \underline{\vec{A}} - \Delta \underline{\vec{A}}$$

$$\Delta \vec{H} = \sigma\mu \frac{\partial \vec{H}}{\partial t} + \varepsilon\mu \frac{\partial^2 \vec{H}}{\partial t^2} \qquad \Delta \vec{E} = \sigma\mu \frac{\partial \vec{E}}{\partial t} + \varepsilon\mu \frac{\partial^2 \vec{E}}{\partial t^2}$$

- time-harmonic case : d/dt = jω
- Laplacian (in cartesian coordiantes) : $\Delta f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$
- separation of the axis of the waveguide, e.g., z
- transverse-Laplacian (in cartesian coordiantes) : $\Delta_T = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$

$$\Delta_T \vec{E} + \frac{\partial^2 \vec{E}}{\partial z^2} = -\varepsilon\mu\omega^2 \vec{E}$$

- simplification: propagating mode in ±z direction : $\vec{E}(z) \sim e^{\mp j\beta z}$

$$\Delta_T \vec{E} + k_T^2 \vec{E} = 0 \qquad k_T^2 = k_0^2 - \beta^2 = \varepsilon\mu\omega^2 - \beta^2 \qquad k_T^2 = k_x^2 + k_y^2$$

$$\Delta_T \vec{H} + k_T^2 \vec{H} = 0$$

- if two out of the six scalar field components fulfill these equations, then the other four will do, too → chose Ez , Hz

$$(\Delta_T + k_T^2)\begin{pmatrix} E_z \\ H_z \end{pmatrix} = 0$$

… equation of Helmholtz

- trigonometric functions are typically used as a solution of these 2nd order differential equations (Ansatz using separation of variables):

$$\overrightarrow{E_z}(x,y,z,t) = \overrightarrow{e_z}\, E_{z0}\, e^{j\omega t} e^{-j\beta z} (A\cos k_x x + B \sin k_x x)\big(C \cos k_y y + D \sin k_y y\big)$$

$$\overrightarrow{H_z}(x,y,z,t) = \overrightarrow{e_z}\, H_{z0}\, e^{j\omega t}\, e^{-j\beta z}\, (E\cos k_x x + F \sin k_x x)\big(G \cos k_y y + H \sin k_y y\big)$$

… where the unknowns A,B,C,D,E,F,G,H are determined applying boundary conditions,
… the remaining four field components are derived from Ez and Hz

- once Ez and Hz are known, the other four components are readily found:

$$\text{curl}\,\vec{E} = \begin{vmatrix} \overrightarrow{e_x} & \overrightarrow{e_y} & \overrightarrow{e_z} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ E_x & E_y & E_z \end{vmatrix} = -j\omega\mu\,\vec{H} \qquad \text{curl}\,\vec{H} = \begin{vmatrix} \overrightarrow{e_x} & \overrightarrow{e_y} & \overrightarrow{e_z} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ H_x & H_y & H_z \end{vmatrix} = j\omega\varepsilon\,\vec{E}$$

$$\frac{\partial}{\partial z} \equiv -j\beta \qquad \frac{\partial^2}{\partial z^2} \equiv -\beta^2$$

… example Hx :

$$\overrightarrow{H_x} = H_x\,\overrightarrow{e_x}$$

$$H_x = \frac{1}{-j\omega\mu}\left(\frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z}\right) \qquad \frac{\partial E_y}{\partial z} = -j\beta E_y$$

$$E_y = \frac{1}{j\omega\varepsilon}\left(\frac{\partial H_x}{\partial z} - \frac{\partial H_z}{\partial x}\right) \qquad \frac{\partial H_x}{\partial z} = -j\beta H_x$$

$$H_x =$$

$$H_x = \frac{j}{\omega^2\varepsilon\mu - \beta^2}\left(\omega\varepsilon\frac{\partial E_z}{\partial y} - \beta\frac{\partial H_z}{\partial x}\right)$$

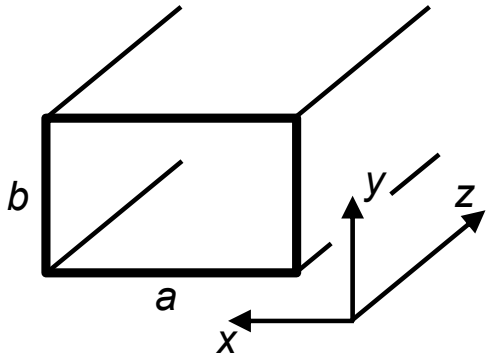
… similarly for Hy, Ex, Ey :

$$H_y = -\frac{j}{\omega^2\varepsilon\mu - \beta^2}\left(\omega\varepsilon\frac{\partial E_z}{\partial x} + \beta\frac{\partial H_z}{\partial y}\right)$$

$$E_x = -\frac{j}{\omega^2\varepsilon\mu - \beta^2}\left(\beta\frac{\partial E_z}{\partial x} + \omega\mu\frac{\partial H_z}{\partial y}\right)$$

$$E_y = \frac{j}{\omega^2\varepsilon\mu - \beta^2}\left(-\beta\frac{\partial E_z}{\partial y} + \omega\mu\frac{\partial H_z}{\partial x}\right)$$

- boundary conditions are applied and will allow to find out the unknown constants



$$\overrightarrow{E_z}(x=0)=0 \quad \rightarrow \quad A=0$$

$$\overrightarrow{E_z}(y=0)=0 \quad \rightarrow \quad C=0$$

$$\overrightarrow{E_z}(x=a)=0 \quad \rightarrow \quad k_x=\frac{m\,\pi}{a}$$

$$\overrightarrow{E_z}(y=b)=0 \quad \rightarrow \quad k_y=\frac{n\,\pi}{b}$$

$$\overrightarrow{E_z}(x,y,z,t)=\overrightarrow{e_z}\,E_{z0}\,e^{j(\omega t-\beta z)}\sin\frac{m\,\pi\,x}{a}\sin\frac{n\,\pi\,y}{b}$$

… that is, the Ez field is defined without any conditions for Hz
→ Ez and Hz are independent, for example,
Hz = 0, a so-called TM mode (transversal magnetic)

- TM_{mn} mode („E modes“) fields :

$$E_z(x,y,z,t)=E_{z0}\,e^{j\omega t}e^{-j\beta z}\sin\frac{m\,\pi\,x}{a}\sin\frac{n\,\pi\,y}{b}$$

$$E_x(x,y,z,t)=E_{z0}\,\frac{m\pi}{a}\,\frac{\beta}{k_T^2}\,e^{j\omega t}e^{-j\beta z-j\frac{\pi}{2}}\cos\frac{m\,\pi\,x}{a}\sin\frac{n\,\pi\,y}{b}$$

$$E_y(x,y,z,t)=E_{z0}\,\frac{n\pi}{b}\,\frac{\beta}{k_T^2}\,e^{j\omega t}e^{-j\beta z-j\frac{\pi}{2}}\sin\frac{m\,\pi\,x}{a}\cos\frac{n\,\pi\,y}{b}$$

$$H_x(x,y,z,t)=E_{z0}\,\frac{n\pi}{b}\,\frac{\omega\varepsilon}{k_T^2}\,e^{j\omega t}e^{-j\beta z+j\frac{\pi}{2}}\sin\frac{m\,\pi\,x}{a}\cos\frac{n\,\pi\,y}{b}$$

$$H_y(x,y,z,t)=E_{z0}\,\frac{m\pi}{a}\,\frac{\omega\varepsilon}{k_T^2}\,e^{j\omega t}e^{-j\beta z-j\frac{\pi}{2}}\cos\frac{m\,\pi\,x}{a}\sin\frac{n\,\pi\,y}{b}$$

$$H_z(x,y,z,t)\equiv 0$$

- TE_{mn} mode („H modes“) fields :

$$H_z(x,y,z,t) = H_{z0}\, e^{j\omega t}\, e^{-j\beta z}\, \cos\frac{m\,\pi\,x}{a} \cos\frac{n\,\pi\,y}{b}$$

$$H_x(x,y,z,t) = H_{z0}\, \frac{m\pi}{a}\, \frac{\beta}{k_T^2}\, e^{j\omega t}\, e^{-j\beta z + j\frac{\pi}{2}} \sin\frac{m\,\pi\,x}{a} \cos\frac{n\,\pi\,y}{b}$$

$$H_y(x,y,z,t) = H_{z0}\, \frac{n\pi}{b}\, \frac{\beta}{k_T^2}\, e^{j\omega t}\, e^{-j\beta z + j\frac{\pi}{2}} \cos\frac{m\,\pi\,x}{a} \sin\frac{n\,\pi\,y}{b}$$

$$E_x(x,y,z,t) = H_{z0}\, \frac{n\pi}{b}\, \frac{\omega\mu}{k_T^2}\, e^{j\omega t}\, e^{-j\beta z + j\frac{\pi}{2}} \cos\frac{m\,\pi\,x}{a} \sin\frac{n\,\pi\,y}{b}$$

$$E_y(x,y,z,t) = H_{z0}\, \frac{m\pi}{a}\, \frac{\omega\mu}{k_T^2}\, e^{j\omega t}\, e^{-j\beta z - j\frac{\pi}{2}} \sin\frac{m\,\pi\,x}{a} \cos\frac{n\,\pi\,y}{b}$$

$$E_z(x,y,z,t) \equiv 0$$

- TM_{mn} mode fields, cont‘d :

  … m=0 and/or n=0 does not exist (all fields zero) → TM_{00}, TM_{10}, TM_{01}

- TE_{mn} mode fields, cont‘d :

  … m=n=0 does not exist (all fields zero) → TE_{00}

  … m=1 and n=0 : TE_{10} :

$$H_z(x,y,z,t)|_{\mathrm{TE10}} = H_{z0}\, e^{j\omega t}\, e^{-j\beta z}\, \cos\frac{\pi x}{a}$$

$$H_x(x,y,z,t)|_{\mathrm{TE10}} = H_{z0}\, \frac{a\beta}{\pi}\, e^{j\omega t}\, e^{-j\beta z + j\frac{\pi}{2}} \sin\frac{\pi x}{a}$$

$$E_y(x,y,z,t)\big|_{\mathrm{TE10}} = H_{z0}\, \frac{a\omega\mu}{\pi}\, e^{j\omega t}\, e^{-j\beta z - j\frac{\pi}{2}} \sin\frac{\pi x}{a}$$

  … all Hy , Ez , Ex = 0 for TE_{10} mode

cutoff frequencies and mode dispersion in the rectangular waveguide

- which modes propagate at a given frequency ?
- how fast is propagation with a given mode at a given frequency ?

speed of propagation: … phase velocity : $v_{phase} = \frac{\omega}{\beta}$

… group (signal) velocity :

… dispersion $v_{group} = \frac{\partial \omega}{\partial \beta}$

$$\beta^2 = \left(\frac{2\pi}{\lambda_g}\right)^2 = \omega^2 \varepsilon\mu - k_T^2 = \omega^2 \varepsilon\mu - \left(\frac{m\pi}{a}\right)^2 - \left(\frac{n\pi}{b}\right)^2$$

Dispersion equation of the m,n-th mode of the rectangular waveguide

- for a propagating mode, β is real, and with $k_T > 0$, there is a minimum frequency ω below which there is no propagation → **cutoff**

- at **cutoff**: $\beta = 0$

$\lambda_g \to \infty$

$v_{phase} \to \infty$

$\omega_{cutoff}^2 \varepsilon\mu = k_T^2$ ***… the wave resonates in the transverse plane***

… for modes of the rectangular waveguide :

$$f_{c,mn} = \frac{c_0}{\lambda_{c,mn}} = \frac{c_0}{\sqrt{\varepsilon_r \mu_r}} \frac{\sqrt{m^2 b^2 + n^2 a^2}}{2ab}$$

… the lowest f_c for $a>b$ has the mode with m=1, n=0 : TE_{10}

… for TE_{10} mode :

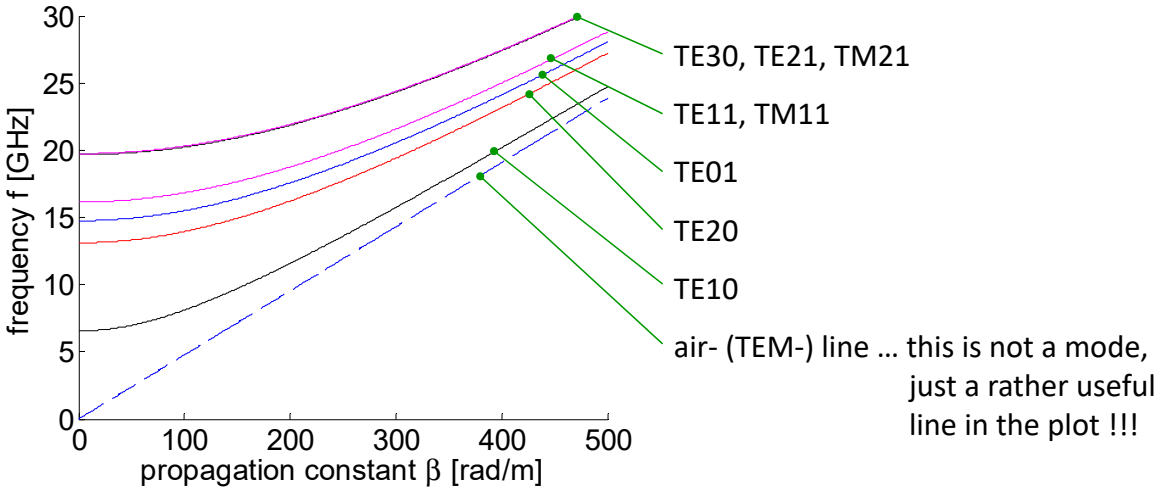
$$f_{c,TE10} = \frac{c_0}{\lambda_{c,TE10}} = \frac{c_0}{\sqrt{\varepsilon_r \mu_r}} \frac{1}{2a}$$

… only Hz and Ey are non-zero. Both without variation in z-direction

- dispersion diagram

… practically most useful is the single- (mono-) mode regime

… in multi-mode waveguides, energy often couples between modes, the more the closer the modes‘ propagation constants



… WR90 standard X-Band waveguide (22.86 x 10.16 mm): monomode freq range 6.56 … 13.12 GHz; technical freq range of operation 8.2 … 12.4 GHz

- TM_{mn} mode fields, cont‘d :

… there is a fixed ratio between some E and H field components in the cross-sectional plane

→ field wave impedance for TM modes („E modes“) :

$$Z_{FE} = \frac{E_x}{H_y} = -\frac{E_y}{H_x} = \frac{\beta}{\omega\varepsilon} \quad , \quad Z_{FE} = \frac{1}{\lambda_g f \varepsilon} = \sqrt{\frac{\mu_0\,\mu_{rel}}{\varepsilon_0\,\varepsilon_{rel}}}\sqrt{1-\left(\frac{f_c}{f}\right)^2}$$

- TE_{mn} mode fields, cont‘d :

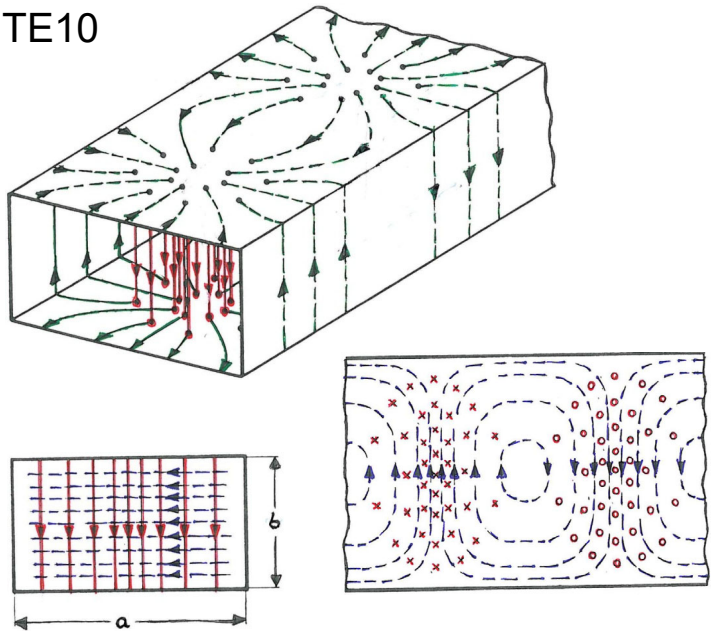
… there is a fixed ratio between some E and H field components in the cross-sectional plane

→ field wave impedance for TE modes („H modes“) :

$$Z_{FH} = \frac{E_x}{H_y} = -\frac{E_y}{H_x} = \frac{\omega\mu}{\beta} \quad , \quad Z_{FH} = \lambda_g f \mu = \sqrt{\frac{\mu_0\,\mu_{rel}}{\varepsilon_0\,\varepsilon_{rel}}} \Big/ \sqrt{1-\left(\frac{f_c}{f}\right)^2}$$

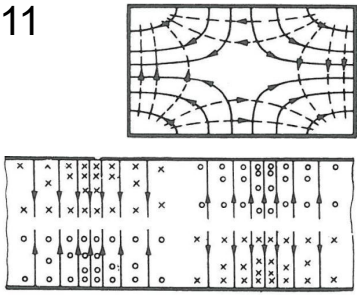
field distribution and wall currents in the rectangular waveguide

TE10

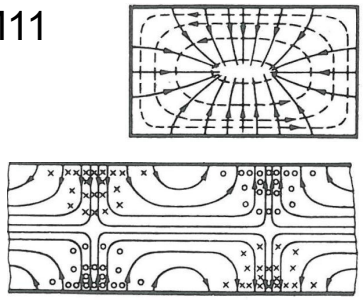


E-field (solid red)
H-field (dashed blue)
wall currents (green)

TE11



TM11



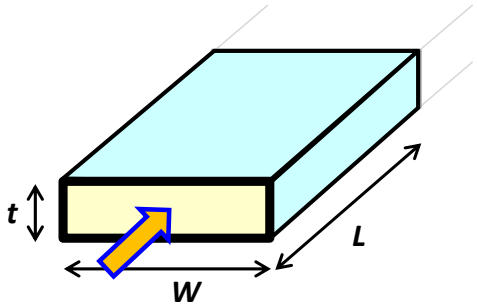
[Zinke, Brunswig]

wall currents and loss in the rectangular waveguide (1/2)

- calculation of conductor loss using power loss method — use wall currents of loss-less waveguide to calculate conductor loss and compare with the Poynting vector integrated over the cross section:

… for TE_{10} mode :

$$\alpha_{\mathrm{TE10}} = R_{\square} \frac{1 + 2\,\frac{b}{a}\left(\frac{f_c}{f}\right)^2}{b\sqrt{\frac{\mu_0\,\mu_{rel}}{\varepsilon_0\,\varepsilon_{rel}}}\sqrt{1-\left(\frac{f_c}{f}\right)^2}}$$



$$R_{\square} = \sqrt{\frac{\pi\, f\, \mu_0\, \mu_{rel}}{\sigma}} \qquad [\Omega]\,,\,[\Omega/\square]$$

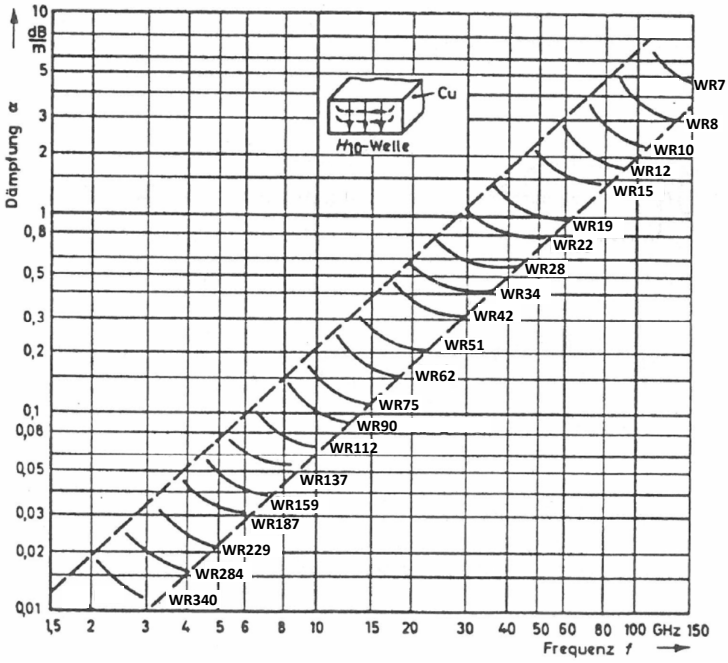
$$R = \frac{1}{\sigma}\frac{L}{Wt} = R_{\square}\frac{L}{W} \qquad [\Omega]$$

… does not work for degenerate modes (like TE_{11} and TM_{11})

wall currents and loss in the rectangular waveguide (2/2)

- rectangular waveguide loss depends on frequency
- realistic loss is related to material conductivity and surface roughness
- TE_{10} loss (Cu, vacuum):

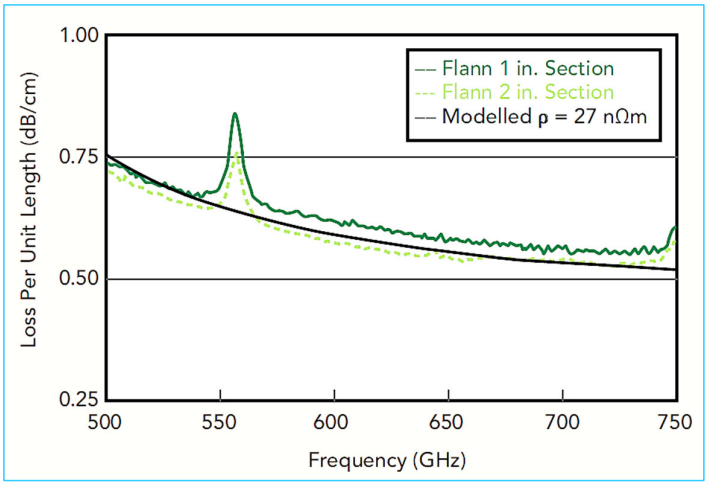
… graph for smooth metal: any roughness will increase loss!

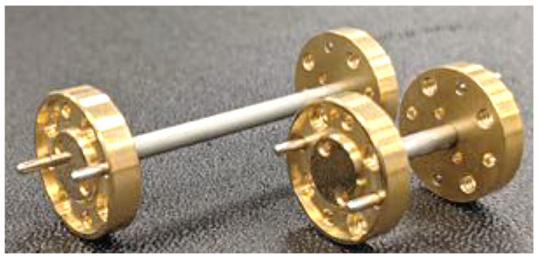


[modified after: W. Jansen, Streifenleiter und Hohlleiter, Hüthig 1992, S. 87]

wall currents and loss in the rectangular waveguide — example:

- realistic loss is related to material conductivity and surface roughness





waveguide inner dimensions
0.38 mm x 0.19 mm

- atmospheric H_2O attenuation peaks at 560 GHz, 750 GHz
- best fit with resistivity 27 nΩm ≅ 37.0 x 10^6 S/m (64% of pure Cu σ)
- datasheet loss was stated 1.1 dB/cm @ 500 GHz

[J. Skinner et al., „Benchmarking Seamless Sub-mmWave and Terahertz Rectangular Metallic Waveguide“, Microwave Jornal, Nov. 2023, pp. 60-68]

Rectangular waveguide designations and cross sections

typical bandwidth is half an octave
designations according to US (WR…), UK (WG…), European (R…) rules

microwave types:

TE_{10} cutoff [GHz]	frequency use range [GHz]	inner dimensions [mm]	band designations	waveguide designation
0.909	1.14 – 1.73	165.1 x 82.55	L	WR650
1.37	1.72 – 2.61	109.22 x 54.61	LS	WR430
2.08	2.60 – 3.95	72.14 x 34.04	S	WR284
3.15	3.9 – 6.0	47.55 x 22.15	C	WR187
6.56	8.2 – 12.5	22.86 x 10.16	X	WR90
9.49	11.9 – 18.0	15.80 x 7.90	Ku	WR62
14.06	17.6 – 26.7	10.67 x 4.32	K	WR42
21.09	26.4 – 40.0	7.112 x 3.556	Ka	WR28
39.9	50 – 75	3.759 x 1.880	V	WR15
48.4	60 – 90	3.048 x 1.524	E	WR12
59.1	75 – 110	2.540 x 1.270	W	WR10

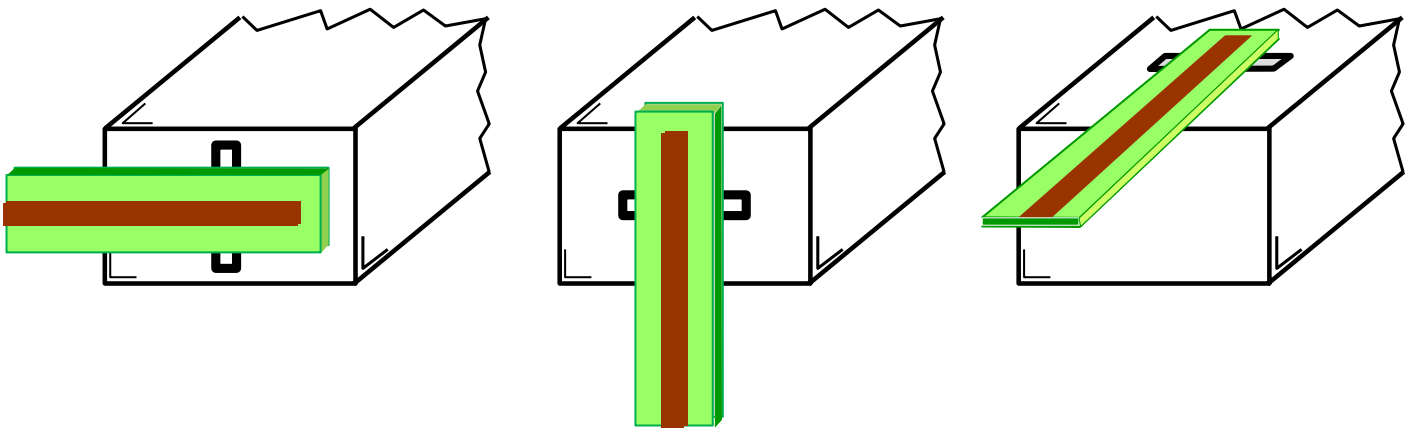
millimeter-wave types:

WAVEGUIDE DIMENSIONS (METRIC)		
WAVEGUIDE	- A -	- B -
„V“ WR-15	3759 micron	1880 micron
„E“ WR-12	3099 micron	1549 micron
„W“ WR-10	2540 micron	1270 micron
„F“ WR-8.0	2032 micron	1016 micron
„D“ WR-6.5	1651 micron	825.5 micron
„G“ WR-5.1	1295 micron	647.5 micron
WR-4.3	1092 micron	546 micron
WR-3.4	864 micron	432 micron
WM-710 (WR-2.8)	710 micron	355 micron
WM-570 (WR-2.2)	570 micron	285 micron
WM-470 (WR1.9)	470 micron	235 micron
WM-380 (WR1.5)	380 micron	190 micron
WM-310 (WR-1.2)	310 micron	155 micron
WM-250 (WR-1.0)	250 micron	125 micron
WM-200 (WR-0.8)	200 micron	100 micron
WM-164 (WR-0.65)	164 micron	82 micron

excitation of rectangular waveguide

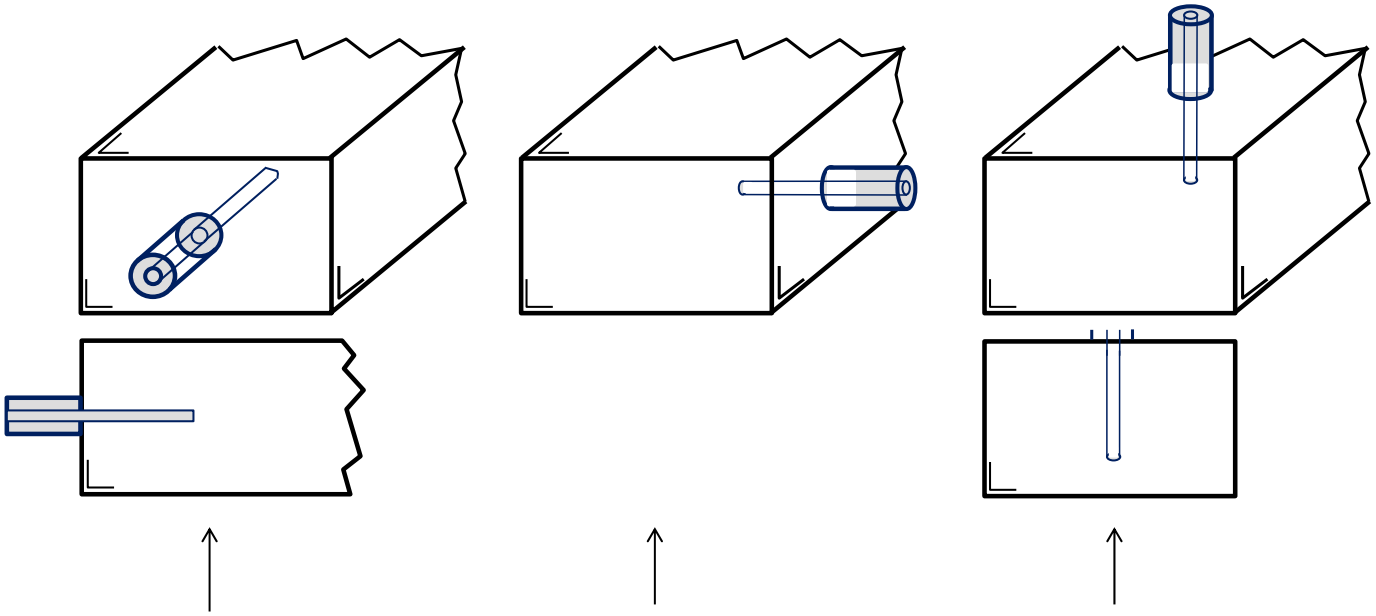
the excitation must create fields (or currents) which are **not orthogonal** to the waveguide mode‘s fields (or currents)

- current coupling using slot in wall



impossible | ok | ok if not λg/4 from the end

- probe coupling using electric probe

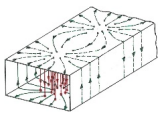


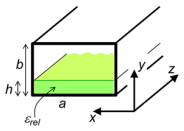
impossible

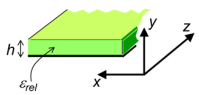
practically impossible;
bad if very asymmetric

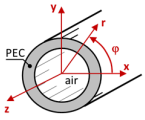
ok (about λg/4 from the end)

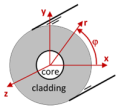
Waveguides — specific cases & application areas:

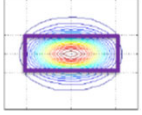
1)  rectangular waveguide (metal, air-filled) — *high power, low loss*

2)  slab-filled rectangular waveguide (metal, dielectric&air-filled) — key for understanding cases 3 & 5 ! — *few to none*

3)  slab waveguide (dielectric, infinite x-section) — *circuit board*

4)  circular waveguide (metal, air-filled) — *antenna feed*

5)  dielectric rod (infinite x-section) — *optical fiber*

6)  other dielectric waveguides — *few to none*

rectangular waveguide partially filled with a dielectric slab

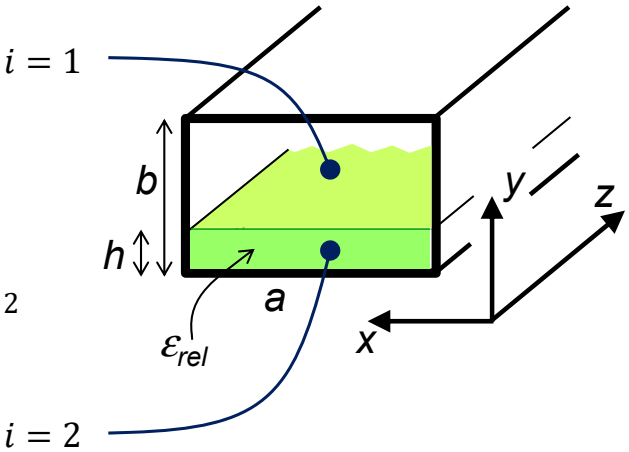
$$\left(\Delta_T + k_{Ti}^2\right)\begin{pmatrix}\overrightarrow{E_{zi}}\\ \overrightarrow{H_{zi}}\end{pmatrix} = 0 \quad , \quad i = 1,2$$

$i = 1$:

$$k_{T1}^2 = k_x^2 + k_{y1}^2 = k_{01}^2 - \beta^2 = \varepsilon_0\mu_0\omega^2 - \beta^2$$

$i = 2$:

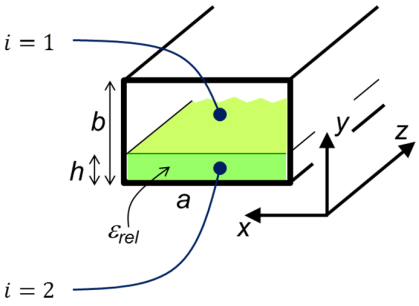
$$k_{T2}^2 = k_x^2 + k_{y2}^2 = k_{02}^2 - \beta^2 = \varepsilon_0\varepsilon_{\mathrm{rel}}\mu_0\omega^2 - \beta^2$$



Ansatz: separation of variables : x :

$$\overrightarrow{E_{z1,2}}(x,y,z,t) \sim \left(A_{1,2}\cos k_x x + B_{1,2}\sin k_x x\right)$$

$$\overrightarrow{H_{z1,2}}(x,y,z,t) \sim \left(E_{1,2}\cos k_x x + F_{1,2}\sin k_x x\right)$$



… boundary conditions at x=0 and x=a :

$$A_{1,2} = 0 \qquad \overrightarrow{E_{z1,2}}(x,y,z,t) \sim \sin\frac{m\,\pi\,x}{a}$$

$$F_{1,2} = 0$$

$$k_x = \frac{m\,\pi}{a} \qquad \overrightarrow{H_{z1,2}}(x,y,z,t) \sim \cos\frac{m\,\pi\,x}{a}$$

… separation of variable y and boundary conditions at y=0, y=b :

$$\overrightarrow{E_{z2}}(y=0) = 0 \quad \rightarrow \quad \overrightarrow{E_{z2}} \sim \sin(k_{y2}y) \qquad \overrightarrow{E_{z1}}(y=b) = 0 \quad \rightarrow \quad \overrightarrow{E_{z1}} \sim \sin(k_{y1}(b-y))$$

$$\left.\frac{\partial\overrightarrow{H_{z2}}}{\partial y}\right|_{y=0} = 0 \quad \rightarrow \quad \overrightarrow{H_{z2}} \sim \cos(k_{y2}y) \qquad \left.\frac{\partial\overrightarrow{H_{z1}}}{\partial y}\right|_{y=b} = 0 \quad \rightarrow \quad \overrightarrow{H_{z1}} \sim \cos(k_{y2}(b-y))$$

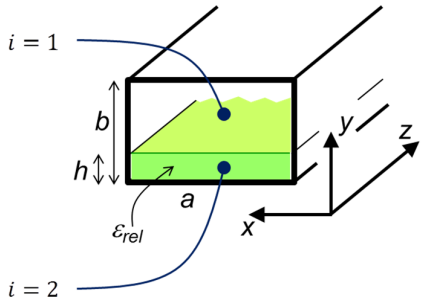
$$E_{z1}(x,y) = K \cdot \sin k_x x \cdot \sin\big(k_{y1}(b-y)\big)$$

$$E_{z2}(x,y) = L \cdot \sin k_x x \cdot \sin k_{y2} y$$

$$H_{z1}(x,y) = M \cdot \cos k_x x \cdot \cos\big(k_{y1}(b-y)\big)$$

$$H_{z2}(x,y) = N \cdot \cos k_x x \cdot \cos k_{y2} y$$

$$k_x = \frac{m\,\pi}{a}$$



Note that here, for m=0, Ez≡0 (and Hz≠0). But for m≠0, both Ez and Hz exist and are coupled. This is a general property of dielectric interfaces in a waveguide‘s cross-section.

… continuity of tangential fields at the air-dielectric interface, y=h :

$$E_{z1}(x, y=h) = E_{z2}(x, y=h)$$

$$\hookrightarrow \quad K \cdot \sin\Big(k_{y1}(b-h)\Big) - L \cdot \sin k_{y2} h = 0 \quad , \quad \sin k_x x \neq 0$$

$$E_{x1}(x, y=h) = E_{x2}(x, y=h) \quad , \quad E_x = -\underbrace{\frac{j}{\omega^2\varepsilon\mu - \beta^2}}_{k_{T1,2}^2}\left(\beta\frac{\partial E_z}{\partial x} + \omega\mu\frac{\partial H_z}{\partial y}\right)$$

$$\hookrightarrow \quad \cos k_x x \neq 0$$

$$\frac{K\beta k_x}{k_{T1}^2}\sin\Big(k_{y1}(b-h)\Big) - \frac{L\beta k_x}{k_{T2}^2}\sin\big(k_{y2}h\big) + \frac{M\omega\mu k_{y1}}{k_{T1}^2}\sin\Big(k_{y1}(b-h)\Big) + \frac{N\omega\mu k_{y2}}{k_{T2}^2}\sin\big(k_{y2}h\big) = 0$$

… similarly for Hz and Hx at y=h :

$$M \cdot \cos\Big(k_{y1}(b-h)\Big) - N \cdot \cos k_{y2} h = 0 \quad , \quad \cos k_x x \neq 0$$

$$H_{x1,2} = \frac{j}{k_{T1,2}^2}\left(\omega\varepsilon\frac{\partial E_{z1,2}}{\partial y} - \beta\frac{\partial H_{z1,2}}{\partial x}\right)$$

$$\frac{K\omega\varepsilon_0 k_{y1}}{k_{T1}^2}\cos\Big(k_{y1}(b-h)\Big) + \frac{L\omega\varepsilon_0\varepsilon_{\mathrm{rel}}k_{y2}}{k_{T2}^2}\cos\big(k_{y2}h\big) - \frac{M\beta k_x}{k_{T1}^2}\cos\Big(k_{y1}(b-h)\Big) - \frac{N\beta k_x}{k_{T2}^2}\cos\big(k_{y2}h\big) = 0$$

… written in matrix form :

$$\begin{bmatrix} \sin\left(k_{y1}(b-h)\right) & -\sin(k_{y2}h) & 0 & 0 \\ \frac{\beta k_x}{k_{T1}^2}\sin\left(k_{y1}(b-h)\right) & -\frac{\beta k_x}{k_{T2}^2}\sin(k_{y2}h) & \frac{\omega\mu k_{y1}}{k_{T1}^2}\sin\left(k_{y1}(b-h)\right) & \frac{\omega\mu k_{y2}}{k_{T2}^2}\sin(k_{y2}h) \\ \frac{\omega\varepsilon_0 k_{y1}}{k_{T1}^2}\cos\left(k_{y1}(b-h)\right) & \frac{\omega\varepsilon_0\varepsilon_{\mathrm{rel}} k_{y2}}{k_{T2}^2}\cos(k_{y2}h) & -\frac{\beta k_x}{k_{T1}^2}\cos\left(k_{y1}(b-h)\right) & -\frac{\beta k_x}{k_{T2}^2}\cos(k_{y2}h) \\ 0 & 0 & \cos\left(k_{y1}(b-h)\right) & -\cos(k_{y2}h) \end{bmatrix} \begin{pmatrix} K \\ L \\ M \\ N \end{pmatrix} = 0$$

… together with : $k_x^2 + k_{y1}^2 = k_{01}^2 - \beta^2 = \varepsilon_0\mu_0\omega^2 - \beta^2$

$$k_x^2 + k_{y2}^2 = k_{02}^2 - \beta^2 = \varepsilon_0\varepsilon_{\mathrm{rel}}\mu_0\omega^2 - \beta^2$$

$$k_x = \frac{m\,\pi}{a}$$

… one solution is : K=L=M=N=0

… one solution is : chose ω , β and calculate K,L,M,N – not physical!

… one solution is : $\beta^2 = \varepsilon_0\mu_0\omega^2$ – not physical!

… the physically meaningful solution requires one undetermined parameter (representing the physical power/magnitude of the mode). This is achieved by chosing ω , β such that the matrix becomes singular.

matrix determinant is zero [matlab: **`det(A)`**]
condition of matrix is infinite [matlab: **`cond(A)`**]
matrix‘ smallest singular value is zero [matlab: **`svd(A)`**]

… once the ω , β are found such that the matrix becomes singular, three out of the four unknowns K,L,M,N can be determined and from those, all the field components will be known (all proportional to a single common real parameter).

… this specific solution is mathematically an Eigen-solution and physically a resonance (here: in the transverse plane of the waveguide).

… ω as a function of β (or vice versa) for many (theoretically: infinite number of) modes can be plotted in the dispersion diagram

… at the **cutoff frequencies**, β=0, there are two equation systems with two unknowns each. They must become singular :

$$\begin{bmatrix} \sin\left(k_{y1}(b-h)\right) & -\sin(k_{y2}h) & 0 & 0 \\ \frac{\beta k_x}{k_{T1}^2}\sin\left(k_{y1}(b-h)\right) & -\frac{\beta k_x}{k_{T2}^2}\sin(k_{y2}h) & \frac{\omega\mu k_{y1}}{k_{T1}^2}\sin\left(k_{y1}(b-h)\right) & \frac{\omega\mu k_{y2}}{k_{T2}^2}\sin(k_{y2}h) \\ \frac{\omega\varepsilon_0 k_{y1}}{k_{T1}^2}\cos\left(k_{y1}(b-h)\right) & \frac{\omega\varepsilon_0\varepsilon_{rel} k_{y2}}{k_{T2}^2}\cos(k_{y2}h) & -\frac{\beta k_x}{k_{T1}^2}\cos\left(k_{y1}(b-h)\right) & -\frac{\beta k_x}{k_{T2}^2}\cos(k_{y2}h) \\ 0 & 0 & \cos\left(k_{y1}(b-h)\right) & -\cos(k_{y2}h) \end{bmatrix} \begin{pmatrix} K \\ L \\ M \\ N \end{pmatrix} = 0$$

… the determinants of the 2x2 matrices are written easily as :

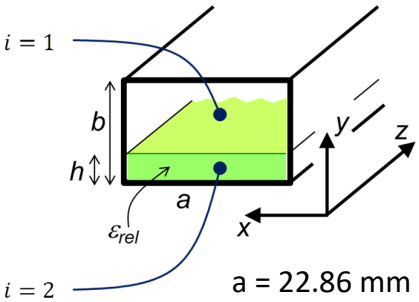
$$\varepsilon_{rel}\,\frac{k_{y2}}{k_{T2}^2}\cos(k_{y2}h)\,\sin\left(k_{y1}(b-h)\right) + \frac{k_{y1}}{k_{T1}^2}\sin(k_{y2}h)\,\cos\left(k_{y1}(b-h)\right) = 0$$

$$\frac{k_{y1}}{k_{T1}^2}\cos(k_{y2}h)\,\sin\left(k_{y1}(b-h)\right) + \frac{k_{y2}}{k_{T2}^2}\sin(k_{y2}h)\,\cos\left(k_{y1}(b-h)\right) = 0$$

$$\left(\frac{m\pi}{a}\right)^2 + k_{y1}^2 = \varepsilon_0\mu_0\omega^2 \quad , \quad \left(\frac{m\pi}{a}\right)^2 + k_{y2}^2 = \varepsilon_0\varepsilon_{rel}\mu_0\omega^2$$

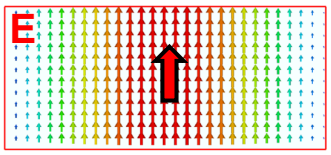
… dispersion diagram, example (1/8) :

(fundamental) mode with m=1 (n=0):



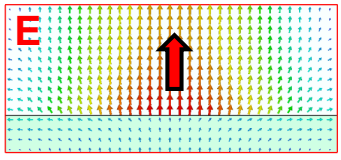
a = 22.86 mm
b = 10.16 mm
h = 2.54 mm
ε_{rel} = 10

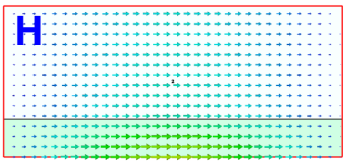
… without slab → TE10



… with slab:

E_{normal} at interface:
D_{normal} continuous

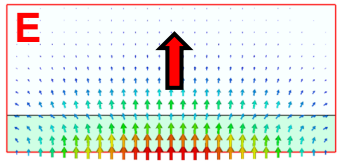


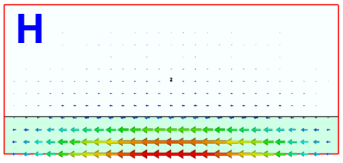


f = 6.3 GHz & β = 85 rad/m : ε_{eff} = 0.41 & v_{ph} = 1.55 c

at lower freq: field mostly in air section

at higher freq: field more & more in slab (high ε_{rel}) section

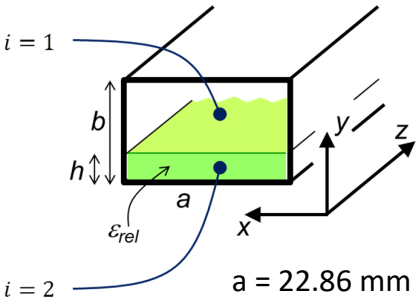




f = 15 GHz & β = 790 rad/m : ε_{eff} = 6.32 & v_{ph} = 0.40 c

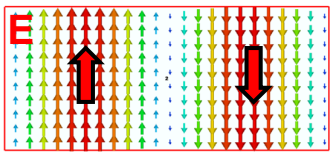
… dispersion diagram, example (2/8) :

(fundamental) mode with m=2 (n=0):



a = 22.86 mm
b = 10.16 mm
h = 2.54 mm
ε_{rel} = 10

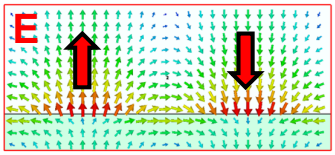
… without slab → TE20

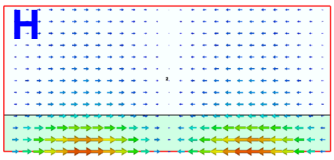


… with slab:

E_{normal} at interface:
D_{normal} continuous

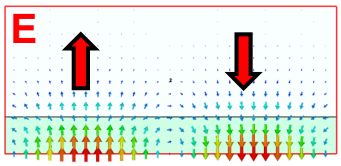
at lower freq: field mostly in air section

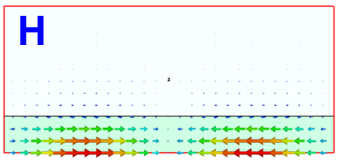




f = 8.8 GHz & β = 30 rad/m : ε_{eff} = 0.03 & v_{ph} = 6.14 c

at higher freq: field more & more in slab (high ε_{rel}) section

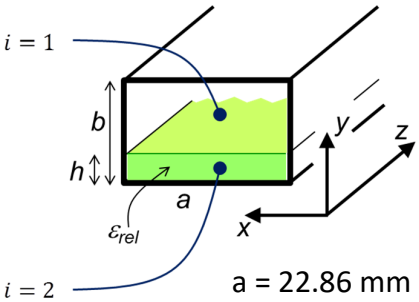




f = 18 GHz & β = 997 rad/m : ε_{eff} = 6.99 & v_{ph} = 0.38 c

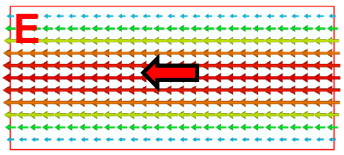
… dispersion diagram, example (3/8) :

(fundamental) mode with m=0 (n=1):



a = 22.86 mm
b = 10.16 mm
h = 2.54 mm
ε_{rel} = 10

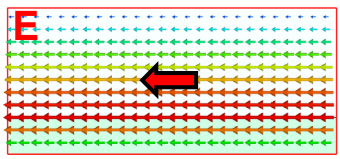
… without slab → TE01

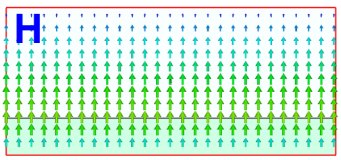


… with slab:

at interface:
$E_{tangential}$ continuous

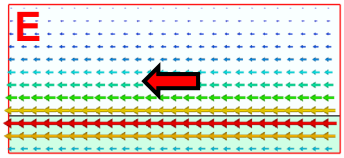
at lower freq: field mostly in air section

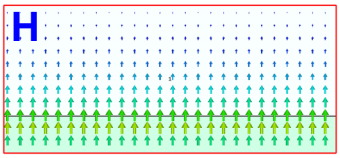




β = 100 rad/m

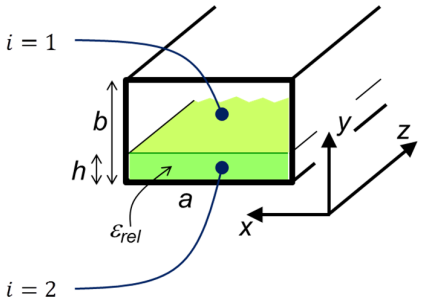
at higher freq: field more & more in slab (high ε_{rel}) section





β = 500 rad/m

… dispersion diagram, example (4/8) :



(fundamental) mode with m=0 (n=1) cont'd:

Ex(y) variation governed by
k_{y1} (in air) and k_{y2} (in slab)

$$k_{y1}^2 = k_0^2 - k_x^2 - \beta^2 \qquad k_{y2}^2 = \varepsilon_{rel} k_0^2 - k_x^2 - \beta^2$$

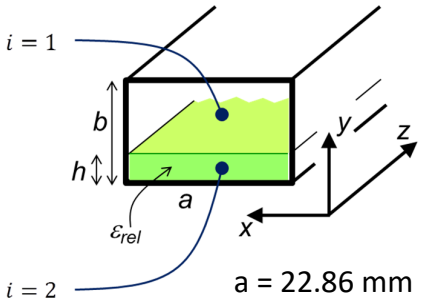
… $k_{y1}^2 > 0$ for small β … $k_{y2}^2 > 0$ for small & medium β

⇨ exp(jk$_y$y) with real k → sin(k$_y$y), cos(k$_y$y)

… $k_{y1}^2 < 0$ for larger β … $k_{y2}^2 < 0$ for larger β **?** … this does not exist

⇨ exp(jk$_y$y) with imaginary k → sinh(k$_y$y), cosh(k$_y$y)

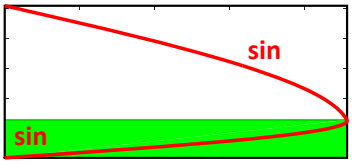
… dispersion diagram, example (5/8) :



(fundamental) mode with m=0 (n=1) cont'd:

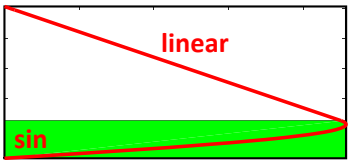
|Ex(y)| :

a = 22.86 mm
b = 10.16 mm
h = 2.54 mm
ε_{rel} = 10



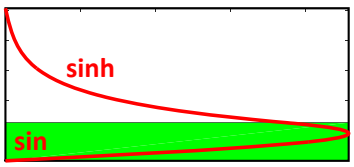
f = 9.6 GHz , β = 65 rad/m

k_{y1} = 190 rad/m , k_{y2} = 633 rad/m



f = 11 GHz , β = 231 rad/m

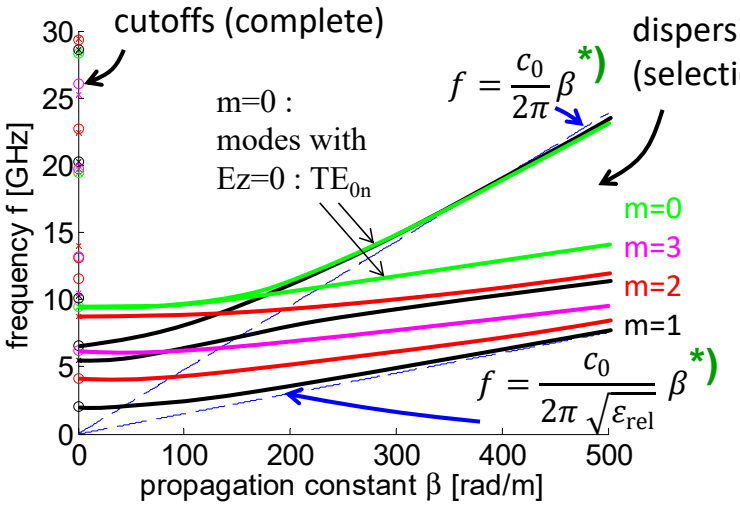
k_{y1} ~ 0 rad/m , k_{y2} = 692 rad/m

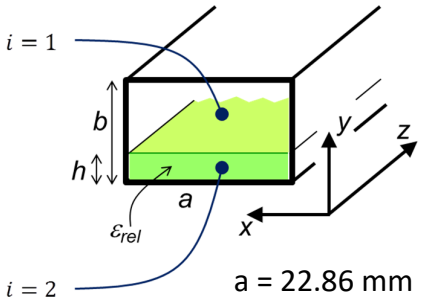


f = 16.7 GHz , β = 700 rad/m

k_{y1} = 605 j rad/m , k_{y2} = 860,06 rad/m

… dispersion diagram, example (6/8) :



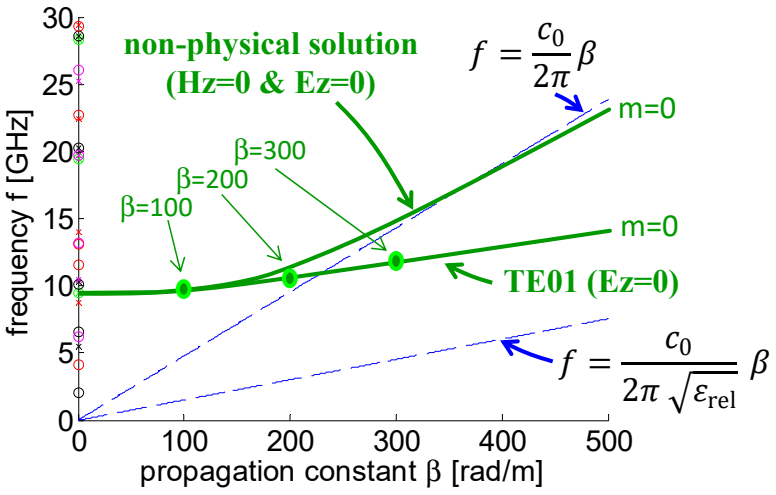


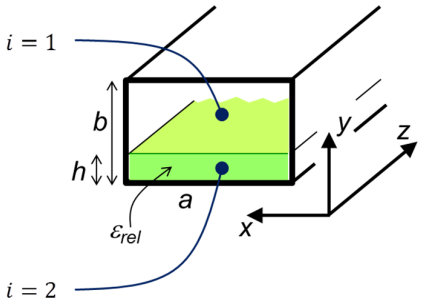
a = 22.86 mm
b = 10.16 mm
h = 2.54 mm
ε_{rel} = 10

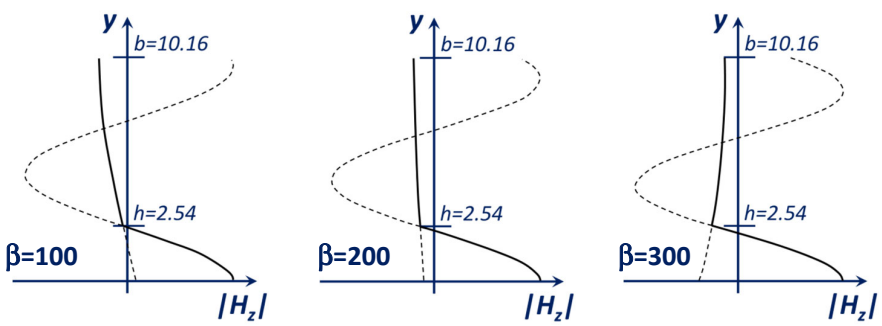
the „air lines“ *) separate the dispersion diagram in 3 regions:

- … $k_{y1}^2 > 0$ for small β
- … $k_{y1}^2 < 0$ for larger β
- … $k_{y2}^2 > 0$ for small & medium β
- … $k_{y2}^2 < 0$ for larger β

… dispersion diagram, example (7/8) :







β=100, **β=200**: $k_{y1}^2 > 0$ & $k_{y2}^2 > 0$

β=300: $k_{y1}^2 < 0$ & $k_{y2}^2 > 0$

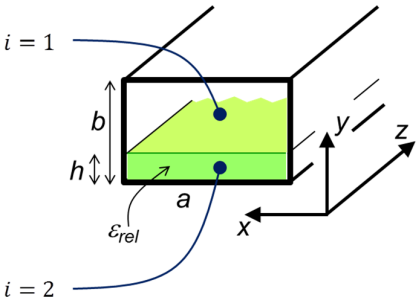
$\sin(jy) = j\sinh y$

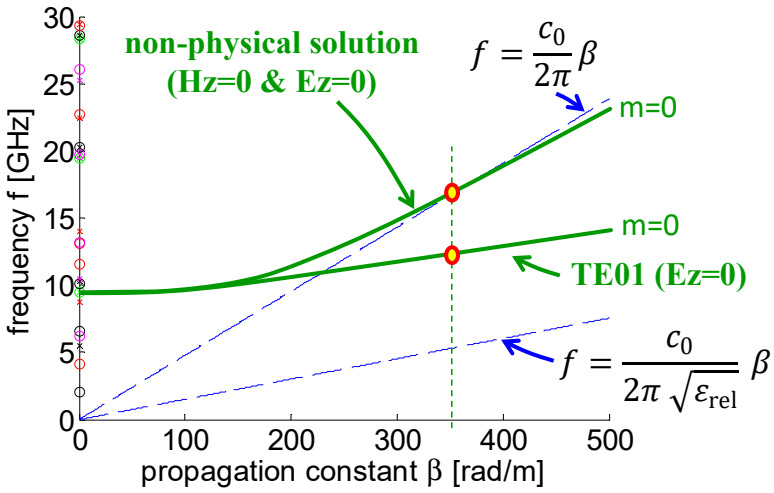
$\cos(jy) = \cosh y$

$k_{y1}^2 < 0$ & $k_{y2}^2 < 0$

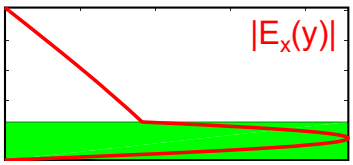
… this does not exist

… dispersion diagram, example (8/8) :



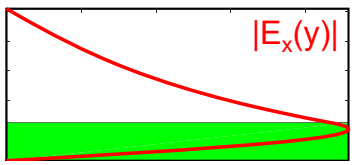


non-physical solution :



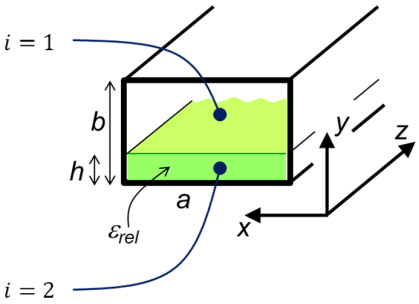
β = 350 rad/m , f = 17.07 GHz
k_{y1} = 73.9 rad/m , k_{y2} = 1075 rad/m

physical solution :



β = 350 rad/m , f_{sol} = 12.42 GHz
k_{y1} = 234 rad/m , k_{y2} = 744.8 rad/m

… other field components of TE_{01} :



m=0 → kx=0 → Ez~sin(k_xx) → Ez=0

m=0 → kx=0 → Hz~cos(k_xx) → dHz/dx=0

Hx=0 , Ey=0

Hy~dHz/dy

Ex~dHz/dy

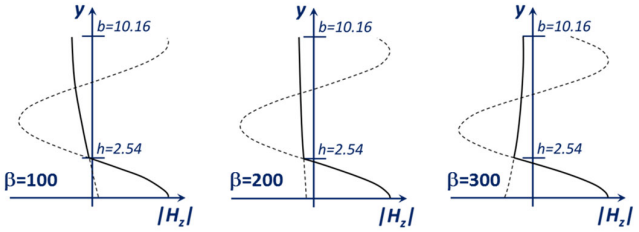
boundary condition:
Ex=0 @ y=0 , y=b :
dHz/dy=0 @ y=0 , y=b

$$H_x = \frac{j}{\omega^2\varepsilon\mu - \beta^2}\left(\omega\varepsilon\cancel{\frac{\partial E_z}{\partial y}} - \beta\cancel{\frac{\partial H_z}{\partial x}}\right)$$

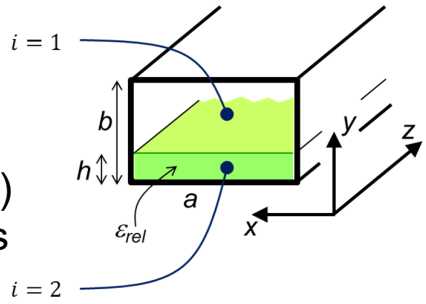
$$H_y = -\frac{j}{\omega^2\varepsilon\mu - \beta^2}\left(\omega\varepsilon\cancel{\frac{\partial E_z}{\partial x}} + \beta\frac{\partial H_z}{\partial y}\right)$$

$$E_x = -\frac{j}{\omega^2\varepsilon\mu - \beta^2}\left(\beta\cancel{\frac{\partial E_z}{\partial x}} + \omega\mu\frac{\partial H_z}{\partial y}\right)$$

$$E_y = \frac{j}{\omega^2\varepsilon\mu - \beta^2}\left(-\beta\cancel{\frac{\partial E_z}{\partial y}} + \omega\mu\cancel{\frac{\partial H_z}{\partial x}}\right)$$



… waveguides with dielectric interfaces in their cross section:



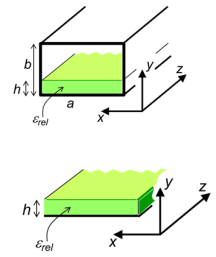
- do not have constant (over cross section) ratios between E and H field components (wave impedance concept of homogeneous waveguides does not apply)

- have modes with both E_Z and H_Z (called „hybrid modes“), only some exceptional modes may still be TE or TM

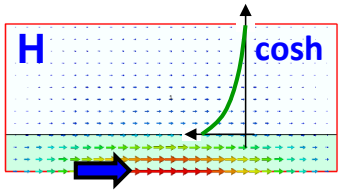
- dispersion curves may cross each other: at some point, more than a single mode may have identical frequency ω and β, called degenerate modes

- phase velocity (ε_{eff}) reaches asymptotically the velocity of a plane wave in the medium of the highest permittivity in the cross section ($\varepsilon_{eff} \rightarrow \max\{\varepsilon_{rel}\}$)

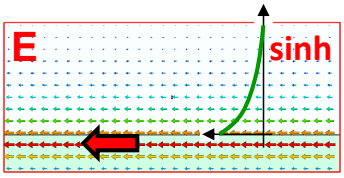
from slab loaded rectangular waveguide to grounded dielectric slab → surface waves



… in slab loaded waveguide, above a certain frequency, the wave is concentrated in the dielectric slab: variation in the air section is governed by sinh & cosh functions

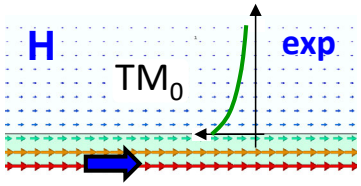


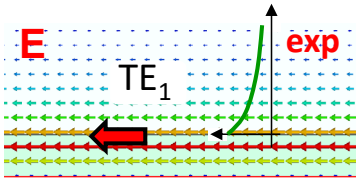
mode with m=1 and transverse magnetic field



mode with m=0 and transverse electric field

… in grounded slab, above a certain frequency, the wave is concentrated in the dielectric slab: variation in the air section is governed by exponential decay

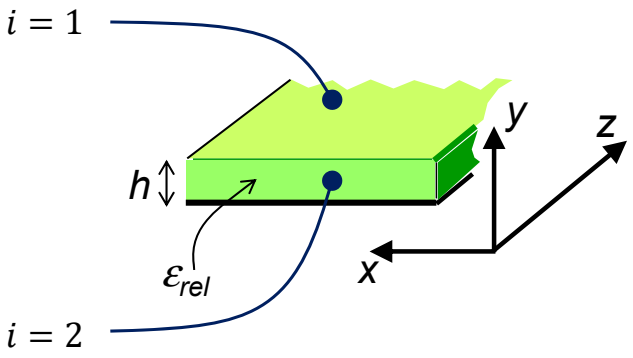




surface waves on grounded dielectric slab

$$\left(\Delta_T + k_{Ti}^2\right)\begin{pmatrix}\overrightarrow{E_{zi}}\\ \overrightarrow{H_{zi}}\end{pmatrix} = 0 \quad , \quad i = 1,2$$

$$\frac{\partial}{\partial x} \equiv 0 \qquad k_x = 0$$



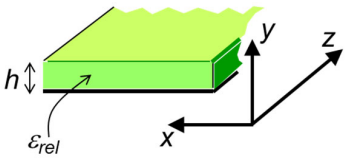
… there are two (independent) sets of modes :

$$E_x = -\frac{j}{\omega^2\varepsilon\mu - \beta^2}\left(\beta\frac{\partial E_z}{\partial x} + \omega\mu\frac{\partial H_z}{\partial y}\right) \qquad H_x = \frac{j}{\omega^2\varepsilon\mu - \beta^2}\left(\omega\varepsilon\frac{\partial E_z}{\partial y} - \beta\frac{\partial H_z}{\partial x}\right)$$

$$E_y = \frac{j}{\omega^2\varepsilon\mu - \beta^2}\left(-\beta\frac{\partial E_z}{\partial y} + \omega\mu\frac{\partial H_z}{\partial x}\right) \qquad H_y = -\frac{j}{\omega^2\varepsilon\mu - \beta^2}\left(\omega\varepsilon\frac{\partial E_z}{\partial x} + \beta\frac{\partial H_z}{\partial y}\right)$$

TE modes (Ez ≡ 0 , Ey ≡ 0 , Hx ≡ 0) ⟺ TM modes (Hz ≡ 0 , Ex ≡ 0 , Hy ≡ 0)

TM modes (Hz ≡ 0 , Ex ≡ 0 , Hy ≡ 0) :



$$k_{y1}^2 = \varepsilon_0\mu_0\omega^2 - \beta^2 \quad , \quad k_{y2}^2 = \varepsilon_0\varepsilon_{rel}\mu_0\omega^2 - \beta^2 \qquad (1)$$

$$E_{zi}(y,z,t) = e^{j\omega t}\, e^{-j\beta z}\left(A_i\, e^{+j k_{yi} y} + B_i\, e^{-j k_{yi} y}\right)$$

$$E_{z2}(y=0) = 0 \quad \rightarrow \quad k_{y2}^2 > 0 \qquad (2)$$ … where k_{y2} is chosen positive real

$$E_{z2}(y) = C \sin k_{2y} y \quad , \quad C := j\,(A_2 - B_2)$$

$$E_{z1}(y \rightarrow \infty) = 0 \quad \rightarrow \quad k_{y1}^2 < 0 \quad \rightarrow \quad \alpha_y^2 = \beta^2 - \varepsilon_0\mu_0\omega^2 \qquad (3)$$

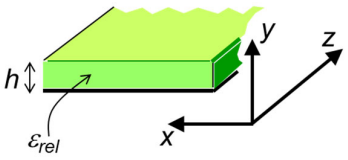
$$E_{z1}(y) = A_1\, e^{-\alpha_y y} \qquad (4)$$ … where α_y is chosen positive real

… continuity of the tangential fields at the boundary y=h :

$$E_{z1}(y=h) = E_{z2}(y=h) \quad : \quad A_1\, e^{-\alpha_y h} = C \sin k_{y2} h \qquad (5x)$$

$$H_{x1} = \frac{-j}{\alpha_y^2}\left(\omega\varepsilon_0\frac{\partial E_{z1}}{\partial y}\right) \quad , \quad H_{x1} = \frac{jA_1\omega\varepsilon_0}{\alpha_y}\, e^{-\alpha_y y}$$

$$H_{x2} = \frac{j}{k_{y2}^2}\left(\omega\varepsilon_0\varepsilon_{rel}\frac{\partial E_{z2}}{\partial y}\right) \quad , \quad H_{x2} = \frac{jC\omega\varepsilon_0\varepsilon_{rel}}{k_{y2}}\cos k_{y2}y$$



$$H_{x1}(y=h) = H_{x2}(y=h) \quad : \quad \frac{A_1}{\alpha_y}\,e^{-\alpha_y h} = \frac{C\varepsilon_{rel}}{k_{y2}}\cos k_{y2}h \qquad \textbf{5y}$$

… these (5x, 5y) are 2 equations with each a „single unknown“, A_1 / C

… for a non-trivial solution (i.e., non-zero fields), the characteristic equation is obtained (by equalling out A_1 / C):

$$\frac{A_1}{C}\,e^{-\alpha_y h} = \sin k_{y2}h = \frac{\alpha_y\,\varepsilon_{rel}}{k_{y2}}\cos k_{y2}h \quad \rightarrow \quad \textbf{5}: \;\; \alpha_y\,\varepsilon_{rel} = k_{y2}\tan k_{y2}h$$

1: $k_{y2}^2 = \varepsilon_0\varepsilon_{rel}\mu_0\omega^2 - \beta^2$

3: $\alpha_y^2 = \beta^2 - \varepsilon_0\mu_0\omega^2$

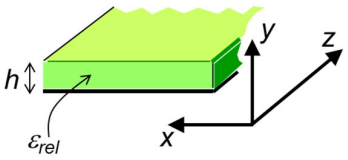
5: $\alpha_y\,\varepsilon_{rel} = k_{y2}\tan k_{y2}h$

… for a given frequency ω, 3 equs allow solving for β, α_y, k_{y2} → dispersion relation (ω,β)

… field distribution is by α_y, k_{y2} and the ratio A_1/C, thus C (or A_1) remains free to chose

… cutoff being defined as the lowest frequency where the mode becomes „guided“, that is,

$\alpha_y = 0$ (and not: $\beta = 0$):



… a purely transverse resonance, that is, β=0, is impossible because this would imply wave propagation (not resonance!) to y→∞ (alternatively, and impossibly, perfect reflection at the slab-air-interface would be required)

5: $\sin k_{y2}h = \frac{\alpha_y\,\varepsilon_{rel}}{k_{y2}}\cos k_{y2}h \;\rightarrow\; \sin k_{y2}h = 0 \;\rightarrow\; k_{y2}h = \frac{m\pi}{2}$, $m = 0,2,4\ldots$

1: $k_{y2}^2 = \varepsilon_0\varepsilon_{rel}\mu_0\omega^2 - \beta^2$

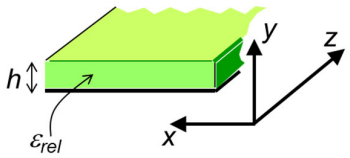
3: $\alpha_y^2 = \beta^2 - \varepsilon_0\mu_0\omega^2$

$$\rightarrow \quad k_{y2} = \frac{2\pi f_{cutoff}}{c}\sqrt{\varepsilon_{rel} - 1}$$

… **TM mode cutoff** ($\alpha_y = 0$): $f_{cutoff,TMm} = \frac{mc}{4h\sqrt{\varepsilon_{rel} - 1}}$, $m = 0,2,4\ldots$

(TM_m mode)

TE modes (Ez ≡ 0 , Ey ≡ 0 , Hx ≡ 0) :



$$k_{y1}^2 = \varepsilon_0 \mu_0 \omega^2 - \beta^2 \quad , \quad k_{y2}^2 = \varepsilon_0 \varepsilon_{rel} \mu_0 \omega^2 - \beta^2$$

$$H_{zi}(y,z,t) = e^{j(\omega t - \beta z)} \left(A_i \, e^{+j k_{yi} y} + B_i \, e^{-j k_{yi} y} \right)$$

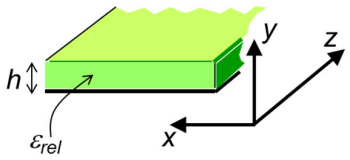
$$E_x \sim \frac{\partial H_z}{\partial y} \quad \rightarrow \quad \left. \frac{\partial H_z}{\partial y} \right|_{y=0} = 0 \quad \Rightarrow \quad \begin{cases} C := j\,(A_2 - B_2) \;\; , \;\; k_{y2}^2 > 0 \\ H_{z2}(y) = C \cos k_{y2} y \end{cases}$$

$$H_{z1}(y \rightarrow \infty) = 0 \quad \rightarrow \quad \begin{matrix} k_{y1}^2 < 0 \\ \alpha_y^2 = \beta^2 - \varepsilon_0 \mu_0 \omega^2 \end{matrix} \quad \rightarrow \quad H_{z1}(y) = A_1 \, e^{-\alpha_y y}$$

… where both α_y and k_{y2} are chosen positive real

… continuity at dielectric-air interface at y=h :

$$\begin{cases} H_{z1}(y=h) = H_{z2}(y=h) \\ E_{x1}(y=h) = E_{x2}(y=h) \end{cases}$$



$$E_x = \frac{-j\omega\mu}{\omega^2 \varepsilon\mu - \beta^2} \frac{\partial H_z}{\partial y}$$

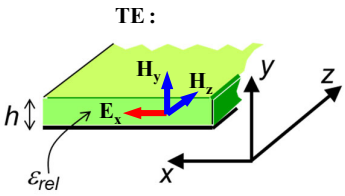
$$\alpha_y = - k_{y2} \cot k_{y2} h$$

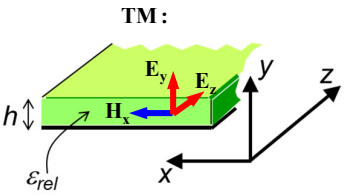
… **TE mode cutoff** ($\alpha_y = 0$): $f_{\text{cutoff,TEm}} = \dfrac{mc}{4h\sqrt{\varepsilon_{\text{rel}} - 1}}$ $m = 1,3,5,\ldots$

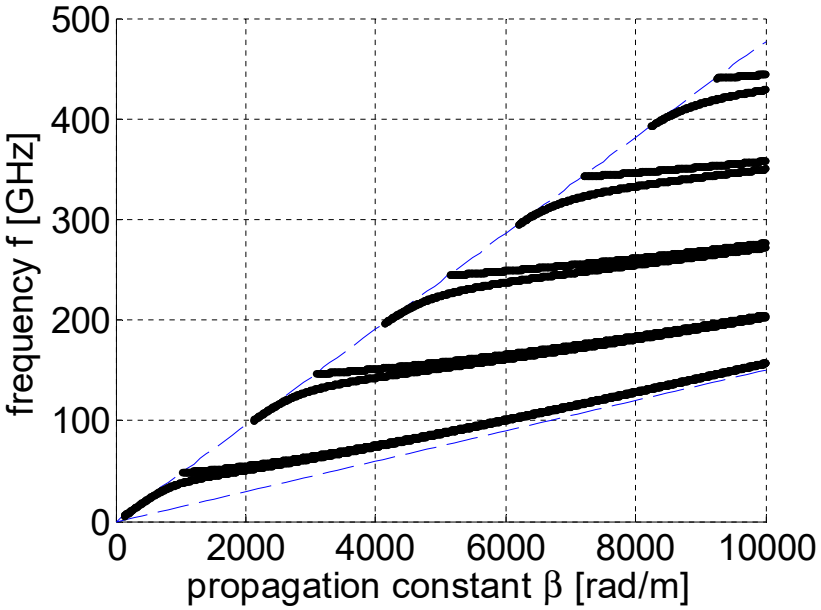
(TE_m mode)

dispersion diagram :

h = 0.51mm, ε_{rel} = 10 :
m=0 : TM_0 : f_{cutoff} = 0
m=1 : TE_1 : f_{cutoff} = 49 GHz
m=2 : TM_2 : f_{cutoff} = 98 GHz
m=3 : TE_3 : f_{cutoff} = 147 GHz
…



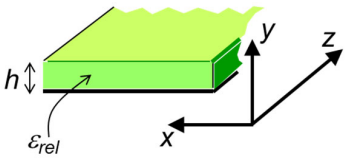




… this is an „open“ waveguide:

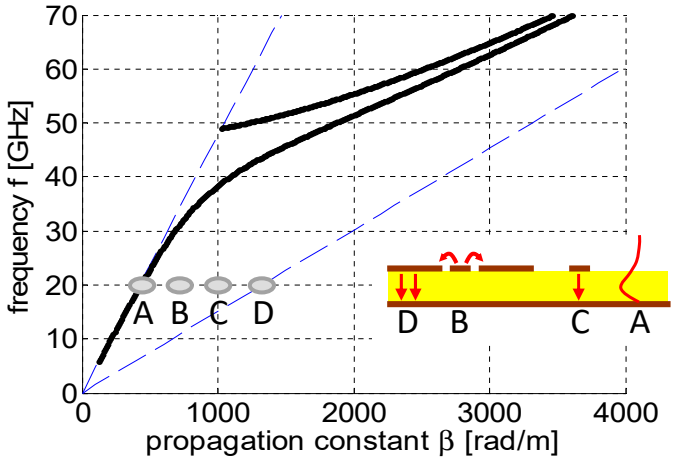
„cutoff“ does not necessarily occur at β = 0, but rather at a finite β, that is, there is some propagation in z direction even at cutoff. The ray-optics equivalent is „total reflection“; while below cutoff, the wave is not „guided“.

about surface waves in planar circuits :

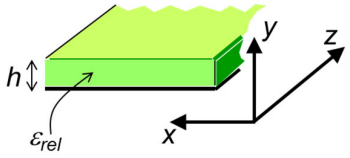


- Planar circuits (microstrip, CPW, GCPW, …) support TEM modes between ground layers or surface wave slab modes, both with zero cutoff

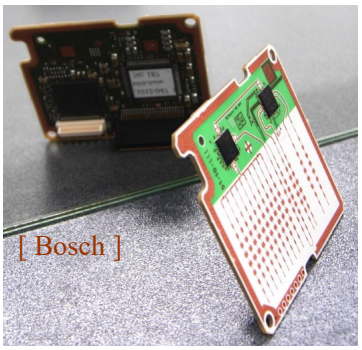
- Coupling to/from the „wanted transmission“ to these modes can be large at some discontinuities. Coupling can be large along line sections if the propagation constant of the „wanted“ mode is close to or larger than the propagation constant of the TEM mode / surface wave



about surface waves in planar circuits :

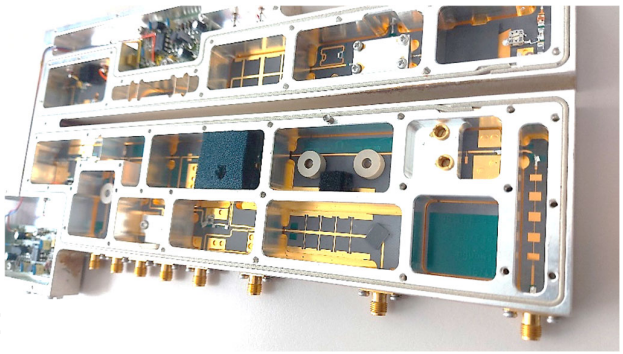


thin, low-eps board:



77 GHz automotive radar

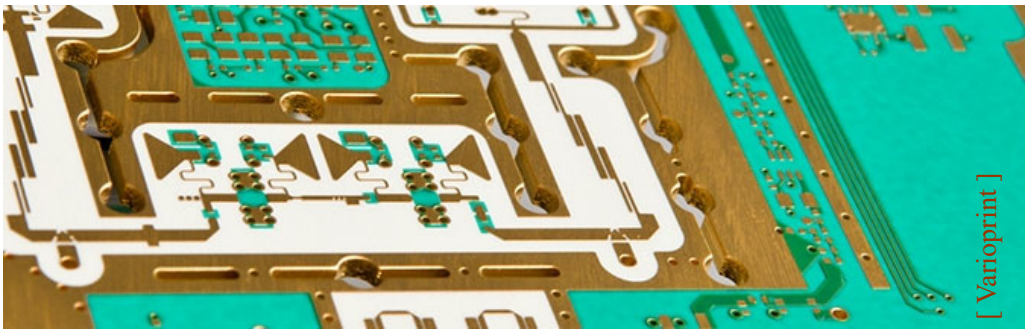
milled/cast metal cavities:



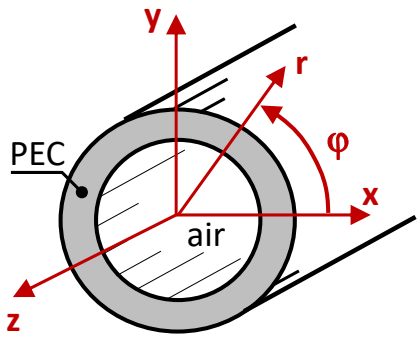
microwave laboratory measurement equipment

through-metal via-holes and slits:

demo board



circular cylindrical waveguide



… TE modes ($E_z = 0$) :

$$(\Delta_T + k_T^2)\,\overrightarrow{H_z} = 0$$

$$k_T^2 = k_0^2 - \beta^2 = \varepsilon\mu\omega^2 - \beta^2$$

$$\Delta_T \cong \frac{\partial^2}{\partial r^2} + \frac{1}{r}\frac{\partial}{\partial r} + \frac{1}{r^2}\frac{\partial^2}{\partial \varphi^2}$$

$$\text{grad}\, f = \hat{\rho}\frac{\partial f}{\partial \rho} + \hat{\varphi}\frac{1}{\rho}\frac{\partial f}{\partial \varphi} + \hat{z}\frac{\partial f}{\partial z}$$

$$\text{div}\,\vec{A} = \frac{1}{\rho}\frac{\partial}{\partial \rho}\left(\rho A_\rho\right) + \frac{1}{\rho}\frac{\partial A_\varphi}{\partial \varphi} + \frac{\partial A_z}{\partial z}$$

$$\text{rot}\,\vec{A} = \text{curl}\,\vec{A} = \hat{\rho}\left(\frac{1}{\rho}\frac{\partial A_z}{\partial \varphi} - \frac{\partial A_\varphi}{\partial z}\right)$$

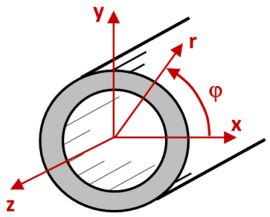
$$+\hat{\varphi}\left(\frac{\partial A_\rho}{\partial z} - \frac{\partial A_z}{\partial \rho}\right) + \hat{z}\frac{1}{\rho}\left(\frac{\partial}{\partial \rho}\left(\rho A_\varphi\right) - \frac{\partial A_\rho}{\partial \varphi}\right)$$

$$= \begin{vmatrix} \hat{\rho}\frac{1}{\rho} & \hat{\varphi} & \hat{z}\frac{1}{\rho} \\ \frac{\partial}{\partial \rho} & \frac{\partial}{\partial \varphi} & \frac{\partial}{\partial z} \\ A_\rho & \rho A\varphi & A_z \end{vmatrix}$$

$$\Delta f = \frac{1}{\rho}\frac{\partial}{\partial \rho}\left(\rho\frac{\partial f}{\partial \rho}\right) + \frac{1}{\rho^2}\frac{\partial^2 f}{\partial \varphi^2} + \frac{\partial^2 f}{\partial z^2}$$

$$= \frac{\partial^2 f}{\partial \rho^2} + \frac{1}{\rho}\frac{\partial f}{\partial \rho} + \frac{1}{\rho^2}\frac{\partial^2 f}{\partial \varphi^2} + \frac{\partial^2 f}{\partial z^2}$$

… separation of variables :



$$H_z(r,\varphi) = R(r)\cdot P(\varphi)$$

$$\underbrace{\frac{r^2}{R}\frac{\partial^2 R}{\partial r^2} + \frac{r}{R}\frac{\partial R}{\partial r} + r^2 k_T^2}_{\text{const.}} + \underbrace{\frac{1}{P}\frac{\partial^2 P}{\partial \varphi^2}}_{\text{const.}} = 0$$

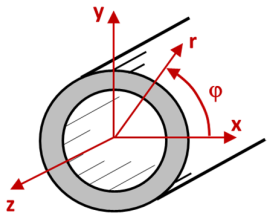
$$:= -k_\varphi^2$$

$$\frac{\partial^2 P}{\partial \varphi^2} + P k_\varphi^2 = 0$$

$$P(\varphi) = A\cos(k_\varphi \varphi) + B\sin(k_\varphi \varphi)$$

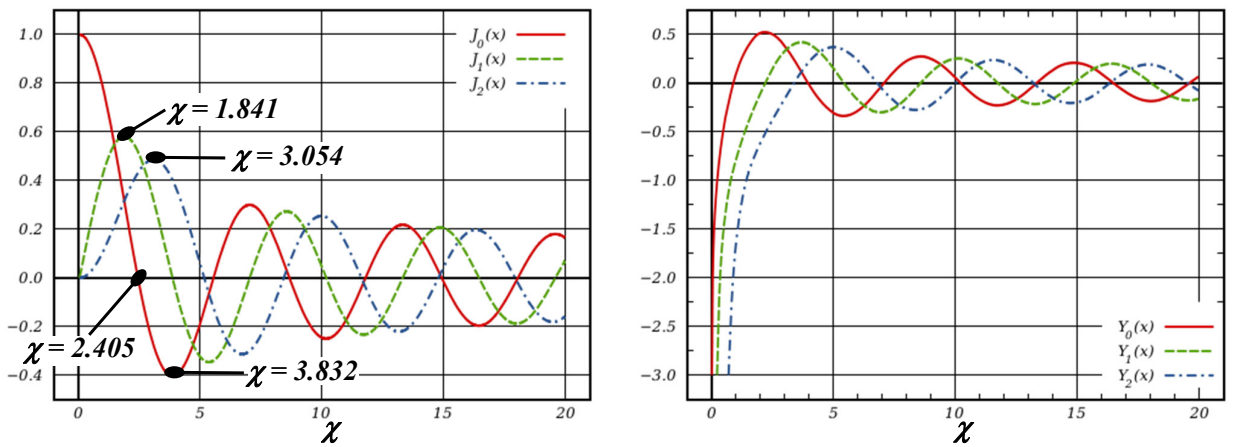
$$k_\varphi = m\pi \quad , \quad m = 0, \pm 1, \pm 2, \ldots$$

$$r^2\frac{\partial^2 R}{\partial r^2} + r\frac{\partial R}{\partial r} + R\underbrace{\left(r^2 k_T^2 - k_\varphi^2\right)}_{m^2} = 0$$



… this is Bessel's differential equation, solution :

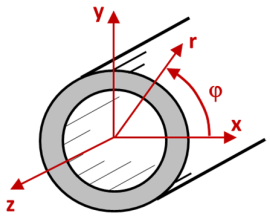
$$R(r) = C\,J_m(k_T r) + D\,Y_m(k_T r)$$



[wikipedia commons]

→ $D = 0$

$$H_z(r,\varphi) = J_m(k_T r)\,[A\cos(m\varphi) + B\sin(m\varphi)]$$



… boundary condition at $r=a$:

$$\left.\frac{\partial H_z(r,\varphi)}{\partial r}\right|_{r=a} = 0 \quad \rightarrow \quad \left.\frac{\partial J_m(k_T r)}{\partial r}\right|_{r=a} = 0$$

TE

… where the n-th zero of $\frac{\partial J_m}{\partial r}$ is designated as p'_{mn} $\quad (\, p'_{mn} = k_T a \,)$

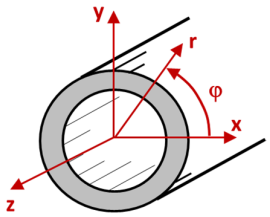
$$\frac{\partial J_0(r)}{\partial r} = 0 \quad : \quad p'_{01} = 3.832 \;,\; p'_{02} = 7.016 \;,\; p'_{03} = 10.173$$

$$\frac{\partial J_1(r)}{\partial r} = 0 \quad : \quad p'_{11} = 1.841 \;,\; p'_{12} = 5.331 \;,\; p'_{13} = 8.536$$

… fundamental mode

$$\frac{\partial J_2(r)}{\partial r} = 0 \quad : \quad p'_{21} = 3.054 \;,\; p'_{22} = 6.706 \;,\; p'_{23} = 9.969$$

… now H_z is defined and thus all other field components



TE

$$H_z(r,\varphi) = J_m(k_T r)\,[A\cos(m\varphi) + B\sin(m\varphi)]$$

… the constants A and B describe the field magitude and the orientation/rotation of the field in φ direction

… TM modes ($H_z = 0$) :

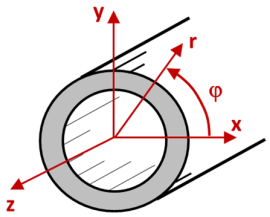
$$(\Delta_T + k_T^2)\,\overrightarrow{E_z} = 0$$

TM

$$E_z(r,\varphi) = J_m(k_T r)\,[A\cos(m\varphi) + B\sin(m\varphi)]$$

$$E_z(r,\varphi)|_{r=a} = 0 \quad \rightarrow \quad J_m(k_T r)|_{r=a} = 0$$

… the TM modes are defined by the ***n***-th zero of the Bessel function (of first kind) of order ***m*** :



TM

$$p_{mn} = k_T a$$

$J_0(r) = 0$: $p_{01} = 2.405$, $p_{02} = 5.520$, $p_{03} = 8.654$

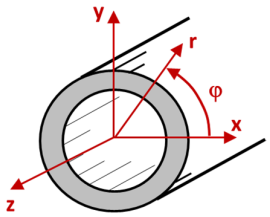
... first higher-order mode

$J_1(r) = 0$: $p_{11} = 3.832$, $p_{12} = 7.016$, $p_{13} = 10.173$

$J_2(r) = 0$: $p_{21} = 5.136$, $p_{22} = 8.417$, $p_{23} = 11.620$

… now E_z is defined and so are all other field components

… the transverse field components in cylindrical coordinates are obtained from the longitudinal components (E_z, H_z) using Maxwell's equations :



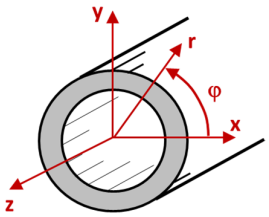
$$E_r = -\frac{j}{k_T^2}\left(\beta\frac{\partial E_z}{\partial r} + \frac{\omega\mu}{r}\frac{\partial H_z}{\partial \varphi}\right)$$

$$E_\varphi = \frac{j}{k_T^2}\left(-\frac{\beta}{r}\frac{\partial E_z}{\partial \varphi} + \omega\mu\frac{\partial H_z}{\partial r}\right)$$

$$H_r = \frac{j}{k_T^2}\left(\frac{\omega\varepsilon}{r}\frac{\partial E_z}{\partial \varphi} - \beta\frac{\partial H_z}{\partial r}\right)$$

$$H_\varphi = -\frac{j}{k_T^2}\left(\omega\varepsilon\frac{\partial E_z}{\partial r} + \frac{\beta}{r}\frac{\partial H_z}{\partial \varphi}\right)$$

cutoff frequencies in circular cylindrical waveguide:



$$k_T^2 = k_0^2 - \beta^2 = \varepsilon\mu\omega^2 - \beta^2 \quad \rightarrow \quad f_c = \frac{c k_T}{2\pi}$$

$$\text{TE}_{mn}: \; f_c = \frac{c}{2\pi}\frac{p'_{mn}}{a} \qquad \text{TM}_{mn}: \; f_c = \frac{c}{2\pi}\frac{p_{mn}}{a}$$

… fundamental mode is TE_{11} : $p'_{11} = 1.841$

… the monomode region is limited by TM_{01} : $p_{01} = 2.405$

ratio of 1.3 only !

lowest mode with rotational symmetric field

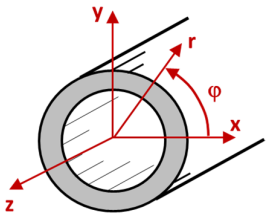
… this ratio is larger in rectangular waveguide (typically 2:1), even larger in ridge waveguide, and goes to infinity in TEM waveguide

… also of interest is TE_{01} : $p'_{01} = 3.832$

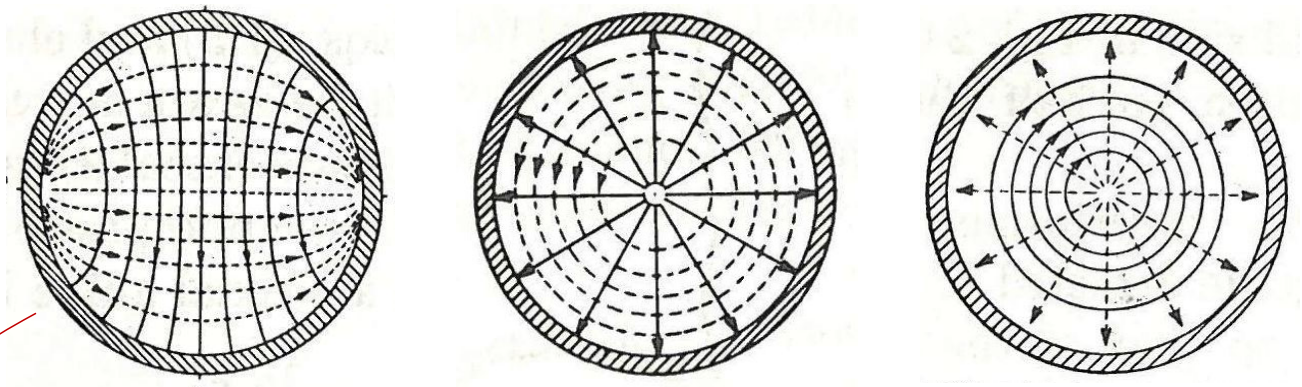
5th mode, degenerated with TM_{11},

E-field is focussing in the center, reduced wall currents, lowest dissipative loss

modes of the circular cylindrical waveguide:



TE_{11} $p'_{11} = 1.841$	TM_{01} $p_{01} = 2.405$	TE_{01} $p'_{01} = 3.832$



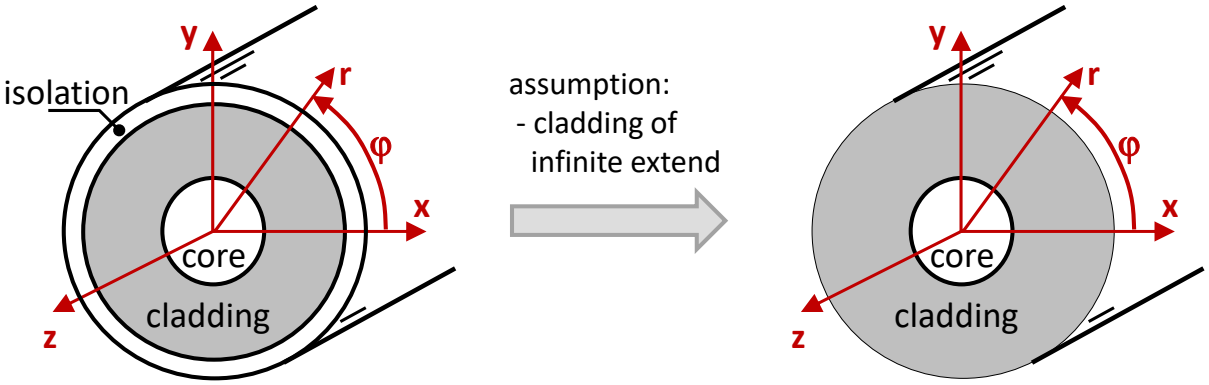
… lowest cutoff, occurs in two orthogonal polarizations

solid lines: E-field *dashed lines: H-field*

dielectric rod waveguide & optical fiber

the cylindrical dielectric rod (embedded in a homogeneous medium of infinite extend) is of limited technical importance (e.g., antennas)

but: the dielectric rod is a quite good simplification of the optical fiber:



the core (radius a) has a higher permittivity (diffraction index) than the cladding, but this „difference“ is small (about 0.1%...2%)

$$\varepsilon_{r,core} > \varepsilon_{r,cladding}$$

the analysis of the modes of the cylindrical dielectric rod is similar to both the circular metallic waveguide and the surface wave slab

in the core region, modes are described
- in radial direction by the Bessel function ***J***
- in angular direction by the sine and cosine functions

in the cladding region, modes are described
- in the radial direction by a decaying function which is also a solution of Bessel‘s diff. equ., that is, the modified Bessel function ***K***

as a result, most modes (with the exeption of modes with angular invariance, m=0) exist in two orthogonal polarizations

modes are TE or TM for m=0, but otherwise „hybrid,“ that is, those modes show all six field components

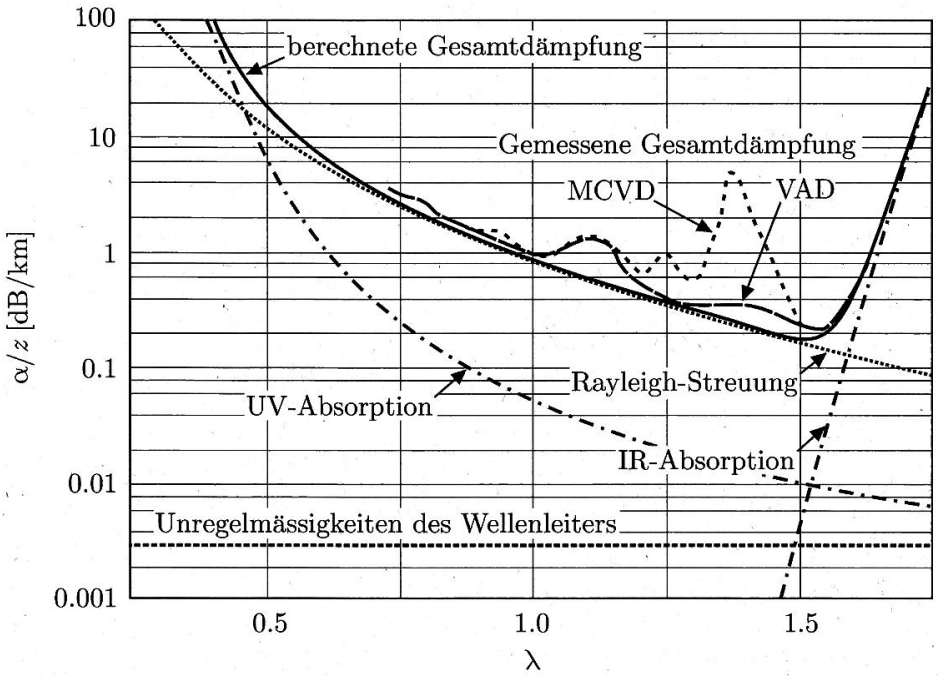
the fundamental mode, HE_{11}, has zero cutoff but at low frequencies, its fields decay very slowly in the cladding. Practically relevant are frequencies just below the cutoff of the first higher-order mode (where then most of energy is carried in the core)

some typical problems related to optical fiber communications:

- polarisation dispersion: the fundamental HE_{11} mode appears in two polarisations. Energy couples at the smallest discontinuity from one mode to the other because of their theoretically identical propagation constant and their similar fields. Due to imperfections (anisotropy of dielectric constant, non-circular core), both modes do not show exactly the same propagation constant (i.e., velocity), causing dispersion.
- material dispersion: the diffraction index (i.e., permittivity) of glass varies over wavelength. A modulated signal has a certain bandwidth, therefore, parts of the signal at one end of the signal (optical) bandwidth travel through a medium of a different permittivity as the parts of the signal at the other end of the (optical) bandwidth do. This leads to dispersion.
- modal dispersion: in a multi-mode fiber, a large-diameter core leads to 1000‘s of modes above cutoff. These modes have different field distributions and therefore different velocity. Energy scintillates between these modes, causing strong dispersion.

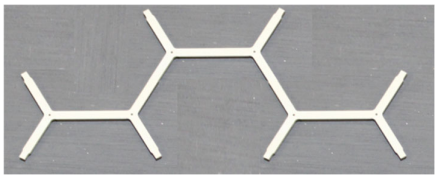
the engineering choice of wavelength for optical fiber communications:

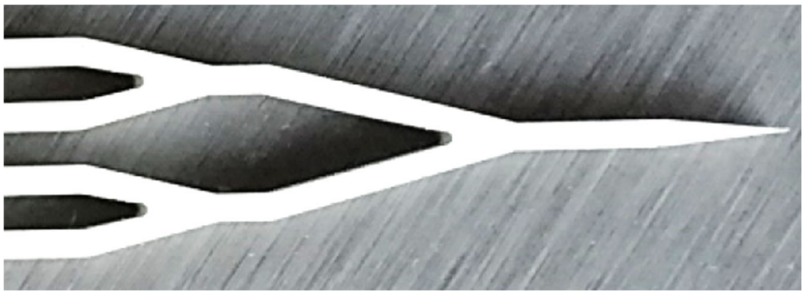
- for min. material dispersion (in SiO_2 based glass), choose $\lambda \approx 1.3\,\mu m$
- for min. attenuation (in SiO_2 based glass), choose $\lambda \approx 1.5\,\mu m$
- to avoid polarisation dispersion, chose a fiber with elliptic core
- for easy in/out coupling, chose a multi-mode fiber
- other issues:
  - laser
  - PD
  - EDFA

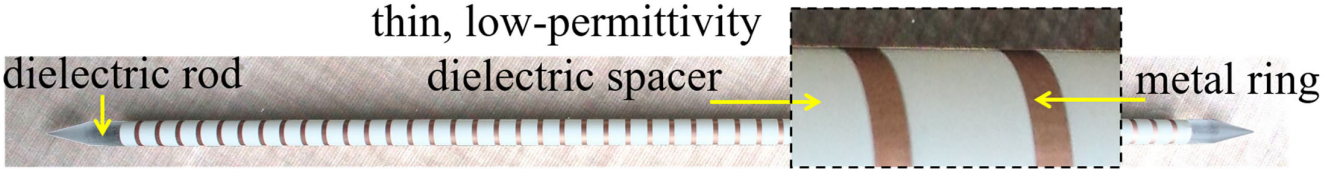


dielectric waveguides for mm-waves

for low-loss transmission and for antennas







dielectric waveguides for mm-waves

most transmission lines (e.g., microstrip, coax) show excessive loss per length at frequencies above 100 GHz
(main reason: high current density)

dielectric-only waveguides do not have conductive loss and can be designed such that large fraction of em-field is in the air and not in the lossy dielectric (remember fundamental surface-wave mode of slab with zero cutoff !)

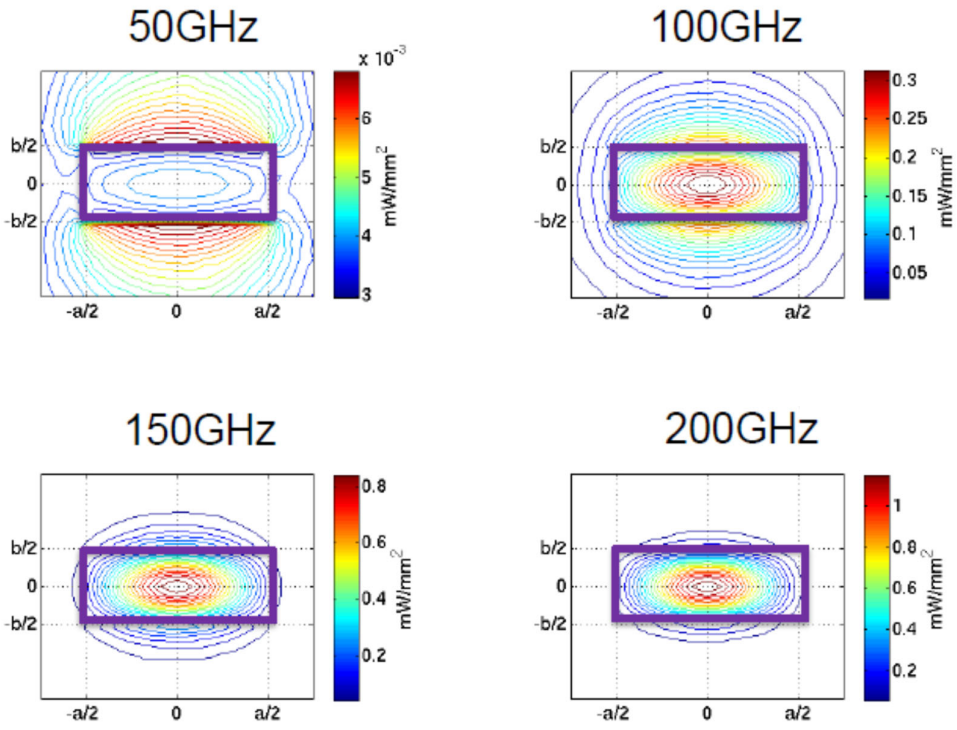
dielectric loss tangent can still be quite small at mm-waves

if most of em-field is in the air, then loss can be arbitrarily small but:
- difficult to support the waveguide mechanically
- strong radiation at waveguide bends
- difficult to transition to „normal“ transmission lines or waveguides

therefore, a tradeoff exists between loss and „practicability“

power density distribution for 2 mm x 1 mm x-section polystyrene strip:

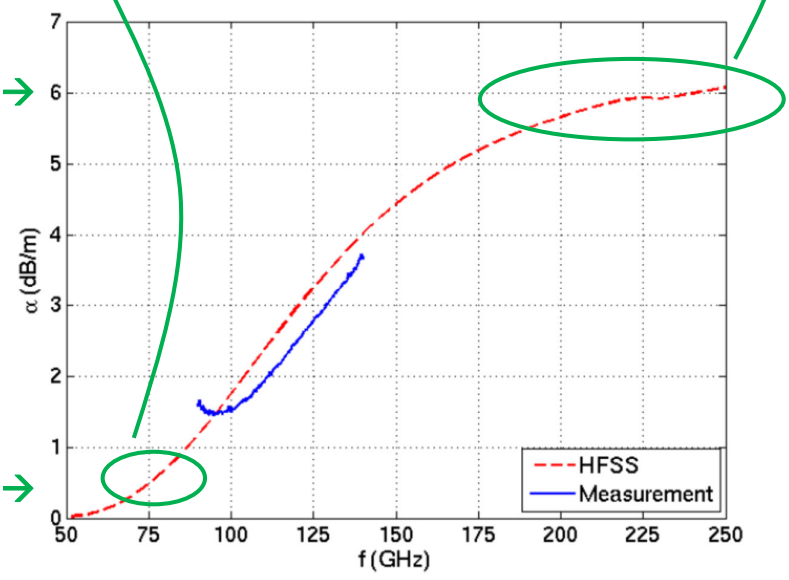
$\varepsilon_{rel} = 2.55$



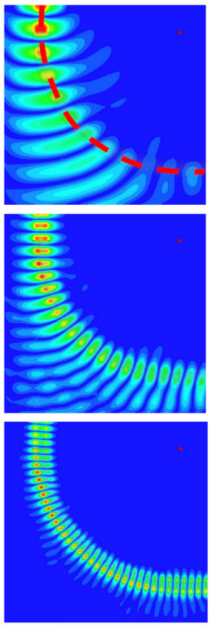
[P. Reynaert, M. De Wit, "Polymer waveguides as an alternative to optical and copper high-speed communication," IEEE Inter. microwave Symp. (IMS), workshop WSA-7, June 2017]

Fields concentrated in the dielectric:
- large dielectric loss contribution
- low radiation at bends

Little field in the dielectric, most in air:
- extremely low transmission loss
- very large leakage at bends



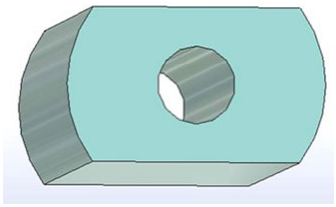
Increased frequency leads to lower radiation at bends:



[P. Reynaert, M. De Wit, "Polymer waveguides as an alternative to optical and copper high-speed communication," IEEE Inter. microwave Symp. (IMS), workshop WSA-7, June 2017]

Flexible polymer (PTFE, $\varepsilon_r = 2.1$) fiber:

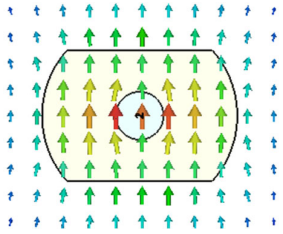
- non-circular for polarization/mode decoupling:



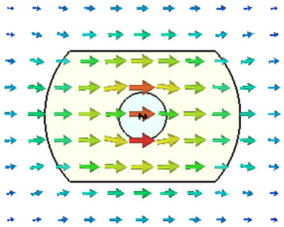
- approx. 1.5 dB/m at 60 GHz

- don't touch ? – foam shield

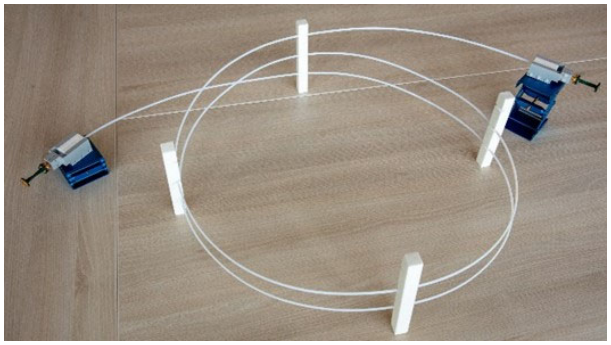
- don't bend ? – radiation is negligible for radius > 20 cm



E-field, mode 1



E-field, mode 2



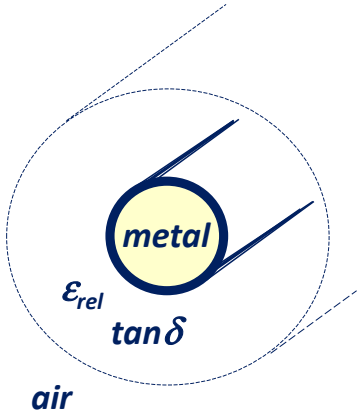
foam shield

Goubau line (or: Harms-Goubau-line)

… F. Harms (1907)
G. Goubau (1951)

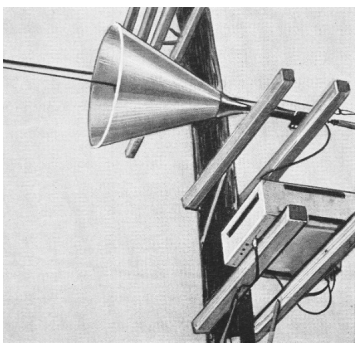
- a „coaxial line without outer conductor"
- a grounded dielectric slab „morphed to circular"

- guides a surface wave with zero cutoff
- low loss (depends on energy fraction in air/dielectric)
- difficult to support mechanically
- radiation at sharp bends

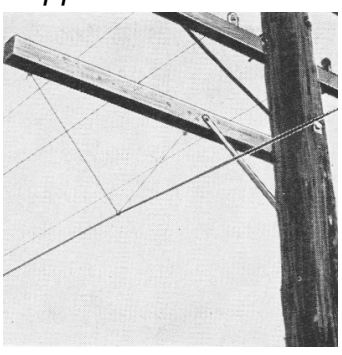


UHF TV transmission/distribution system in the 1950s

coax launcher:



support structure:

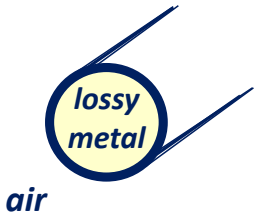


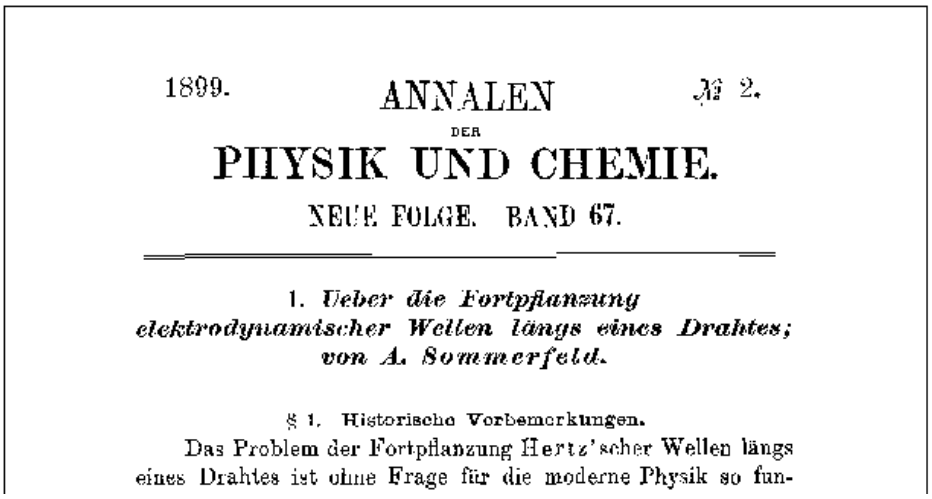
[R.B. Gary, "The G-line community TV system," Radio & Television News, Nov. 1956]

Sommerfeld wire

- similar to Goubau-line, but lossy metal instead of dielectric coating
- note that a single perfectly-conducting wire does not guide any wave !

- guides a wave with zero cutoff
- historic milestone (1899!)





1899. ANNALEN DER PHYSIK UND CHEMIE. № 2.

NEUE FOLGE. BAND 67.

1. *Ueber die Fortpflanzung elektrodynamischer Wellen längs eines Drahtes;* von *A. Sommerfeld.*

§ 1. Historische Vorbemerkungen.

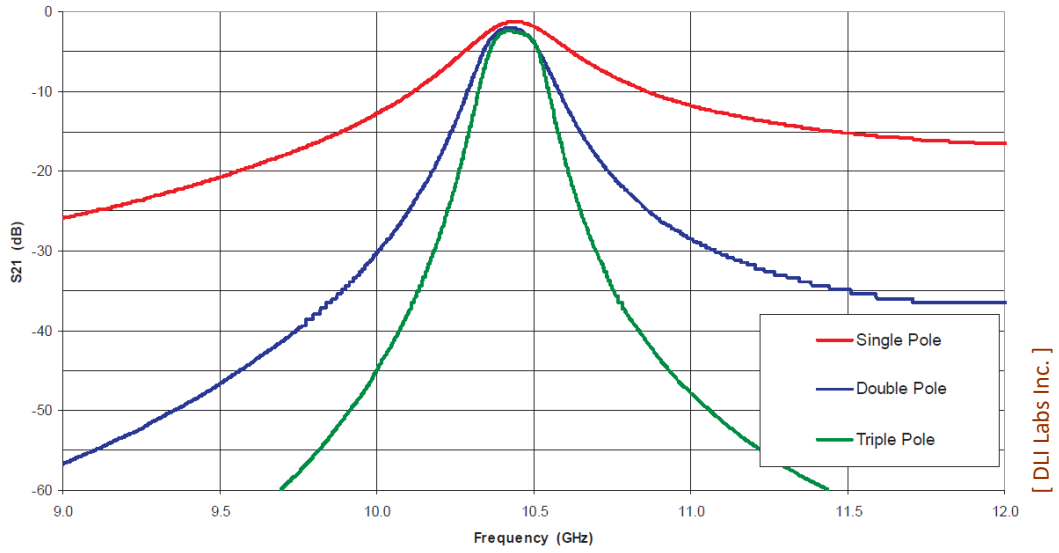
Das Problem der Fortpflanzung Hertz'scher Wellen längs eines Drahtes ist ohne Frage für die moderne Physik so fun-

Radio Frequency Technology — main topics :

- A: introduction & motivation
- B: transmission lines & waveguides
  - line equivalent circuit
  - impedance transformation
  - planar transmission lines
  - rectangular waveguide
  - slab waveguide
  - circular waveguide, dielectric rod & fiber
- **C: resonators**
  - resonators
- D: non-reciprocal components
  - non-reciprocity
  - isolator & circulator
- E: RF measurement techniques
  - about S parameters
  - power measurement
  - network analyzer
  - noise measurement
  - signal analysis

waveguide resonators and filters

- resonators are basic frequency-selective components used as frequency filters or impedance transformers
- several (cascaded) resonators increase selectivity (slope steepness) at the expense of increased pass-band insertion loss (and increased size):



- at low frequencies, resonance circuits are made of L and C

  … lumped resonators

- at higher frequencies, resonantors are made of short sections (e.g., half wavelength section) of transmission lines

  … distributed resonators

- at microwaves & mm-waves, half-wavelength sections of air-filled waveguides can be used

  … cavity resonators

- important are losses attributed to the resonator, as loss increases in-band transmission loss and ultimately reduce the frequency selectivity of the resonator/filter

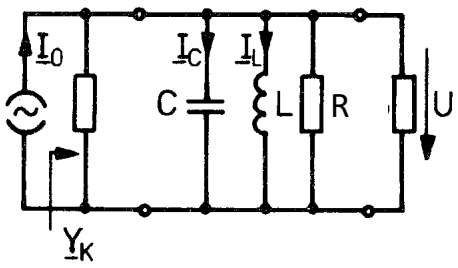
- resonators with low loss require large volume

[mini circuits]

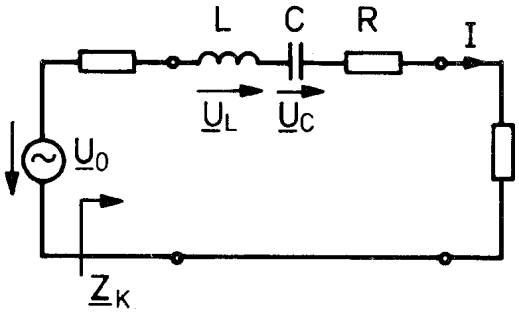
about resonance circuits

- the ratio between stored energy and dissipated energy (per cycle) is the quality factor Q (*dt.*: Güte) (more specifically: $Q = 2\pi \frac{\text{stored Energy}}{\text{dissipated Energy}}$)

parallel resonance circuit

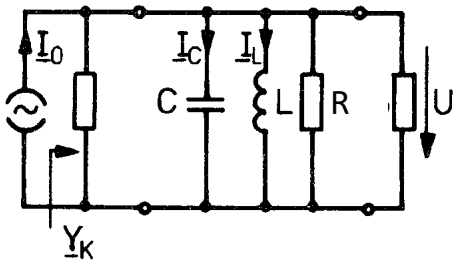
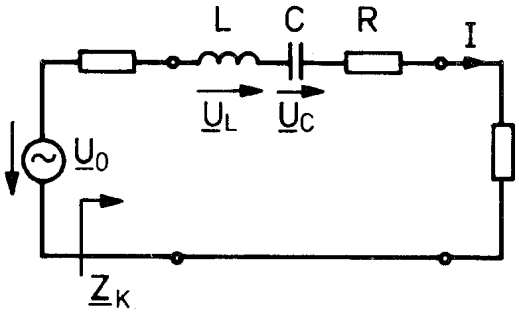


series resonance circuit



- unloaded Q : $Q_{0,\text{parallel}} = R\sqrt{\frac{C}{L}}$ $Q_{0,\text{series}} = \frac{1}{R}\sqrt{\frac{L}{C}}$

- loaded Q : considers source and load („R“ then is the parallel/series connection with source and load resistances)

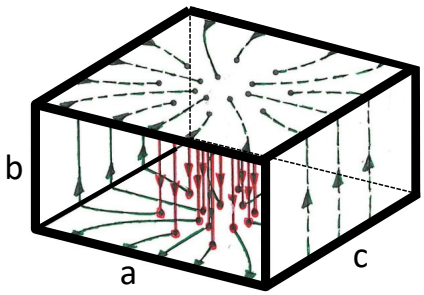
	parallel resonance circuit	*series resonance circuit*
		
- unloaded resonator :	„short“ at low & high frequencies, R at resonance	„open“ at low & high frequencies, R at resonance
- 3dB-bandwidth :	frequency band limited to where output voltage reduces to 0.71 of max	frequency band limited to where output voltage increases to 1.41 of min

$$Q = \frac{f_r}{f_{max} - f_{min}} = \frac{f_r}{\Delta f_{3dB}}$$

half wavelength resonators

- similar to mechanics or acoustics, half-wavelength long structures resonate

<u>rectangular waveguide resonator:</u>



$$\omega^2 \varepsilon \mu = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 + \left(\frac{p\pi}{c}\right)^2$$

m=1 , n=0 , p=1 → TE_{101} (a.k.a. „H_{101}“)

→ f_r does not depend on b !

→ index notation is not unique

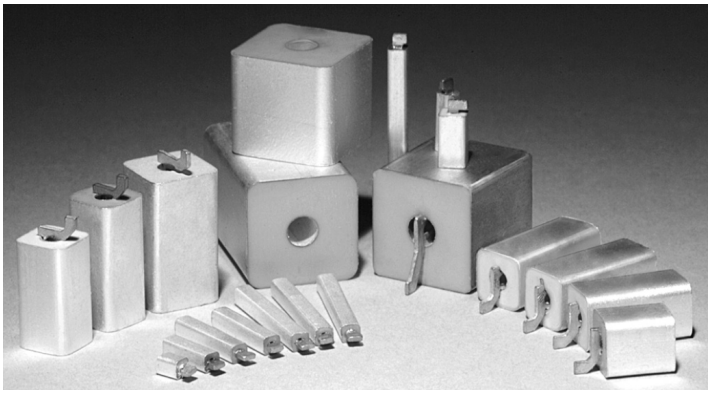
quarter wavelength resonators

- similar to mechanics or acoustics, quarter-wavelength long structures can resonate if one end is kept „loose“/“open“ and the other end is kept „fixed“/“shorted“

- a rectangular waveguide / circular waveguide open end will radiate

coaxial line resonator:

- open cross section much smaller than free-space wavelength
- often filled with high-permittivity dielectric for size reduction
- Q factor still much higher compared to planar transmission-line resonators

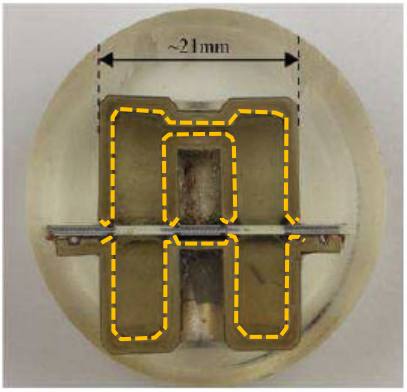


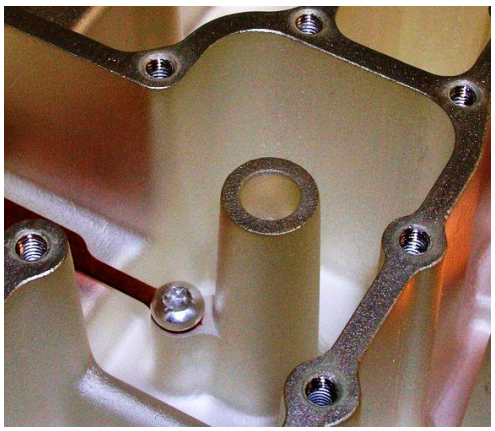
re-entrant resonators

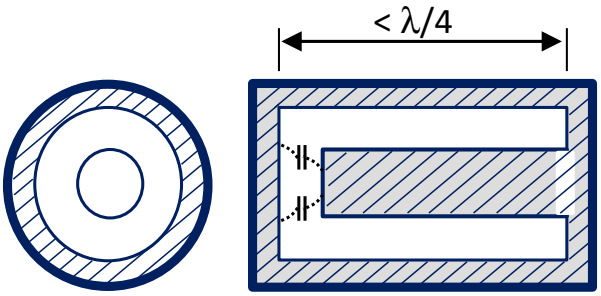
- the length of a resonant transmission-line section can be reduced by use of (quasi-) lumped C or L

re-entrant co-axial line resonator
(*dt.*: Topfkreis):

- a less-than-quarter-wavelength long coaxial resonator is loaded at its open end by an additional, quasi-lumped capacitance

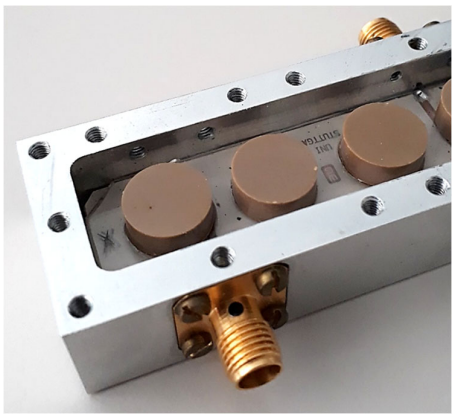


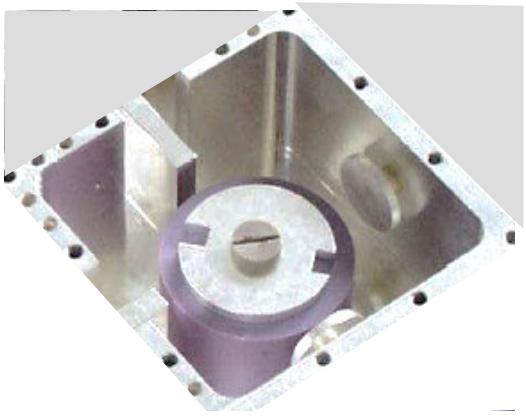




dielectric resonators

- made of low-loss, high-permittivity material in form of puck or ring
- resonates in TE_{011}-like mode (dielectric-air interface behaves like PMC)
- some radiation occurs – in filters, metal enclosure prevents radiation, thereby increases Q





- typical resonator dielectric material is oxide ceramic (single crystals typically have lower tanδ but are *much* more expensive)

e.g., Sapphire (Al203). Note: strongly anisotropic

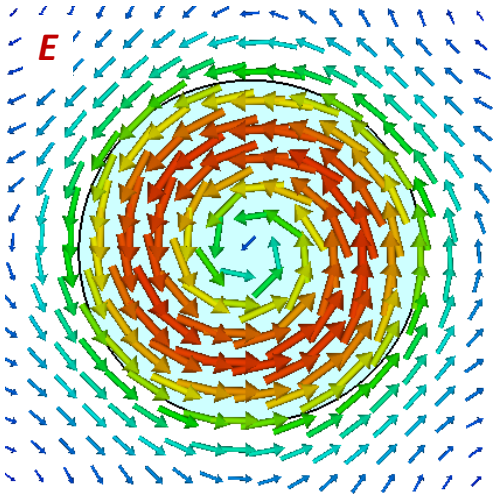
Material Composition	ε	Loss Tangent at 4 GHz	Temperature Coefficient ppm/°C
Ba Ti_4O_9	38	0.0001	+4
$Ba_2Ti_9O_{20}$	40	0.0001	+2
(Zr-Sn) TiO_4	38	0.0001	–4 to + 10 adj.
$Ba(Zn_{1/3}Nb_{2/3})O_2$-$Ba(Zn_{1/3}Ta_{2/3})O_2$	30	0.00004	0 to + 10 adj.
BaO-PbO-Nd_2O_3-TiO_2	90	0.0002 (1 GHz)	+10 to –10 adj.
$MgTiO_3$-$CaTiO_3$	21	0.000007	+10 to –10 adj.

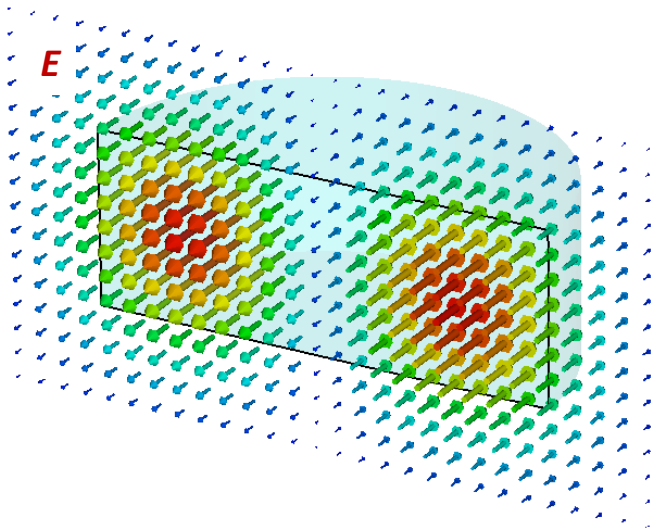
Temperature coefficient:
- Material expansion
- Change of permittivity
- both

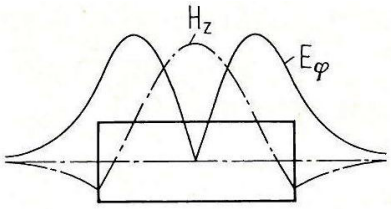
Combined temperature coefficient can be used to compensate temp variation of other parts (e.g., oscillator active stage), resulting in temp-stable operation.

- typically, higher ε_{rel} relates to higher tanδ
- typically, $f \propto \tan\delta$

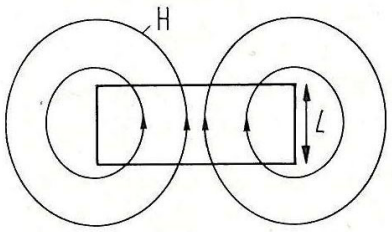
- TE_{011} field distribution

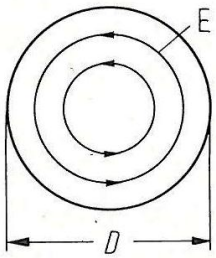






[Meinke Gundlach, p. L49]





resonator loss

- unloaded resonators show dielectric loss, conductor loss, radiation loss

- these loss contributions are difficult to measure separately, but can often easily be simulated separately

- the Q factor contributions can be added:

$$\frac{1}{Q_{\text{total}}} = \frac{1}{Q_{\text{external_coupling}}} + \frac{1}{Q_{\text{dielectric}}} + \frac{1}{Q_{\text{conductor}}} + \frac{1}{Q_{\text{radiation}}}$$

- conductor loss depends on metal conductivity, crystalinity, roughness

  … at f > 20 GHz, mirror-like surface finish is required to reduce conductivity by < 5%

- dielectric loss (tanδ) typically grows with frequency

  … for many ceramic materials, tanδ ~ f in the GHz-frequency range

- unloaded Q of rectangular waveguide TE_{101} resonator:

$$Q_{\mathrm{TE101}} = \frac{b\, c_0}{2\,\delta\, f_r}\,\frac{(a^2+c^2)^{3/2}}{c^3(a+2b)+a^3(c+2b)}$$

height: b
skin depth : δ

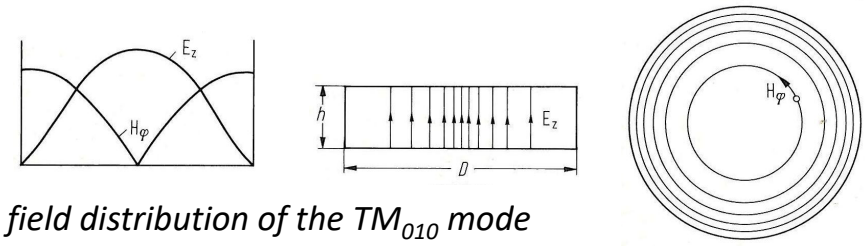
- unloaded Q of circular waveguide TM_{010} resonator:

$$Q_{\mathrm{TM010}} = 0.38\,\frac{c_0}{\delta\, f_r}\,\frac{1}{1+0.50\,D/h}$$

„the morphed version of rectangular TE_{101}“

$$f_{\mathrm{r,TM010}} = \frac{c_0}{1.31\,D}$$

diameter: D
height: h



field distribution of the TM_{010} mode

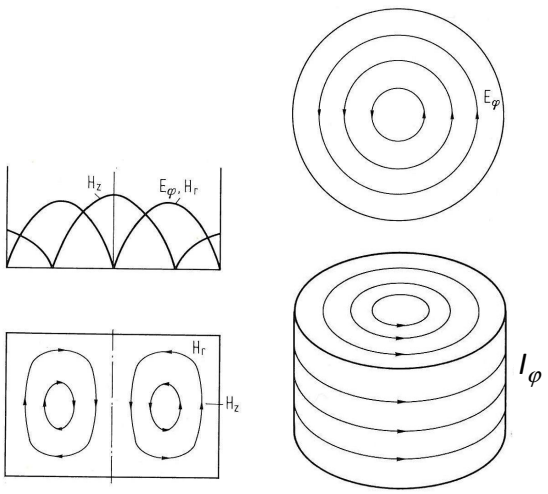
[Meinke Gundlach, p. L46]

- unloaded Q of circular waveguide TE_{011} / TE_{012} / TE_{013} resonator:

$$Q_{\mathrm{TE011}} = 0.61\,\frac{c_0}{\delta\, f_r}\,\frac{(1+0.17\,(D/h)^2)^{3/2}}{1+0.17\,(D/h)^3}$$

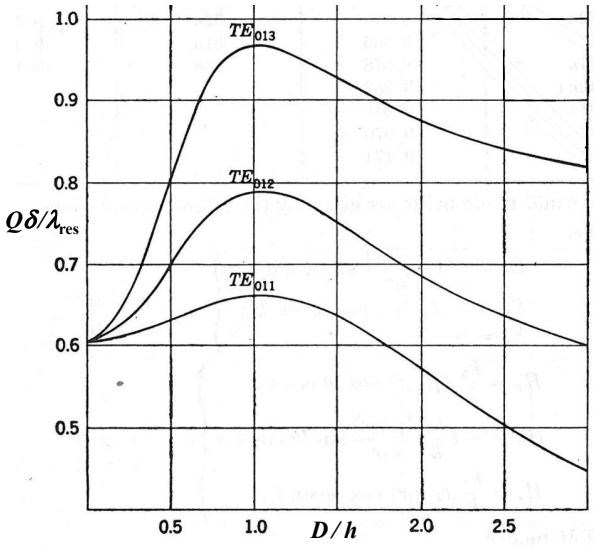
highest Q (but large volume)

circumferential currents only



field distribution of the TE_{011} mode

[Meinke Gundlach, p. L46]



... this plot is for 3 differently-sized resonators showing the same resonance frequency !

Q-factor of cavity resonances TE_{01n} (n = 1,2,3)

[Montgomery, Technique of Microwave Measurement, 1964, p. 300]

resonator input / output coupling

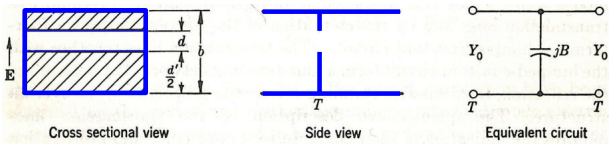
- coupling to transmission lines or to the neighboring resonator

capacitive c.:
open-ended line in high-E-field region

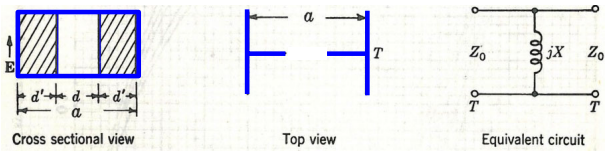
inductive c.:
loop-ended line in high-H-field (high wall current) region

iris c.:
wall opening to the neighboring resonator

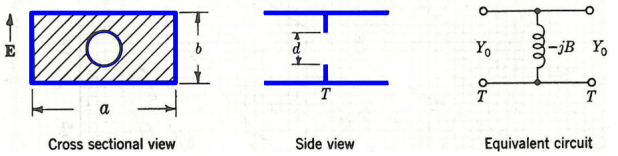
- capacitive iris:
short line section of very low impedance:



- inductive iris:
short below-cutoff (evanescent-mode) line section:

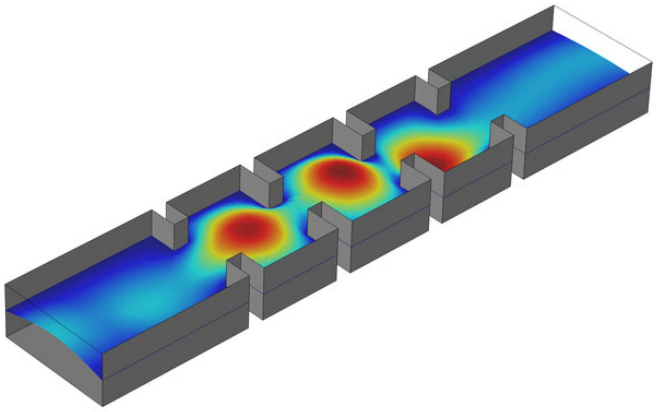


- hole coupling:



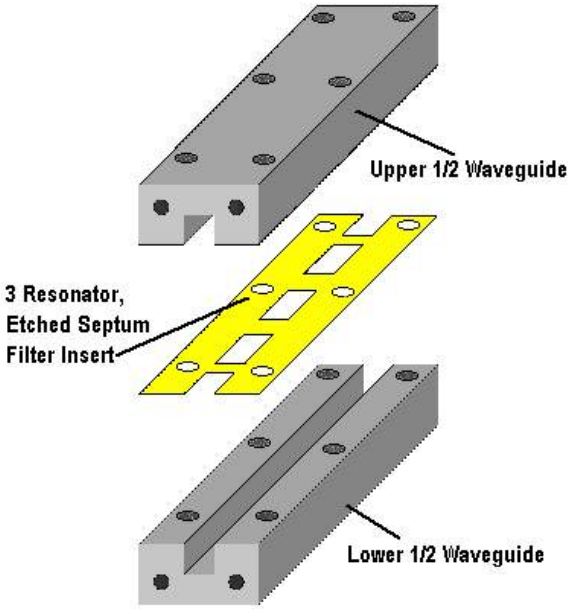
resonator coupling in filters

inductive-iris-coupled three-resonator waveguide filter (E field magntitude plot)



[www.comsol.com]

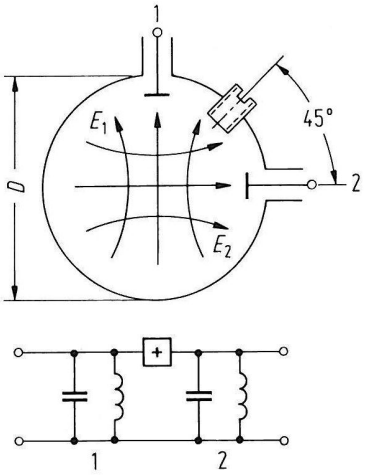
E-plane-septum evanscent-coupled three-resonator waveguide filter

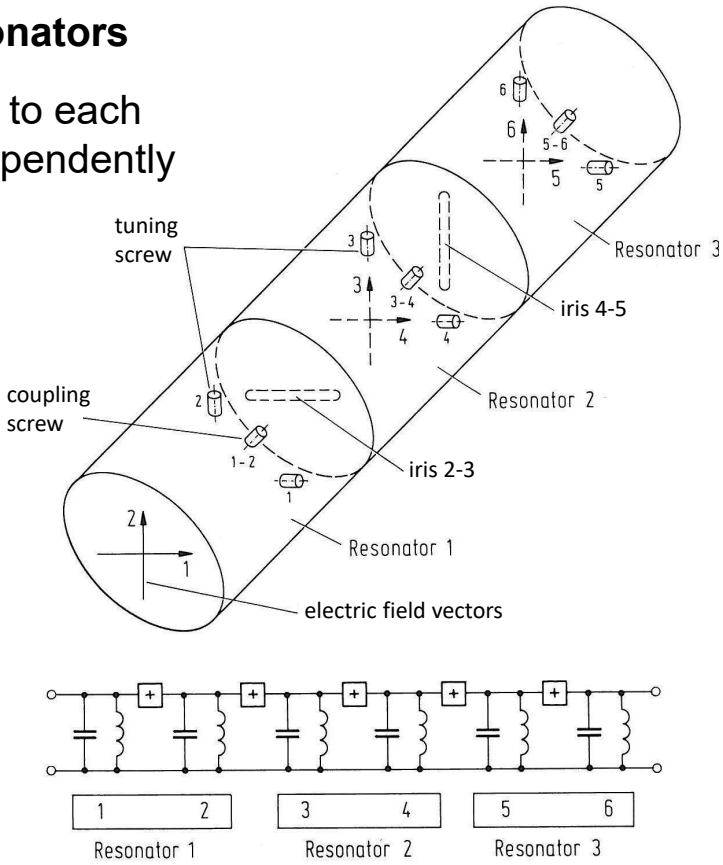


[www.e-fab.com]

dual-mode and triple-mode resonators

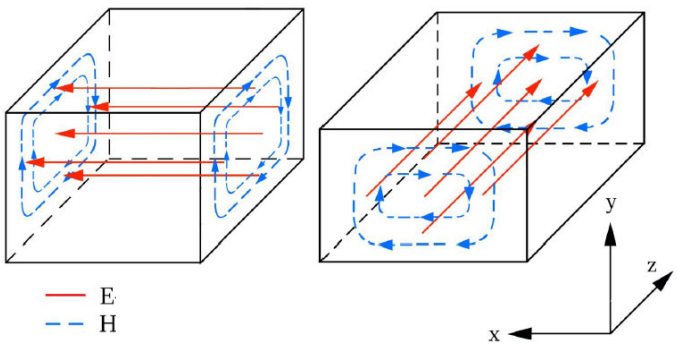
- cavity resoances are orthogonal to each other, that is, they can exist independently from one another (this is in reality an approximation)
- degenerate (or close to) resonances can make use of the same volume:

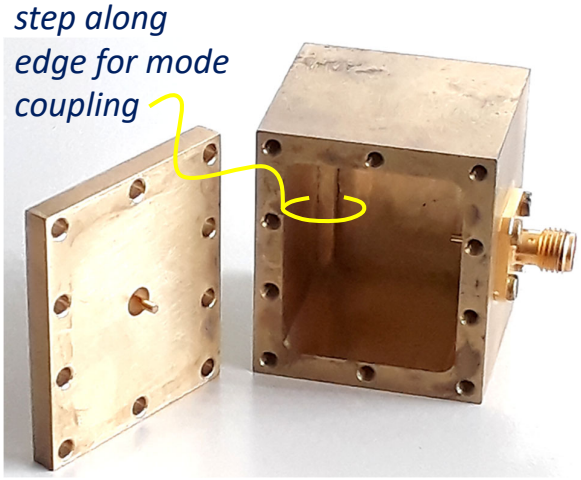


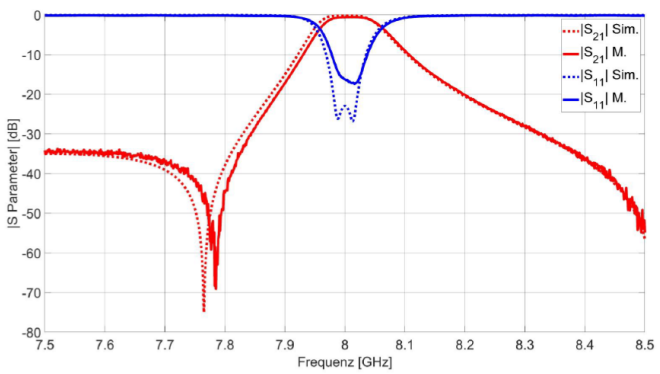


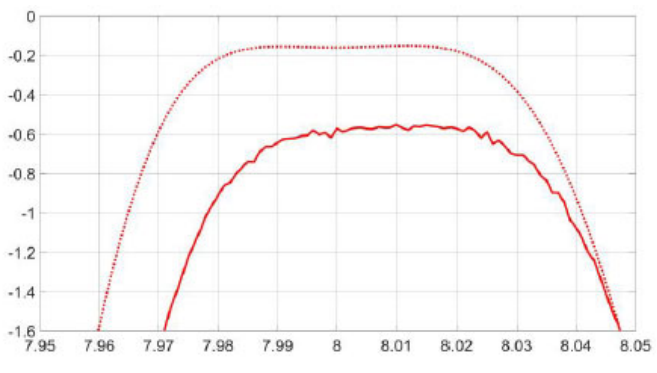
[Zinke Brunswig, 1990, p. 328-9]

TE_{011} / TE_{110} cavity dual-mode filter



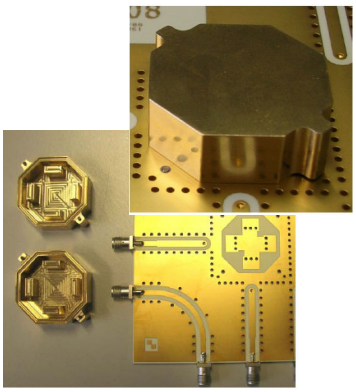


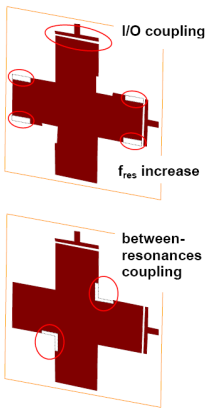


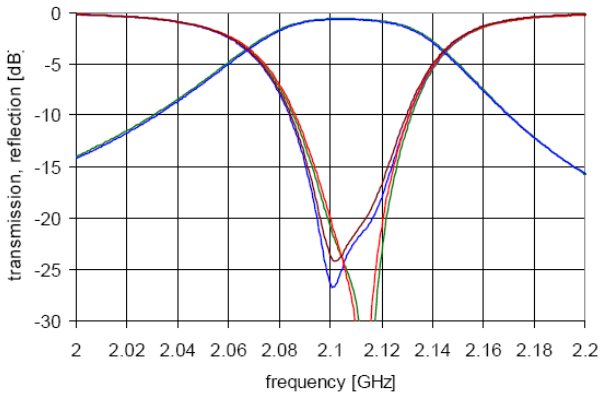


[P. Allbach, 2019]

- dual-mode resonator: surface-mount cavity resonator and filter

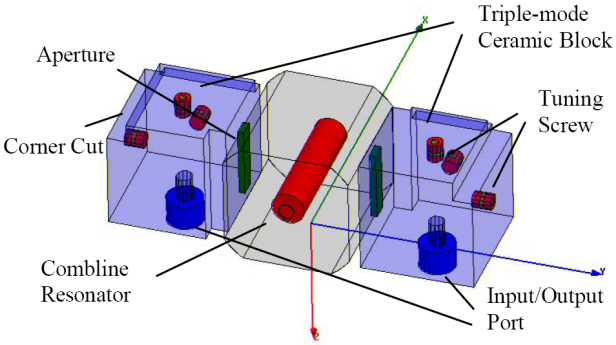


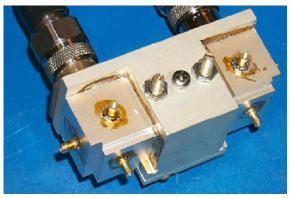


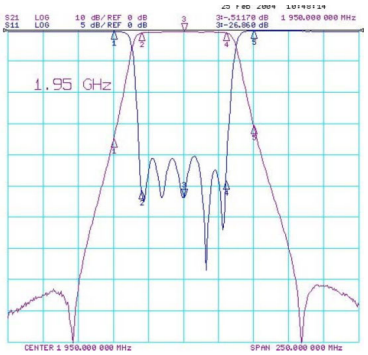


[Hesselbarth, IEEE IMS 2009]

- triple-mode resonator: dielectric cube resonator and (7-pole-) filter



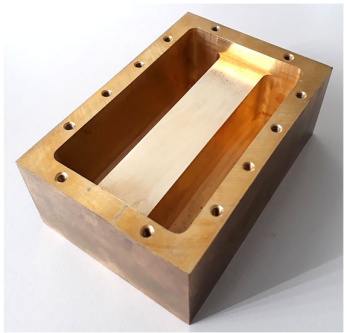




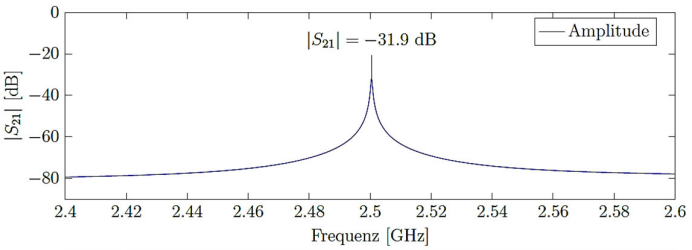
[Rahman, EuMC 2004]

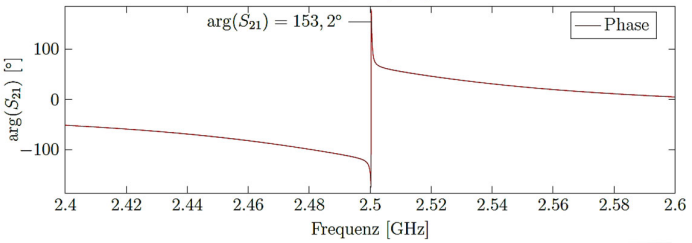
Resonator example: rectangular / cylindrical cavity resonators

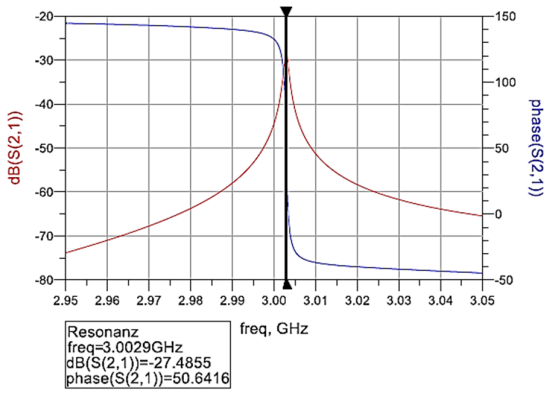
- from student lab course („Fachpraktikum“) microwave oscillator project



rectangular cavity with ridge, 3 GHz







cylindrical cavity with post, 2.5 GHz

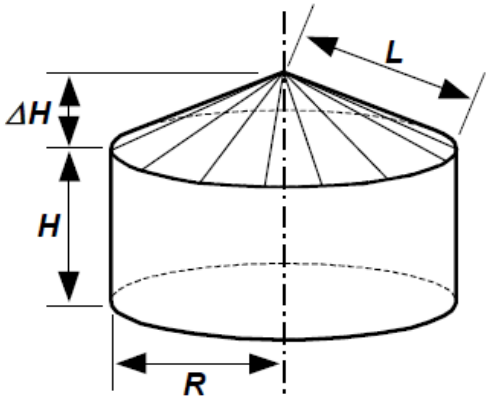
[IHF PüL SS 2017]

Resonator example: cylindrical cavity TM_{010} resonator

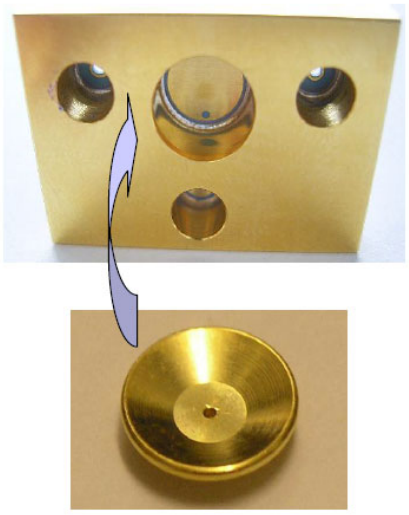
- cylindrical cavity of radius R
- resonance frequency independent of height H

$$f_{R,TM010} = 0.38274 \cdot \frac{c}{R}$$

- dome-shaped top will slightly reduce resonance frequency
- frequency-stabilization over temperature change: if dome-shaped top has higher temp-coefficient of expansion than cylinder



cylinder wall milled & top turned → Q_0 = 2000 @ 32 GHz

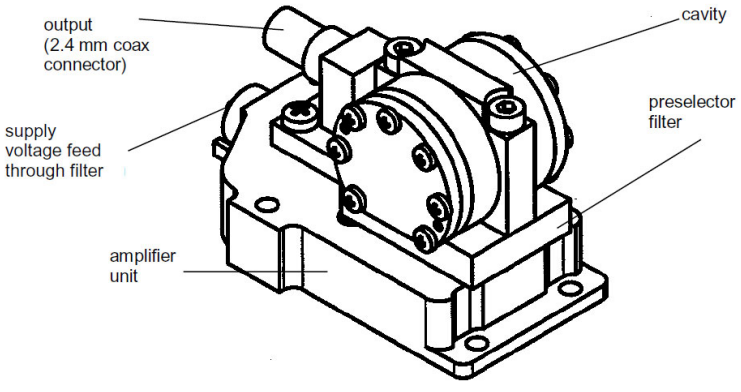


cylinder wall rolled & top high-speed turned → Q_0 = 2500 @ 32 GHz

Resonator example: cylindrical cavity TE_{013} resonator

- higher-order resonances have higher Q (more stored energy per lossy wall)
- issues:
  - + larger volume,
  - + other resonances nearby
  - + degenerate mode resonances

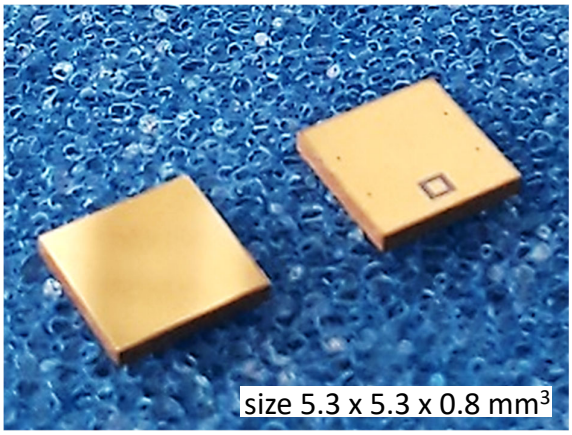
- example: 32 GHz (Ka-band) oscillator stabilized with TE_{013}-mode cavity in the feedback loop
  - + degenerate mode TM_{113} suppressed by lossy trace
  - + other mode resonances suppressed by planar filter
  - + measured Q_L of 18‘000

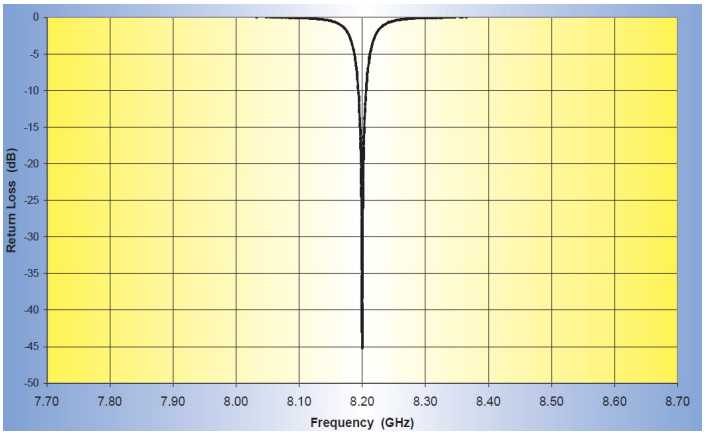


[Geist, Barth, EuMC 2003]

Resonator example: dielectric-filled TE_{110} resonator

- filling the cavity resonator with dielectric material reduces size & volume proportional to $\sqrt{\varepsilon_{rel}}$
- loss increases and Q_0 reduces
- DLI Labs, 8.2 GHz resonator for 50Ω planar feed: $Q_0 = 500$ ($Q_L = 250$)
- $\varepsilon_{rel} \approx 48$





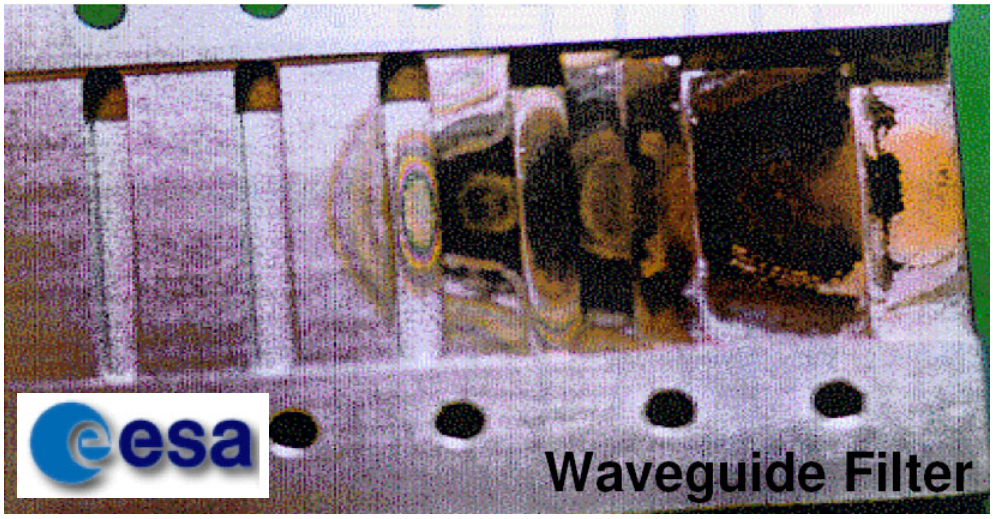
[DLI Labs Inc.]

breakdown in resonators

- filters contain resonators; resonating electromagnetic fields lead to very high field strength particularly in pass band
- some resonators in a filter are more exposed than others; in a series connection of resonators, the first and last resonator often show highest field strength
- narrow-band filters (→ weakly coupled resonators) have lower threshold
- in a resonator, electric field is strongest at edges and tips, to increase breakdown level, small radii convex metal must be avoided (e.g., use rounded-end tuning rods instead of thread-ended tuning screws)
- dielectrics may have a „shielding effect“ and increase the threshold
- breakdown is different whether filters are filled with gas or not
- in gas filled resonators, the breakdown is based on a corona-like effect:
  - heat (e.g., self-heating) of the metal surface leads to electron emission (thus this breakdown depends on both RF power level and time of exposure)
  - accelerated electrons cause more gas heating as well as secondary ionisation

[K. Frigui et al., "Experimental and numerical study of microwave breakdown in a two-channel output multiplexer," Proc. 45th EuMC, 2015, pp. 682-685]

- in vacuum „filled“ resonators, a different „multi-paction breakdown occurs:
  - at high field strengths, electrons will be emitted from metal surface
    → these electrons will accelate and cycle with the RF field
    → fast electrons will hit metal and knock out more electrons
    → runaway effect (a.k.a. breakdown)
  - particularly critical in high vacuum — gases may stop the electrons
  - breakdown occurs almost instantaneously after application of high RF power



temperature compensation

- resonance frequency depends on physical resonator dimensions
  → thermal expansion results in decrease of resonance frequency

$$\alpha_{\text{res}} = -\alpha_{\text{metal}}$$

LINEAR THERMAL EXPANSION COEFFICIENTS OF TYPICAL MATERIALS:

	PLASTICS	ALUMINUM	CERAMIC	KOVAR	SILICON	INVAR
α [ppm/K]	30…100	22.2	5…7	6	3	1.5

- for resonators containing dielectrics, resonance frequency will also vary with temperature-induced change of permittivity
  → increase of permittivity results in a decrease of resonance frequency

$$\alpha_{\text{res}} \sim -\sqrt{\alpha_{\text{permittivity}}}$$

TEMPERATURE COEFFICIENTS OF VARATION OF PERMITTIVITY:

	TEFLON (PTFE)	ALUMINA (AL_2O_3)	RUTILE (TiO_2)	COMMERCIAL ε_{REL} = 35
α [ppm/K]	-500	+100	-900	-10 … +40

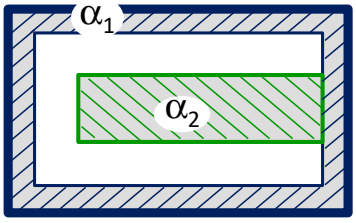
- compensation of temperature influence on resonance frequency

- dielectric resonators: compensated high-eps dielectrics can also account for thermal expansion of metal shielding and thermal variation of in/out-couplings

- metal cavities: homogeneous: Invar.

non-homogeneous: small plastic / rutile part added

non-homogeneous: two parts with different expansion benefit from special field distribution



$\alpha_1 > \alpha_2$

humidity, air pressure, microphony

- air-filled high-Q cavity resonators are sensitive on the permittivity of air, which mainly depends on humidity and pressure:

$$n_{\text{air}} = \sqrt{\varepsilon_{\text{rel,air}}} = 1 + \frac{1}{10^6}\,\frac{77.6}{T}\left(p + 4.81\,\frac{e}{T}\right)$$

T … Temperature (K)

p … air pressure (mbar)
1 mbar = 1 hPa
1013 mbar = 1 atm

e … partial pressure of water vapour (mbar)
water vapour saturation at sea level at 20°C: 23 mbar

example:
dry air @ 1013 mbar @ 20°C: ε_{rel} = 1.000‘54

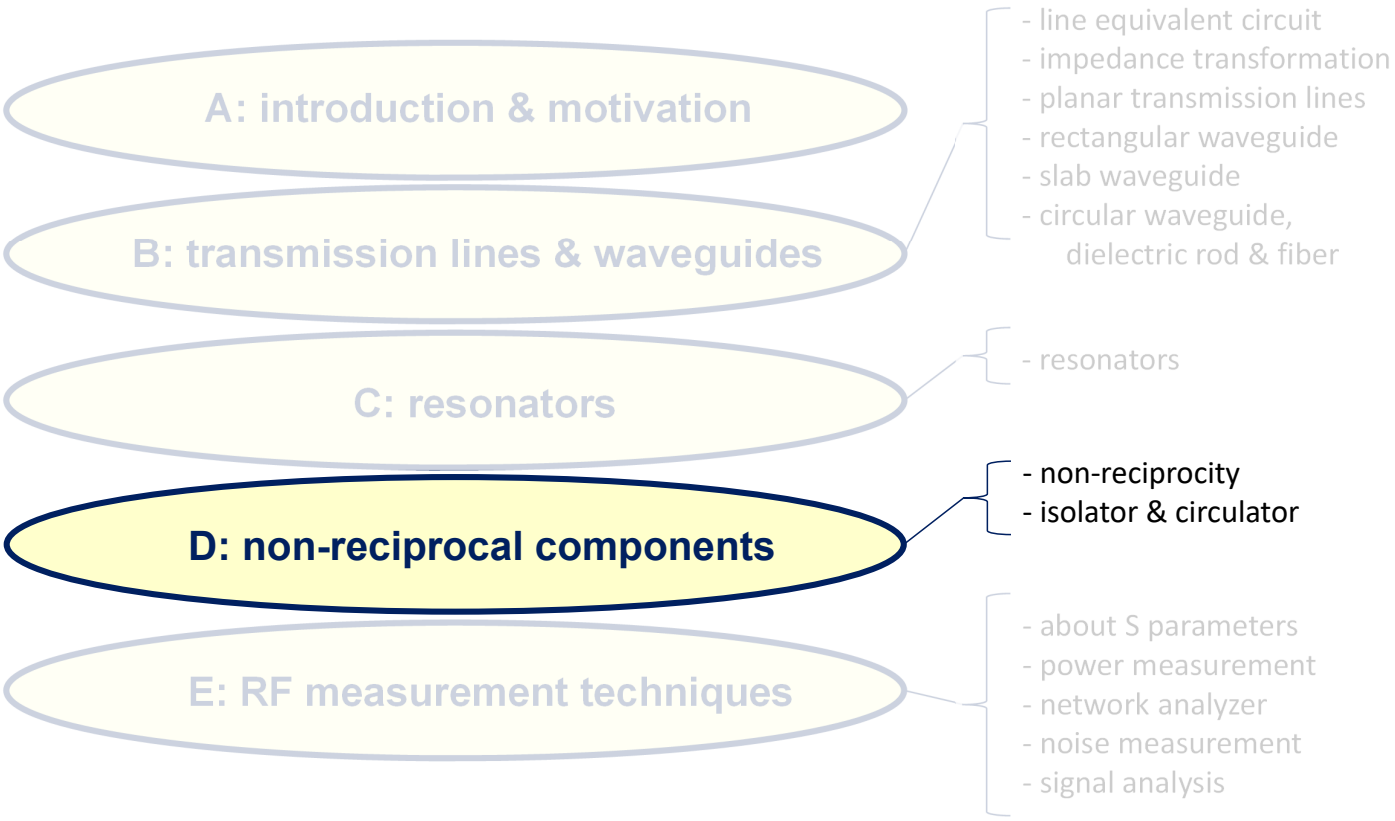
$\varepsilon_{\text{rel}} \approx 81$

- water in liquid form will drastically affect the resonance frequency; drying may be necessary („getters“ such as Zeolite will attract water vapour)

- microphony: in resonator-stabilized, mechanically wobbly, low-noise oscillators, mechanical vibration will appear in the phase-noise spectrum

[E.K. Smith, S. Weintraub, „The constants in the equation for atmospheric refractive index at radio frequencies,“ Proc. IRE, 1953, pp. 1035-7]

Radio Frequency Technology — main topics :

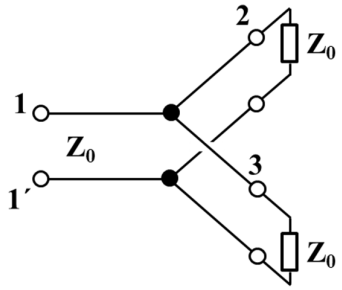


Non-reciprocal microwave components

… so far all passive components were reciprocal: $S_{ij} = S_{ji}$

Examples:

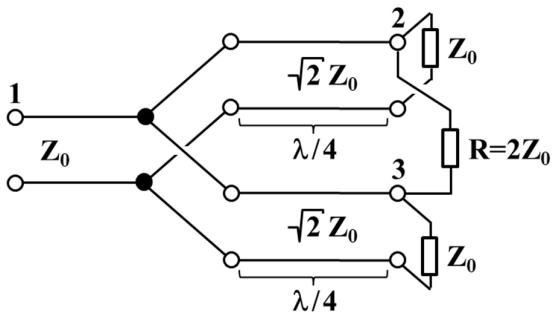
Y-junction divider
loss-less but not matched



$$[S] = \frac{1}{3}\begin{bmatrix} -1 & 2 & 2 \\ 2 & -1 & 2 \\ 2 & 2 & -1 \end{bmatrix}$$

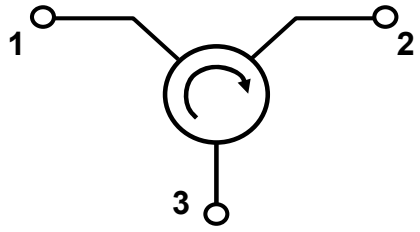
$$1 = |S_{11}|^2 + |S_{21}|^2 + |S_{31}|^2$$

Wilkinson
all ports matched but lossy



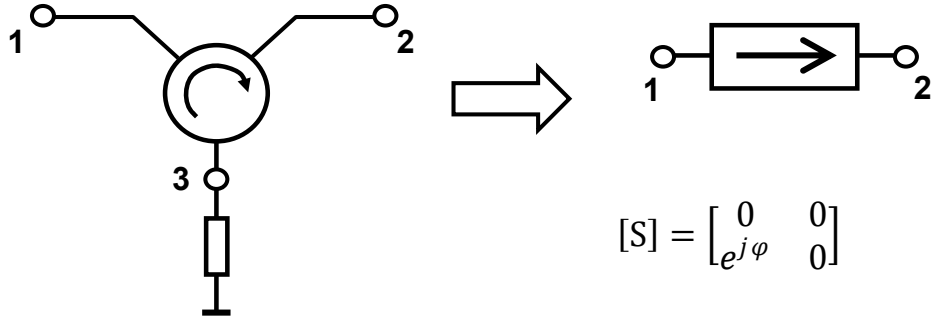
$$[S] = \frac{-j}{\sqrt{2}}\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

New: Circulator: 3-port, matched, loss-less, non-reciprocal: $S_{ij} \neq S_{ji}$



$$[S] = \begin{bmatrix} 0 & 0 & e^{j\varphi} \\ e^{j\varphi} & 0 & 0 \\ 0 & e^{j\varphi} & 0 \end{bmatrix}$$

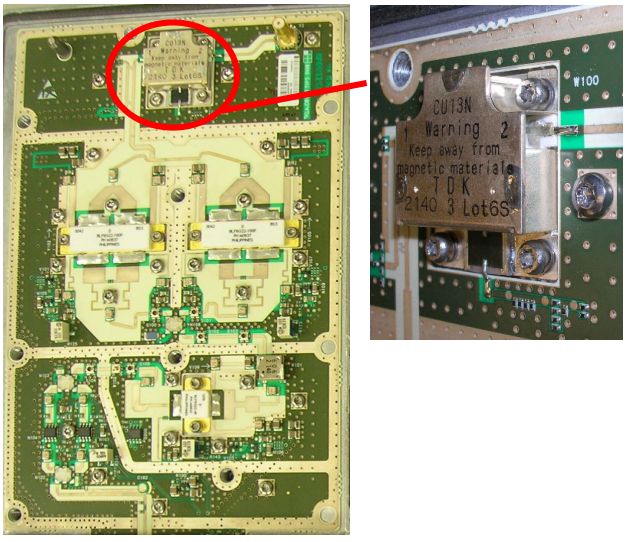
New: Isolator: 2-port, matched, loss-less, non-reciprocal: $S_{12} \neq S_{21}$



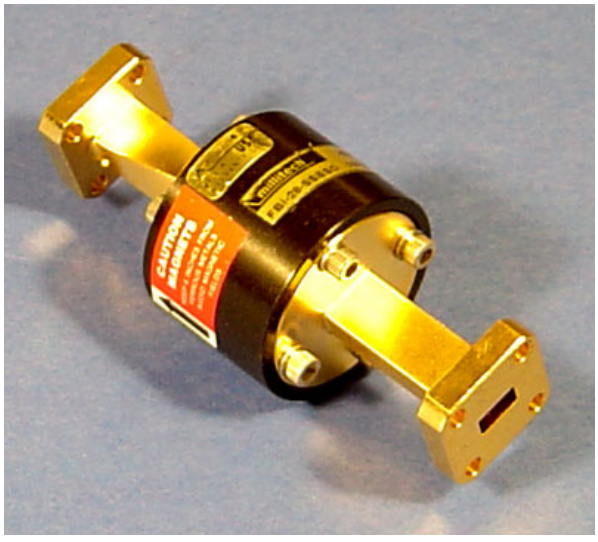
$$[S] = \begin{bmatrix} 0 & 0 \\ e^{j\varphi} & 0 \end{bmatrix}$$

Examples:

Circulator with resistor at third port as protecting device at the output of a base-station power amplifier (SMT, ~ 2 GHz)



Isolator for half-octave bandwidth (based on Faraday effect) (waveguide device, 26.5 GHz – 40 GHz)

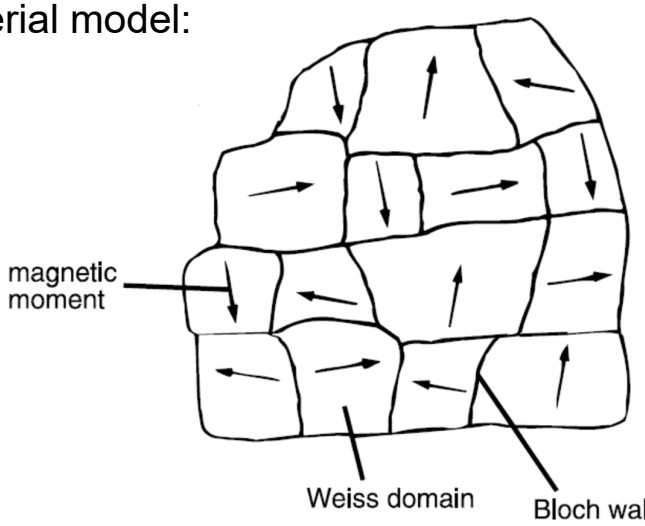


Ferrimagnetic material behaviour and description:

… **ferrimagnetic materials** (aka ferrites) are typically ceramics or crystals

… do not confuse with ferromagnetic materials (some metals) – conductive!

… ferrite material model:



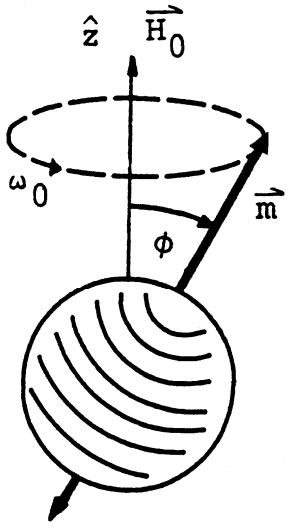
[Philips Application Note AN98035, 1998]

… where the magnetic moments originate from the spinning electrons

… the many Weiss domains are oriented randomly

… application of an external (static) magnetic field of increasing strength will increasingly align all the magnetic moments

… the aligning magnetic moment will perform a precession, until everything is aligned at large external field and the material is said to be saturated



… the mechanical equivalent of the precession is easily demonstrated :

[MIT]

… spinning wheel with originally horizontal axle is hanging asymmetrically. The vertical gravitation force causes the axle end to precess.

… note that most magnetic moment stems from electron spin, and not from orbiting electrons. Overall spin is most often zero, because spins are paired, but in magnetic materials, there are unpaired spins

… magnetic fields generated by permanent magnets will bring ferrites into saturation easily

… the angular frequency of the precession, called **Larmor frequency**, is proportional to the magnetic field bias strength

$$\omega_0 = \mu_0 \, \Gamma H_0 \qquad , \qquad f_0 = \frac{\omega_0}{2\pi} = \frac{\mu_0}{2\pi} \Gamma H_0$$

where the **gyromagnetic ratio** is $\Gamma = \frac{q_e}{m_e} \approx 1.76 \cdot 10^{11} \, \frac{\mathrm{C}}{\mathrm{kg}} = 1.76 \cdot 10^{11} \, \frac{\mathrm{m}^2}{\mathrm{Vs}^2}$

with electron charge q_e, electron mass m_e, $f_0 = \frac{\omega_0}{2\pi} = 35.2\ \mathrm{kHz} \cdot \frac{H_0}{\mathrm{A/m}}$

… with magnetic fields generated by coils, field strengths of up to 1 MA/m will tune the Larmor frequency in the range up to 35 GHz

… a small RF field applied to a strongly biased ferrite will undergo anisotropic variation:

- the component of the RF magnetic field in direction of the bias will enhance / reduce that strong bias by a negligible amount,
- the component of the RF magnetic field in the plane perpendicular to the magnetic bias will cause some change of the precession of the magnetc moments of the material. That is, the component perpendicular to both the bias and the RF field will be changed, too!

example: H_{bias} in z-direction, H_{RF} in x-direction will cause some B in y-direction!

… this is described the permeability tensor. Assuming saturation and H_{bias} in z-direction, the permeability tensor takes the form:

$$\begin{pmatrix} B_x \\ B_y \\ B_z \end{pmatrix} = \begin{bmatrix} \mu_1 & j\mu_2 & 0 \\ -j\mu_2 & \mu_1 & 0 \\ 0 & 0 & \mu_0 \end{bmatrix} \begin{pmatrix} H_x \\ H_y \\ H_z \end{pmatrix}$$

… the parameters μ_1 and μ_2 are both dependent on RF frequency and on magnetic bias field strength. Both are complex numbers to account for loss.

… a material with $\mu_{r,jk} = -\mu_{r,kj}$ or $\varepsilon_{r,jk} = -\varepsilon_{r,kj}$ is called **gyrotropic**.

Besides ferrites and some crystals, also the ionosphere and other magnetized plasmas show this behaviour.

… a comment about usage of units:

in magnetics, the CGS system of units is used often (instead of SI):

magnetic field strength (H): 1 A/m = 4π x 10^{-3} Oe [Oersted]

magnetic flux density (B): 1 T = 1 Wb/m^2 = 10^4 G [Gauss]

Larmor frequency: $f_0 = \frac{\omega_0}{2\pi} = 2.8\ \mathrm{MHz} \cdot \frac{H_0}{\mathrm{Oe}}$

absolute permeability: $\mu_0 = \frac{1\ \mathrm{G}}{1\ \mathrm{Oe}} = 4\pi \cdot 10^{-7}\ \frac{\mathrm{Vs}}{\mathrm{Am}}$

i.e., in CGS system, B and H have the same numerical value in a non-magnetic medium

Faraday effect:

… consider the propagation of a plane wave in infinite ferrite medium: assume magnetic bias in z-direction and also the RF wave propagating in z-direction:

the solution of Maxwell's equation are LHCP and RHCP waves with different propagation constants:

$$\overrightarrow{H_\pm} \sim \left(\overrightarrow{e_x} \mp j\overrightarrow{e_y}\right) e^{j\omega t - \gamma_\pm z}$$

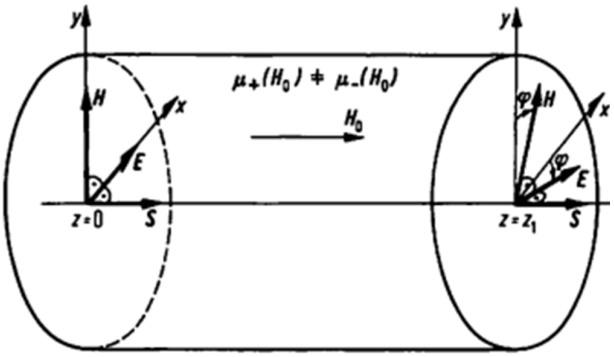
$$\gamma_\pm = \alpha_\pm + j\beta_\pm = j\omega\sqrt{\varepsilon\,\mu_\pm}$$

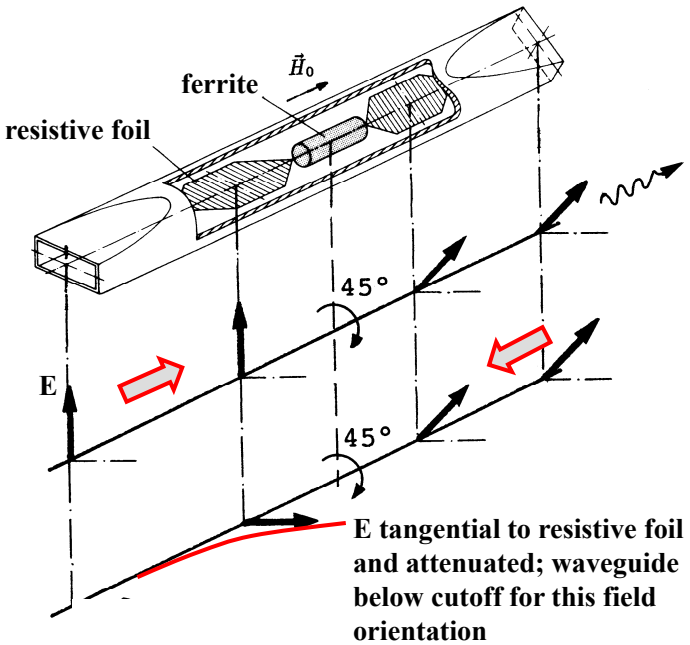
$$\mu_\pm = \mu_0\left(\mu_\pm^{'} - j\mu_\pm^{''}\right) = \mu_1 \pm \mu_2$$

… note that any linear-polarized wave can be seen as a combination of a LHCP wave and a RHCP wave. Since $\mu_+ \neq \mu_-$ if $\mu_2 \neq 0$ (different propagation constants of the two waves), the direction of polarization of the linear-polarized wave rotates along z! This is called Faraday-effect.

… note that magnitude and the direction of rotation depend on μ_+ and μ_- and thus on frequency and on the strength of the magnetic bias

… illustration of the Faraday effect for a linear-polarized wave:





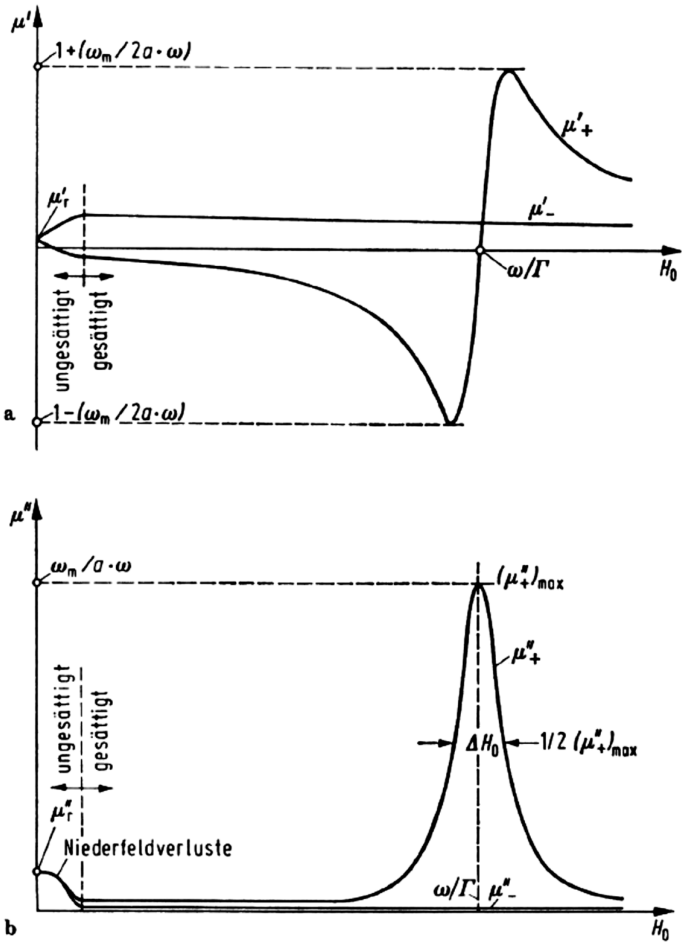
… concept of an isolator (uniline) based on the Faraday effect:
(a 3-port or 4-port device can be realized adding orthomode transducers)

Gyromagnetic resonance and loss considerations:

… below saturation, ferrimagnetic materials are lossy and do not much interact with the RF fields

… above saturation, μ_-' is constant and rather small, while μ_-'' is very small

… around saturation, μ_+' and μ_+'' vary strongly: μ_+' is negative, then zero at the gyromagnetic resonance, peaking at large value and then decaying but still much larger than μ_-' . μ_+'' is always larger than μ_-'' and peaks at the gyromagnetic resonance.

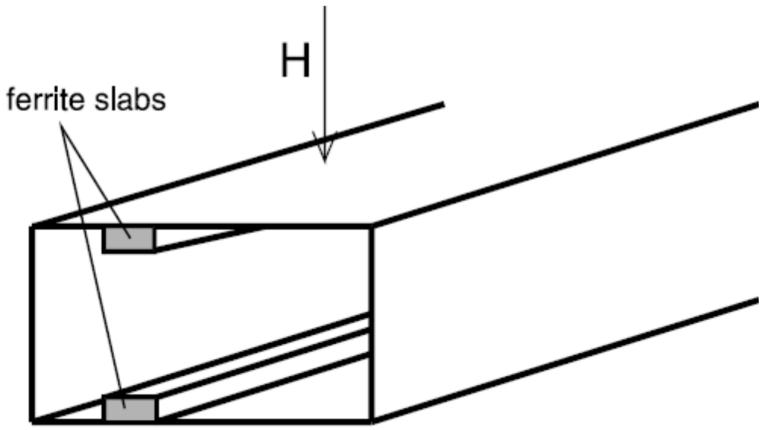


[Zinke, Brunswig, Springer]

Resonance isolator:

… in a rectangular waveguide partly filled with biased (saturated) ferrimagnetic material, a TE10-like mode will experience **large/small** attenuation if its magnetic fields (at the location of the ferrite) form circular movement in the plane perpendicular to the magnetic bias with sense of rotation **similar/against** the rotation of the magnetic moment. This can create an isolator.

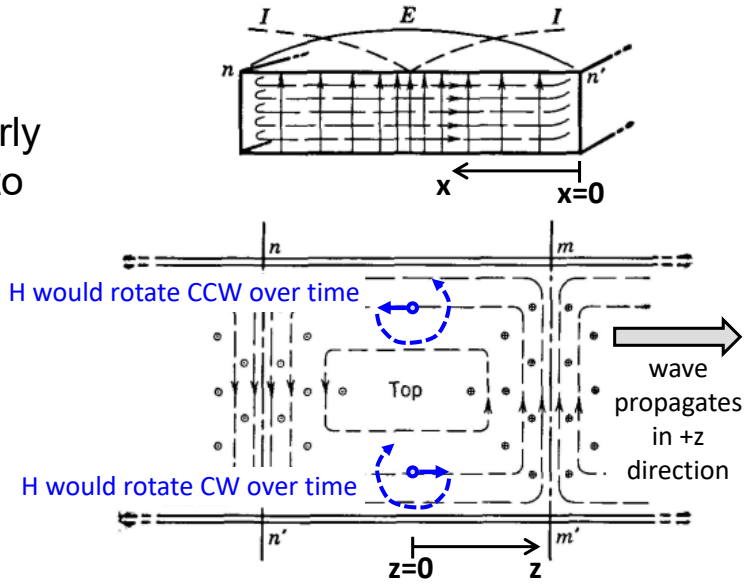
… The device must operate close to the frequency of gyromagnetic resonance in order for the attenuation becoming large. Thus, the device is called resonance isolator with rather small frequency bandwidth (but good for high power as heat removal is effective).



[Pozar, Microwave Engineering 2nd edition, Wiley 1998, chapter 9.6]

Resonance isolator:

… the slab(s) with biased ferrite to be located at position of circularly varying magnetic RF field and to be biased at the gyromagnetic resonance for maximum attenuation



… rectangular waveguide TE_{10}:

$$\beta = \sqrt{k_0^2 - \left(\frac{\pi}{a}\right)^2}$$

$$H_x(x,y,z,t)|_{\mathrm{TE10}} = H_{z0}\,\frac{a\beta}{\pi}\,e^{j\omega t}\,e^{-j\beta z + j\frac{\pi}{2}}\sin\frac{\pi x}{a}$$

$$H_z(x,y,z,t)|_{\mathrm{TE10}} = H_{z0}\,e^{j\omega t}\,e^{-j\beta z}\cos\frac{\pi x}{a}$$

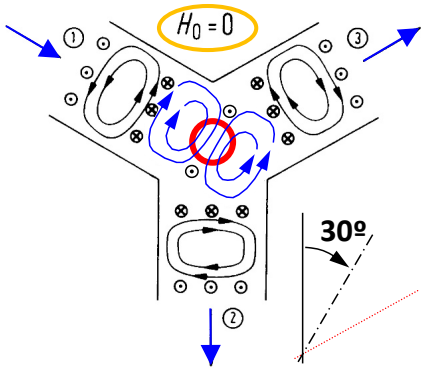
$$\frac{H_x}{H_z} = \frac{a\beta}{\pi}\,j\tan\frac{\pi x}{a}$$

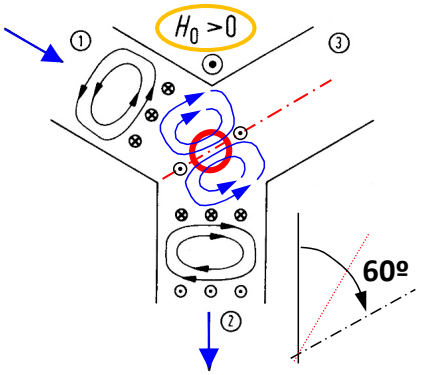
… for slab position x follows: $\frac{H_x}{H_z} = j \;\rightarrow\; \frac{\pi}{a\beta} = \tan\frac{\pi x}{a}$

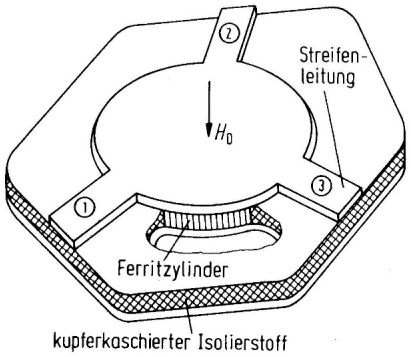
Junction circulator:

… a ferrite disk behaves like a TM_{110} resonator (bottom and top PEC, circumference PMC). In magnetically biased state, two different but strongly coupled resonances. Can be arranged such that one of the „transmit“ ports is decoupled

Operation principle in TE10 rectangular waveguide (top view), but stripline / microstrip is similar:







… a derivation based on Maxwell‘s equations for resonator modes found in:
Pozar, Microwave Engineering 2nd edition, Wiley 1998, chapter 9.6

Junction circulator:

… feed / impedance transformation sections make device narrowband

absorber / matched load at third port

big permanent magnet on top removed for visibility

Birefrigence:

… consider the propagation of a plane wave in infinite ferrite medium:
assume magnetic bias in z-direction and
the RF wave propagating in x-direction:

case where H_{RF} is in z-direction: $\vec{H} \sim \overrightarrow{e_z}\, e^{j\omega t - \gamma_\parallel x}$

$$\gamma_\parallel = j\omega \sqrt{\varepsilon\, \mu_0}$$

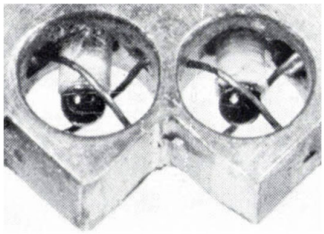
case where H_{RF} is in y-direction: $\vec{H} \sim \overrightarrow{e_y}\, e^{j\omega t - \gamma_\perp x}$

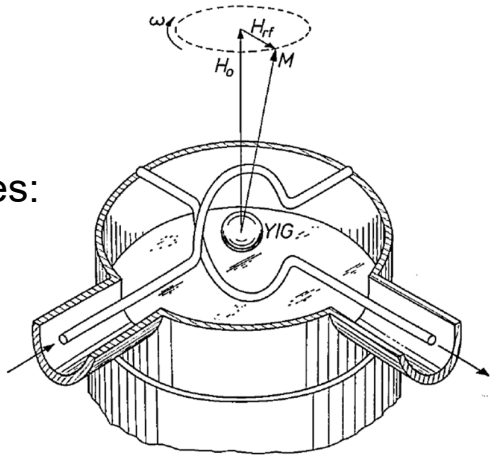
$$\gamma_\perp = j\omega \sqrt{\varepsilon\, \frac{\mu_1^2 - \mu_2^2}{\mu_1}}$$

… therefore, the permeability and thus the propagation constant of the linear-polarized wave depends on the polarization direction

YIG sphere resonator:

… a small sphere of ferrimagnetic material
is magnetically biased and perpendicularly
coupled to input+output RF transmission lines:





… design frequency
0.5 … ~ 40 GHz;
tuning range up to 10:1

… since input coupling loop and output coupling loop are orthogonal, there will be transmission only at the frequency of gyromagnetic resonance (which can be tuned by variation of the magnetic bias field strength)

… note that the sphere is much smaller than the wavelength, thus it does not operate in dielectric resonator resonance mode (ε_{rel} = 16 for YIG)

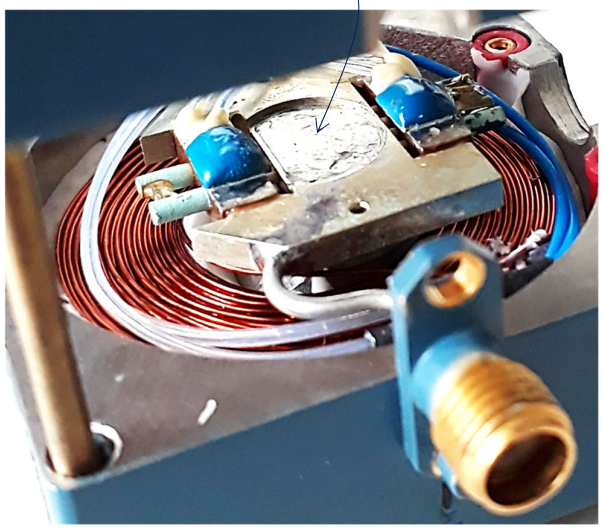
… high-Q tuneable microwave resonators are used for tuneable oscillators and tuneable bandpass filters, among others

[P. Röschmann, „YIG filters," Philips tech. Rev., vol. 32, pp. 322-327, 1971]

YIG sphere tuneable filter example:

… tuneable bandpass 3…26.5 GHz

six YIG spheres with couplings and shielding within here



Practical considerations:

… ferrites are metal oxide compounds including Fe and some other metal

… as sintered ceramics, ferrites show saturation magnetization H_{sat} of 300…500 kA/m, which is readily achieved by using permanent magnets

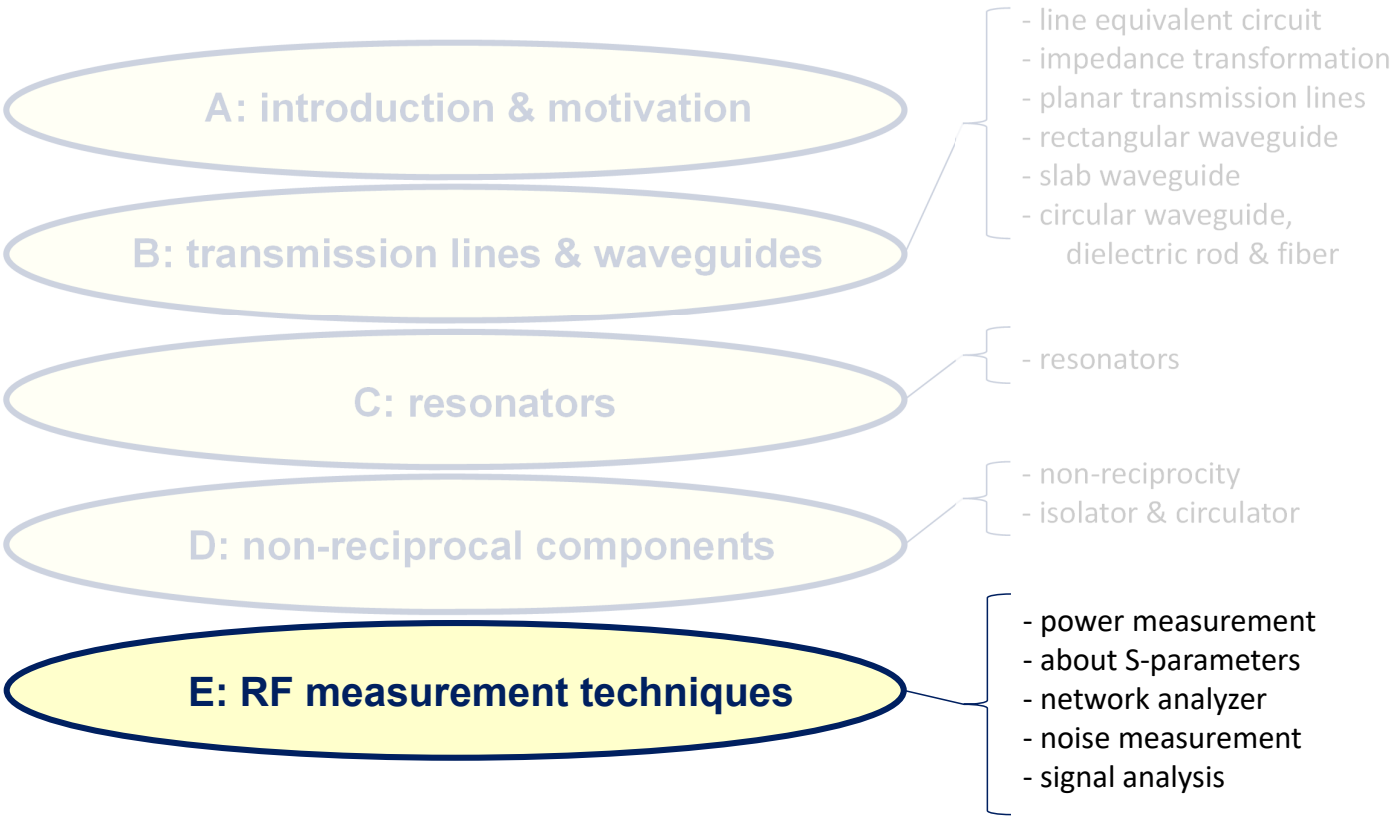
… their permittivity ε_{rel} is in the range of 9…18 and they have some tanδ

… yttrium iron garnet (YIG) is ferrimagnetic but shows another crystal structure. When sintered (and saturated), YIG-resonators show Q-factors of 100…300, while YIG single crystal have Q-factors of > 2000

… YIG and other garnets have lower saturation magnetization than ferrites

… YIGs are operated at the Larmor frequency, which requires, e.g., 140 kA/m magnetic bias for resonance at 5 GHz, but 1.1 MA/m magnetic bias for resonance at 40 GHz. The latter is difficult to achieve and is thus limiting the upper frequency of operation of tuneable YIG devices

Radio Frequency Technology — main topics :



RF measurement techniques — Outline

In the RF / microwave domain, the accurate measurement of voltage, current, electric field strength, magnetic field strength etc. is difficult if not impossible. Accurate measurement of time and frequency is possible. Measurement of power is possible with reasonable accuracy. As a result, power meters, frequency generators and spectrum analyzers are important measurement tools. Combining power measurements with the definition of reference impedance of a transmission line, S-parameters can be defined and used to characterize N-ports. The most important tool to measure S-parameters, that is, to measure power of a signal on a line of a specified impedance, is the network analyzer.

Power measurement

About S parameters

Network analyzer

Noise and noise measurement

Signal analysis

Power measurement

power being a physical quantity detectable „from DC to daylight“

simplest detection principle is „absorption + conversion to heat“

- calorimeters / bolometers determine absorbed power, e.g., by measuring the temperature increase of a water bath heated by the absorbing resistance — high power levels needed, slow and very expensive, preferably used in metrology labs

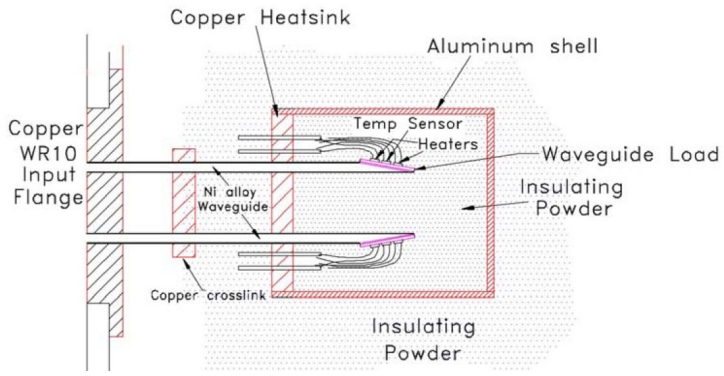
- thermocouple or thermistor measurements: induction of a voltage due to temperature difference (Seebeck effect) or variation of DC resistance due to temperature variation; sensed in a bridge circuit (that is, comparison to reference temperature) — reasonably accurate, not too fast, used for commercial equipment („averaging“)

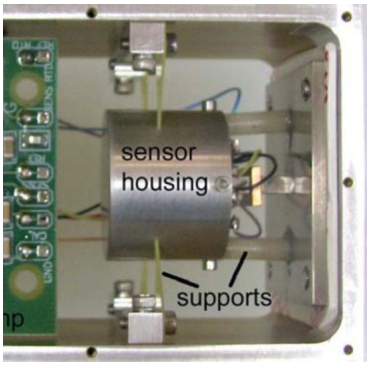
- mixing to DC by non-linear circuits, eg, diodes: fast and very sensitive, but depending on the waveform of the RF signal

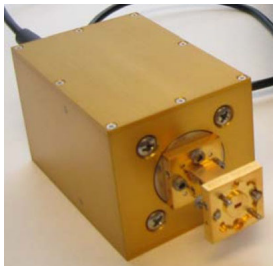
calorimeter / bolometer (ie, devices which heat up through absorption of energy)

measuring the rise in temperature of an absorber due to RF power, comparing with the (DC-) power needed for the same temperature rise

practical problems are
- repeatability of the absorber (in a dual-absorber arrangement),
- thermal insulation (including the connecting RF line, which may be a rectangular waveguide),
- response time (which can be several minutes for metrology lab devices),
- detection of low power (typically at least a few mW needed)



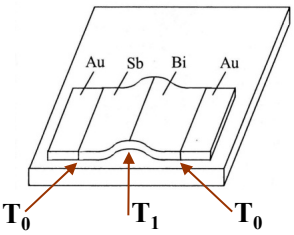




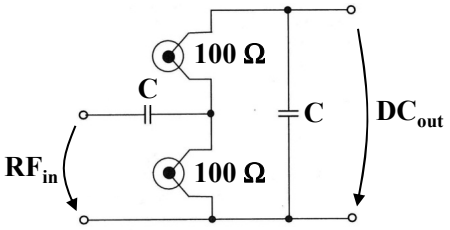
[Erickson PM4, Virgina Diodes Inc.]

thermocouples

Junction between two metals (or metal-to-semiconductor), generates a voltage due to temperature difference at the junction (Seebeck effect)



… old; new designs use semiconductor + thinfilm structures

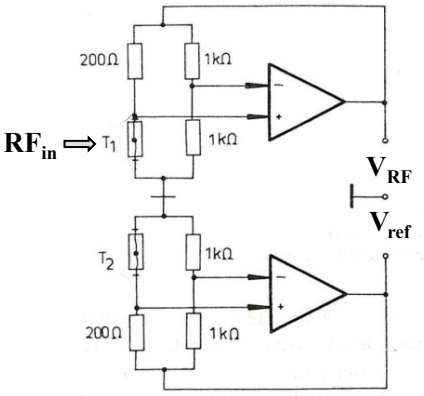


[B. Schiek, Grundlagen der Hochfrequenz-Messtechnik, Springer 1999]

thermistor bridge

An oxide ceramic resistor with negative temperature coefficient. In a bridge circuit, DC current heats up the thermistor until bridge is balanced. Measured RF power will substitute DC power which is then measured. A second, identical bridge without RF used to eliminate changes in environmental temperature.

sensitivity: down to about -35dBm



diode power detector (1/5)

if the RF signal is applied to the exponential i(v) characteristic of a diode, a DC mixing signal (amongst many other mixing products) is generated.

$$I = I_0 \left(\exp\left(\frac{U}{U_{th}}\right) - 1 \right)$$

here, U_{th} is inversely proportional to temperature, but also I_0 depends on temperature in a rather complex way.

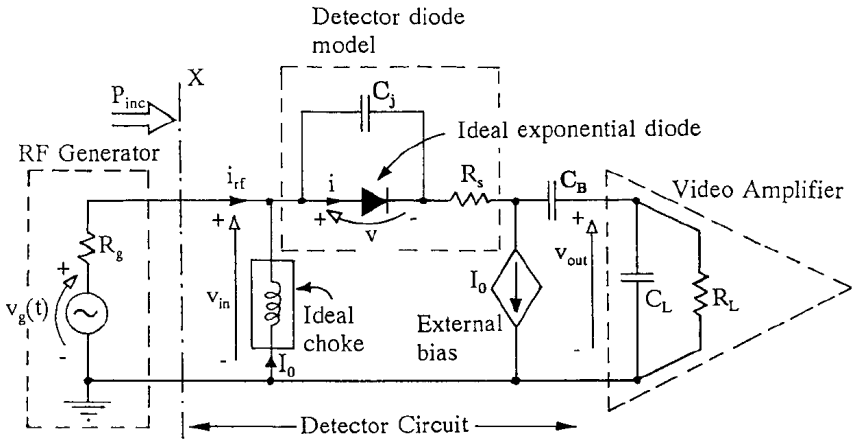
[Agilent]

diode power detectors can detect very small signals, e.g., up to -70 dBm (100pW)

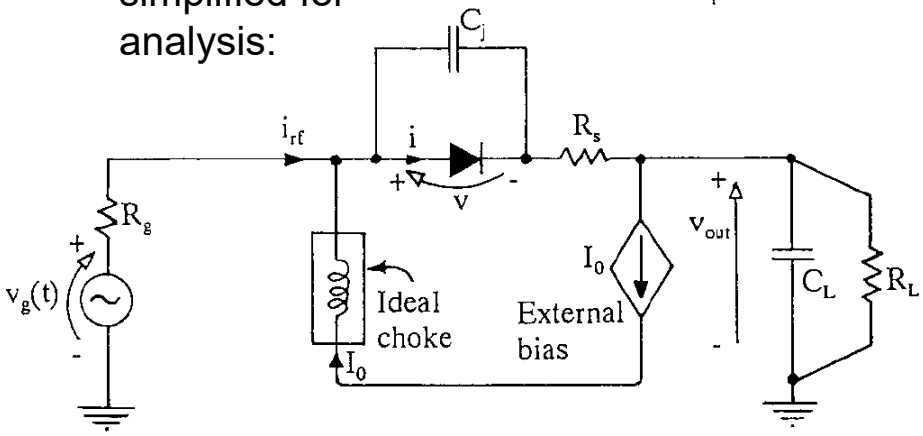
Note: since a diode detector is in fact a diode mixer, both diode detectors and diode mixers exhibit strong temperature dependence of their performance.

diode power detector (2/5)

a simple diode detector circuit:



can be further simplified for analysis:



diode power detector (3/5)

[R.G. Harrison, X. Le Polozec, „Nonsquarelaw behavior of diode detectors analyzed by the Ritz-Galerkin method," IEEE Trans. Microw. Theory Techn., vol. 42, no. 5, May 1994, pp. 840-846]

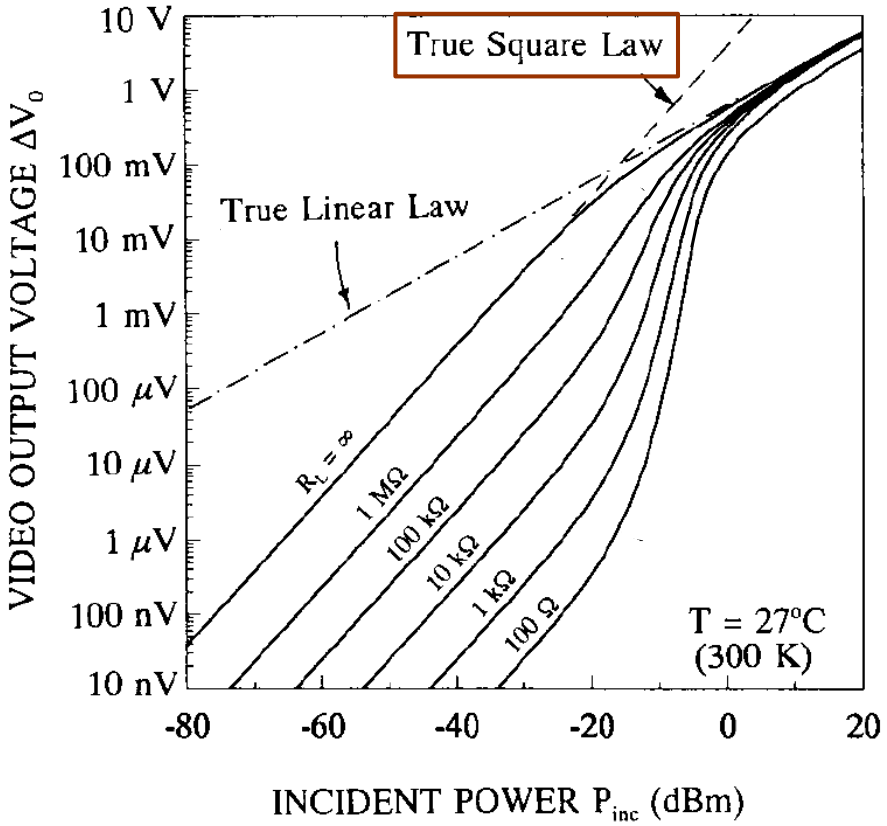
for a large RF input signal, the detector works like a well-known rectifier:

$$v_{\text{out,DC}} \sim v_{\text{inc,RF}} \sim \sqrt{P_{\text{inc,RF}}}$$

for a small RF input signal, the detector works in the so-called **square-law regime**:

$$v_{\text{out,DC}} \sim v_{\text{inc,RF}}^2 \sim P_{\text{inc,RF}}$$

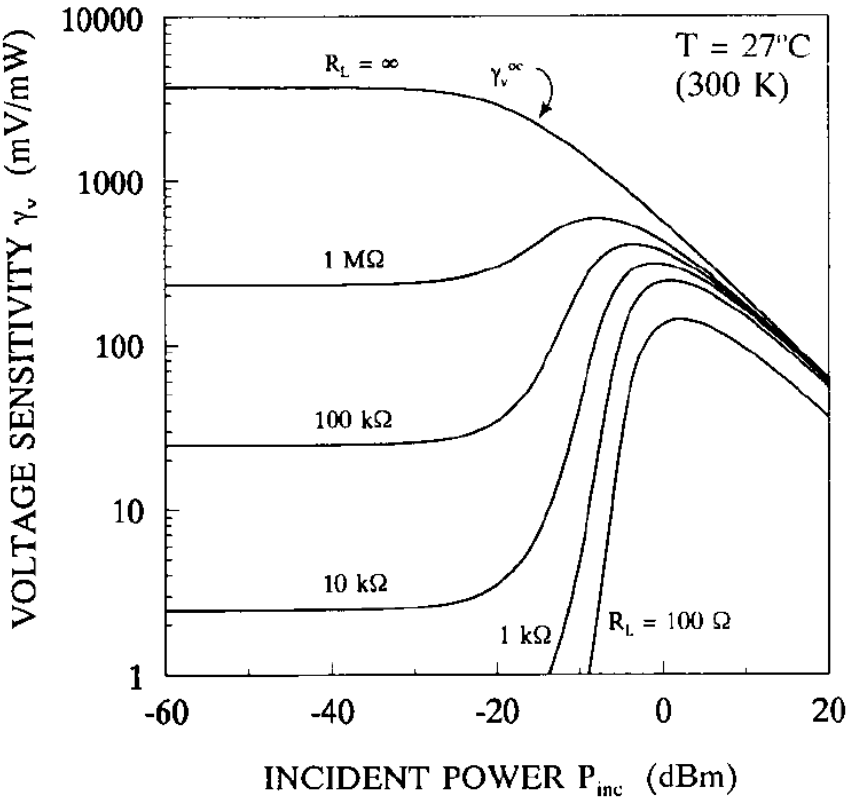
the load resistance is of large importance:



diode power detector (4/5)

sensitivity depends largely on load resistance:

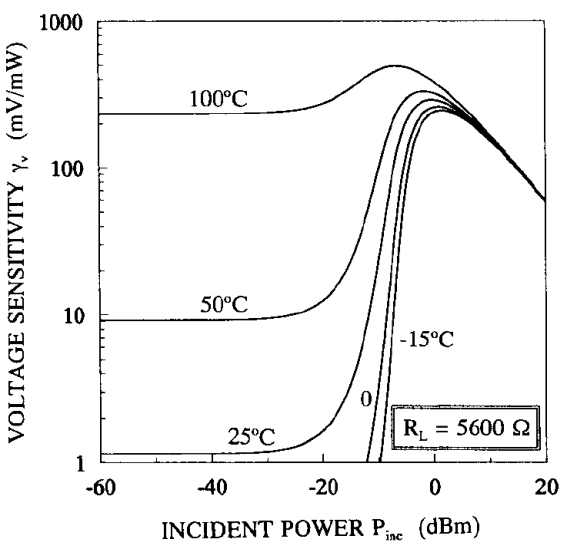
[R.G. Harrison, X. Le Polozec, „Nonsquarelaw behavior of diode detectors analyzed by the Ritz-Galerkin method," IEEE Trans. Microw. Theory Techn., vol. 42, no. 5, May 1994, pp. 840-846]

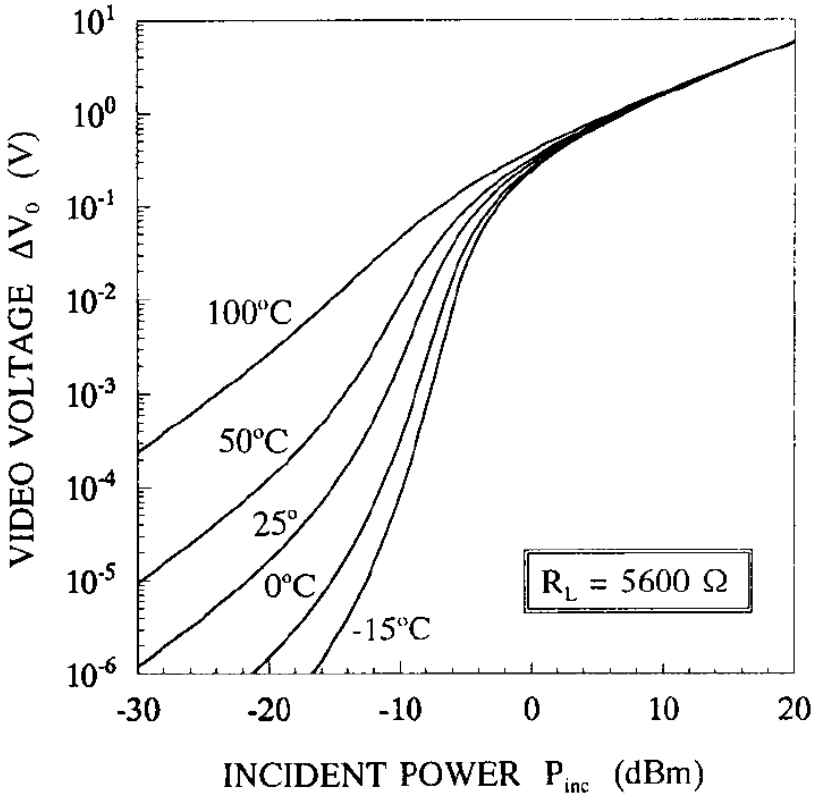


diode power detector (5/5)

temperature dependence is huge and must be accounted for (e.g., using a look-up table)

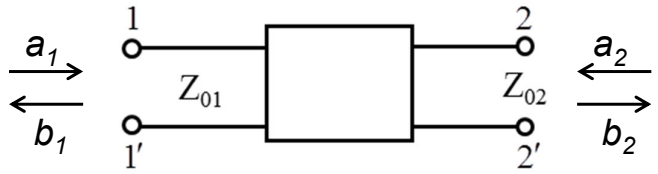
[R.G. Harrison, X. Le Polozec, „Nonsquarelaw behavior of diode detectors analyzed by the Ritz-Galerkin method," IEEE Trans. Microw. Theory Techn., vol. 42, no. 5, May 1994, pp. 840-846]





About S parameters

in the RF domain, n-ports are typically characterized by the **$[S]_{nn}$ matrix**



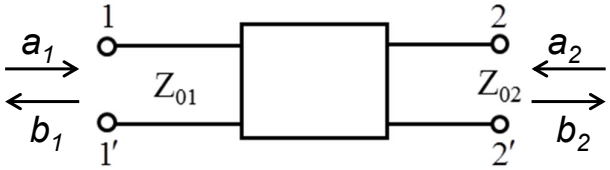
$$a_i = \frac{U_{hi}}{\sqrt{Z_{0i}}} = I_{hi}\sqrt{Z_{0i}} \qquad b_i = \frac{U_{ri}}{\sqrt{Z_{0i}}} = I_{ri}\sqrt{Z_{0i}} \qquad [\sqrt{\text{W}}]$$

$$u_{i,normiert} = \frac{U_{hi} + U_{ri}}{\sqrt{Z_{0i}}} = a_i + b_i \qquad [\sqrt{\text{W}}]$$

$$i_{i,normiert} = (I_{hi} - I_{ri})\sqrt{Z_{0i}} = a_i - b_i \qquad [\sqrt{\text{W}}]$$

$$\begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = [\text{S}] \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$$

in special cases, the **transmission or chain matrix [T]** is used

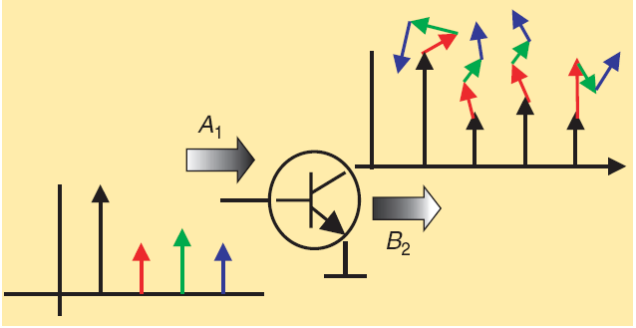


$$\begin{pmatrix} b_1 \\ a_1 \end{pmatrix} = [\text{T}] \begin{pmatrix} a_2 \\ b_2 \end{pmatrix}$$

normally, circuits described by S parameters are considered linear.

for non-linear circuits,

- the operation point / power can be definied for a linearized operation. Then, a specific S of parameters can be defined.
- so-called **X-parameters** were introduced considering nonlinearity in the frequency domain and are supported now by measurement equipment and simulation tools



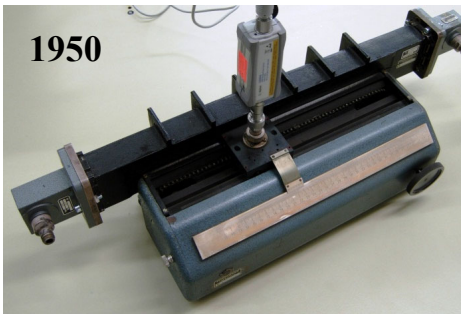
[J. Verspecht, D.E. Root, "Polyharmonic distortion modeling," IEEE Microwave Magazine, vol. 7, no. 3, June 2006, pp. 44-57]

Network Analyzer

S-parameters are the most powerful way to describe the functionality of (linear) RF N-ports. S-parameters can be derived from reflection and transmission measurements. Such measurements are done with **network analyzers**.

S-parameters are referring to a **reference impedance**. Network analyzers are **calibrated** prior to the regular measurements in order to define a **reference impedance** level and a **phase reference** plane.

For **vector network analyzers**, calibration can also **correct for imperfections** (ie, those leading to systematic measurement errors) in the hardware of the network analyzer.



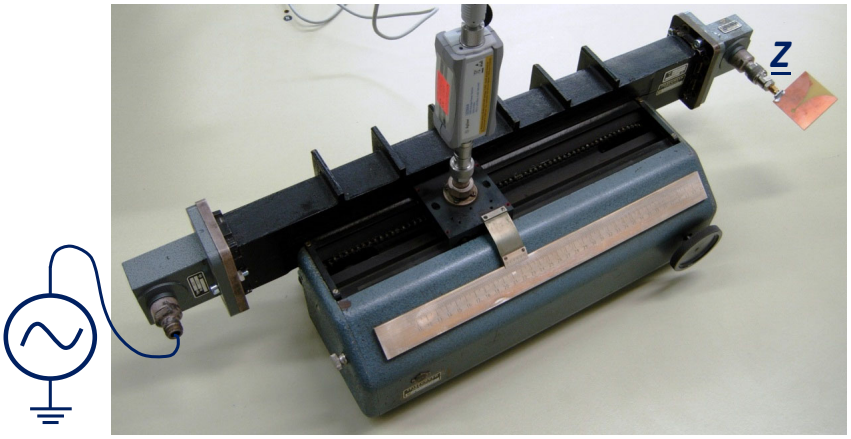
„slotted line“ measurement of reflection coefficient

- a movable, tiny probe senses the standing wave along a transmission line
- the detected voltage is proportional to the sensed power and therefore to the square of the field/voltage magnitude of the standing wave (diode detector)
- the magnitude of the reflection coefficient can be calculated from the minimum and the maximum of the standing wave
- knowing the location of a standing wave maximum with shorted load, and measuring the location of the standing wave maximum of the unknown load, the phase of the reflection coefficient can be calculated

$$v_{\mathrm{det,max}} \cong |v_+|^2 \, (1 + |\Gamma|)^2$$

$$v_{\mathrm{det,min}} \cong |v_+|^2 \, (1 - |\Gamma|)^2$$

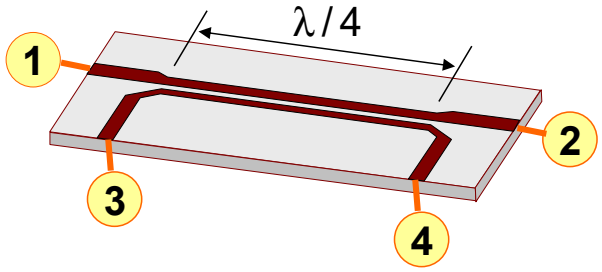
$$|\Gamma| = \frac{1 - \sqrt{v_{\mathrm{det,min}} / v_{\mathrm{det,max}}}}{1 + \sqrt{v_{\mathrm{det,min}} / v_{\mathrm{det,max}}}}$$



measurement of reflection / transmission using directional couplers

a directional coupler can measure the |Γ| of a load:

$$\begin{pmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{pmatrix} = \begin{bmatrix} 0 & T & C & 0 \\ T & 0 & 0 & C \\ C & 0 & 0 & T \\ 0 & C & T & 0 \end{bmatrix} \cdot \begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{pmatrix}$$



… matched ports, $S_{ii} = 0$

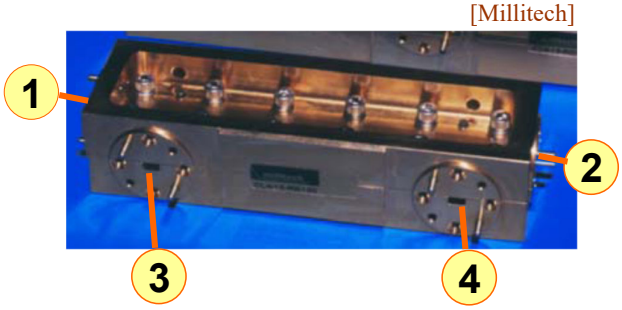
… isolation, $S_{41} = 0$, $S_{32} = 0$

… transmission „T“

… coupling „C“

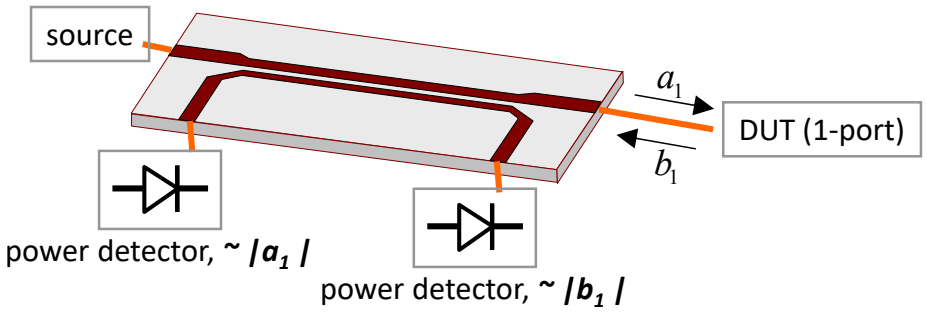
→ directivity: „coupling / isolation“

[Millitech]

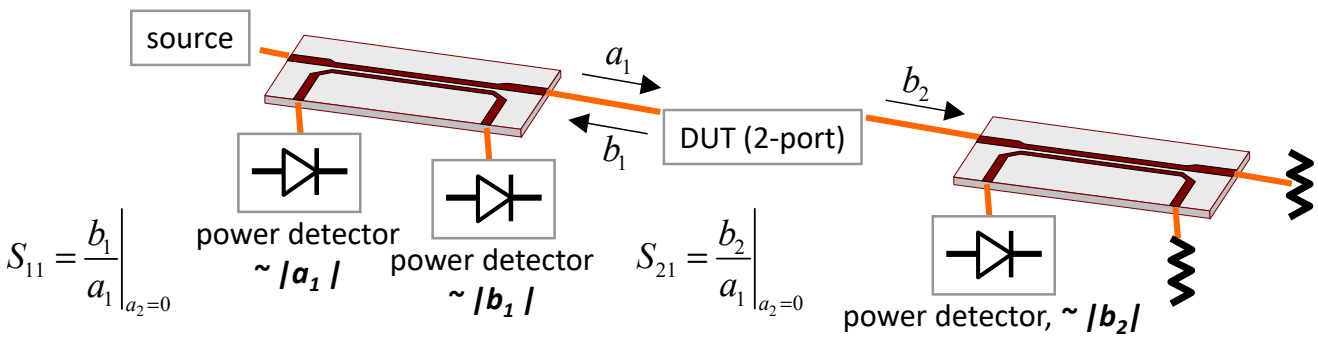


measurement of |Γ| resp $|S_{11}|$ with a directional coupler and two power detectors:

problem for small reflections: limited coupler directivity / non-perfect isolation



measurement of $|S_{11}|$ and $|S_{21}|$ with two directional couplers:



measurement using a receiver instead of using a power detectors:

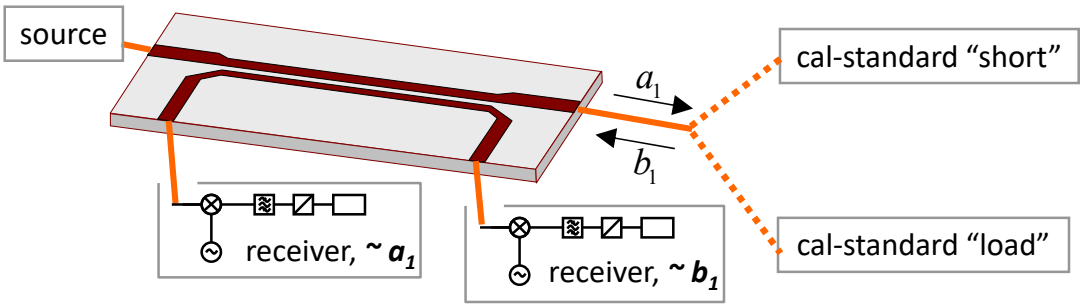
IF a receiver is used to measure magnitude AND phase,

THEN the measurement of a „short“ calibration standard can determine a phase reference (in this case, 180° because of $\Gamma_{short} = -1$)

THEN the complex Γ of the DUT (resp. S_{11}) can be determined

THEN the measurement of a „matched load“ calibration standard determines a complex signal due to limited isolation of the coupler (and due to limited receiver a_1 match), which later can be subtracted (in a vectorial way) from the (vectorial) measurement of the DUT, therefore correcting for finite directivity of the directional coupler

This is calibration !



a typical vector network analyzer:

measures S-parameters of 2-port devices

has a directional coupler „behind“ each port connector

has a receiver „behind“ each port connector & coupler, sensing input to this port connector

has a source with a reference receiver, both together can be switched to either port (therefore allowing to measure S_{21} and S_{12} separately)

a typical calibration:

defines a reference phase plane and a reference impedance level

for 1-port device: „matched load“ defines the reference impedance, and „short“ or „open“ define the reference phase plane. In order to avoid ambiguities, both „open“ and „short“ are required

for 2-port device: „matched load“, „open“ and „short“ are applied to each port separately. Then, both ports are connected together („thru“) which defines a common phase reference plane.

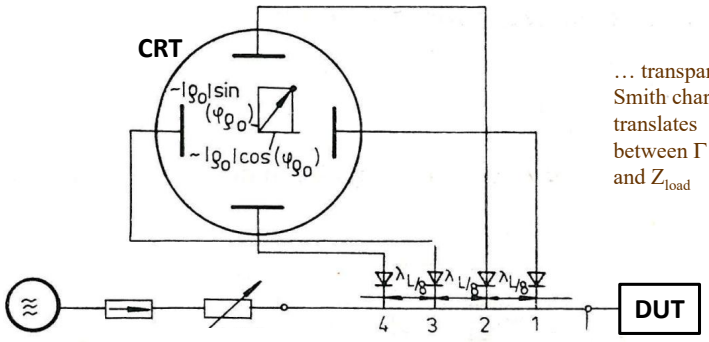
→ there are **many other calibration schemes** for 2-ports

Interferometer

the measurement of calibration standards in magnitude AND phase allows for the vectorial correction of hardware imperfections

a power detector measures magnitude only, whereas a receiver can measure magnitude and phase. a IQ-reciever is in fact based on a 90° hybrid and two amplitude detectors (including negative amplitudes) (plus down-conversion, ADC, etc…).

4 power detectors separated by 90° in the complex phasor plane can measure a complex signal. However, three detectors separated by 120° will do the same.



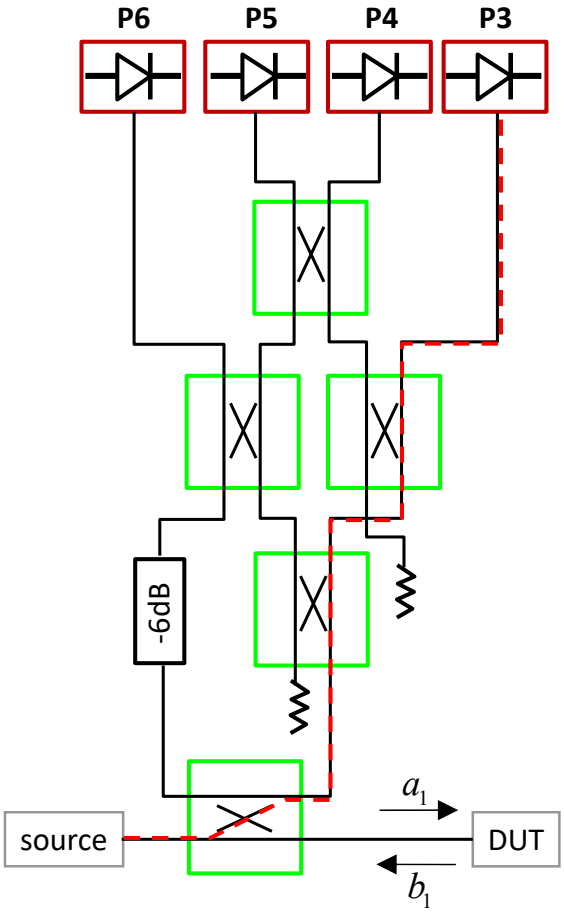
Rohde & Schwarz „Zg-Diagraph“, 300-2400 MHz, 1950-1977:

… transparent Smith chart translates between Γ and Z_{load}

Six-port interferometer

3 power detectors sensing a signal in an interferometric setup, which realizes that the three detector‘s phasors are not collinear (aligned) in the complex plane, are able to measure magnitude and phase of that signal, provided that the respective phase differences between the detectors are known (e.g., from previous calibration).

in a 6-port structure, a fourth power detector senses the source power only



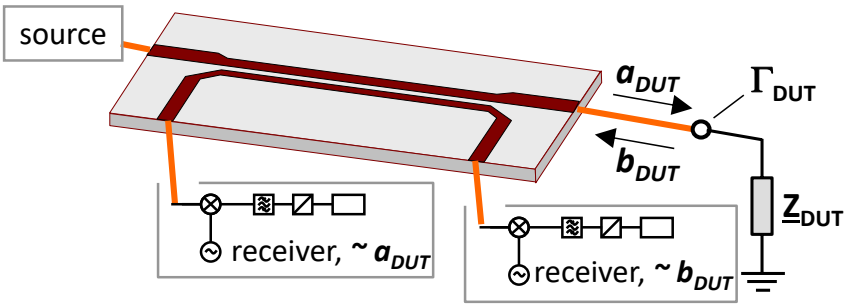
Example circuit of a six-port interferometer

power detector P3 receives a signal proportional to $|a_1|$, but -9 dB smaller, and nothing from b_1

detectors P4, P5, P6 receive each a superposition of signals proportional to both $|a_1|$ and $|b_1|$

VNA error correction

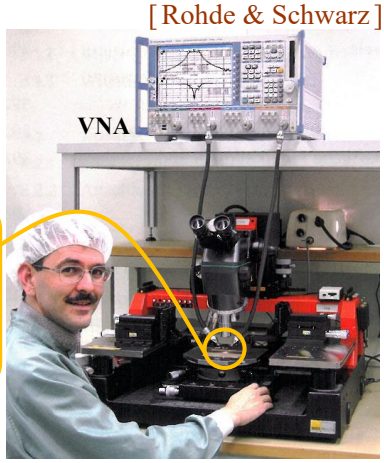
consider the measurement of the complex reflection coefficient of a DUT:

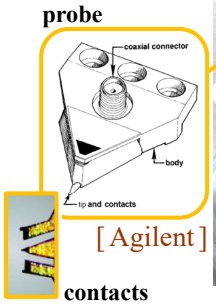


with non-perfect hardware, the directivity of the coupler is limited, and the impedance levels of coupler, source and receivers are different.

there is often a long electrical distance between the coupler and the DUT port (e.g., a cable) and there can be standing waves (e.g., caused by connectors) and losses – all these effects should be separated from the measurement of the DUT

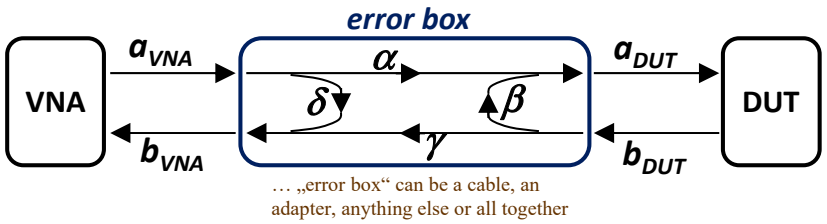
[Rohde & Schwarz]





[Agilent]

linearity of the system allows to put the combined effects of all imperfections (e.g., coupler, connectors, adapters, cables etc.) in an „error box“ an then treat the VNA (source + coupler + receiver) as perfect:



… „error box“ can be a cable, an adapter, anything else or all together

$$\Gamma_{meas} = \frac{b_{VNA}}{a_{VNA}} \qquad \Gamma_{DUT} = \frac{b_{DUT}}{a_{DUT}}$$

suppose **everything perfect**, then $\alpha=\gamma=1$ and $\beta=\delta=0$, then $\Gamma_{meas}=\Gamma_{DUT}$, that is, the measured value is actually the true value of Γ of the DUT.

otherwise:

assume the error box described by its two-port S-parameters, α, β, γ, δ :

$$\begin{pmatrix} a_{DUT} \\ b_{VNA} \end{pmatrix} = \begin{bmatrix} \alpha & \beta \\ \delta & \gamma \end{bmatrix} \begin{pmatrix} a_{VNA} \\ b_{DUT} \end{pmatrix} \quad \Rightarrow \quad \boxed{\Gamma_{meas} = \delta + \frac{\alpha\,\gamma\,\Gamma_{DUT}}{1-\beta\,\Gamma_{DUT}}}$$

the above relation between Γ_{meas} and Γ_{DUT} contains 3 complex unknowns, namely, β, δ and the product αγ.

they can be determined using measurement of 3 (un-correlated) known loads called „standards“.

- measurement of „match“: $\Gamma_{DUT} = \Gamma_M = 0$ $\quad \Gamma_{meas,M} = \delta$

- measurement of „open“: $\Gamma_{DUT} = \Gamma_O$ $\quad \Gamma_{meas,O} = \delta + \frac{\alpha\,\gamma\,\Gamma_O}{1-\beta\,\Gamma_O}$

- measurement of „short“: $\Gamma_{DUT} = \Gamma_S$ $\quad \Gamma_{meas,S} = \delta + \frac{\alpha\,\gamma\,\Gamma_S}{1-\beta\,\Gamma_S}$

the 3 complex unknowns, β, δ, αγ are found from the above 3 equations:

$$\delta = \Gamma_{meas,M}$$

$$\beta = \frac{\Gamma_S(\Gamma_{meas,O} - \Gamma_{meas,M}) - \Gamma_O(\Gamma_{meas,S} - \Gamma_{meas,M})}{\Gamma_O\Gamma_S(\Gamma_{meas,O} - \Gamma_{meas,S})}$$

$$\alpha\,\gamma = \frac{(\Gamma_O - \Gamma_S)(\Gamma_{meas,O} - \Gamma_{meas,M})(\Gamma_{meas,S} - \Gamma_{meas,M})}{\Gamma_O\Gamma_S(\Gamma_{meas,O} - \Gamma_{meas,S})}$$

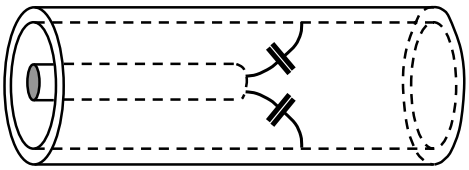
once β, δ, αγ are known, calibrated measurements can be performed:

$$\Gamma_{DUT} = \frac{\Gamma_{meas} - \delta}{\alpha\,\gamma + \beta\,(\Gamma_{meas} - \delta)}$$

the 3 standards do not need to be „open“, „short“ and „match“, but they need to be known exactly. For the purpose of error minimization, they should be distributed over the complex load plane as widely as possible – and this for all frequencies. Therefore, „almost-open“, „almost-short“ and „almost-match“ are a good choice.

deviations from a perfect „open“ and „short“ are small but frequency-dependent, they are stored as correction-capacitance and –inductance in the VNA

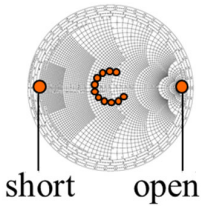
non-perfect „open“ is corrected with a (frequency-dependent) end capacitance



a perfect matched load is difficult to realize at high frequency, and a non-perfect almost-match is difficult to describe over frequency (in particular, in its reflection phase)

instead of measuring a „perfect“ matched load, a so-called sliding load is used.

A sliding load is a moveable small reflection at the end of a low-loss, well-defined transmission line. For various lengths, its input reflection forms a circle centered around the impedance of the transmission line. The impedance of the line can be manufactured at much higher accuracy than a load absorber at high frequency.



example:

broadband fixed load :
[Agilent 85056A / 00901-60003]
up to 4 GHz : $|\Gamma| < 0.008 = -42$ dB
up to 50 GHz : $|\Gamma| < 0.050 = -26$ dB

sliding load :
[Agilent 85056A / 00915-60003]
up to 50 GHz : $|\Gamma| < 0.016 = -36$ dB

[Spinner]

→ two-port and multi-port calibrations are mathematically and practically much more complicated.

Noise measurements at microwave frequency

- in a receiver, even without a useful signal at its input, „something“ will be detected

  → **distortion** (deterministic, but unwanted)

  → **noise** (statistical, unwanted)

- the „something“ at the output of the receiver (or, any other system), comprises a noise contribution of the receiver (or, system) itself

- parameters to describe this noise contribution and methods to measure these parameters are needed

→ **about noise**

parameters: signal-to-noise-ratio, noise figure, noise temperature

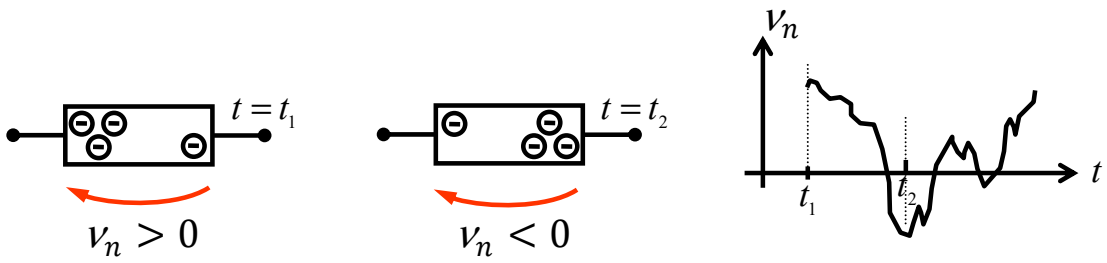
noise figure measurement

oscillator phase noise

phase noise measurement

Noise sources — thermal noise

- „thermal noise“ (a.k.a. Johnson noise) originates from Brownian motion of free electrons in a conductor:



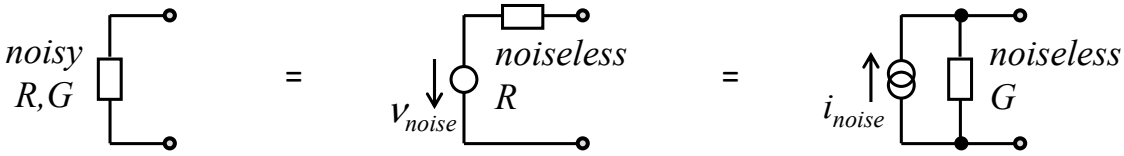
- time average of noise voltage (and current) is zero:

$$\overline{v_n(t)} = 0 \quad , \quad \overline{i_n(t)} = 0$$

- but time average noise power and r.m.s. voltage (current) are not zero:

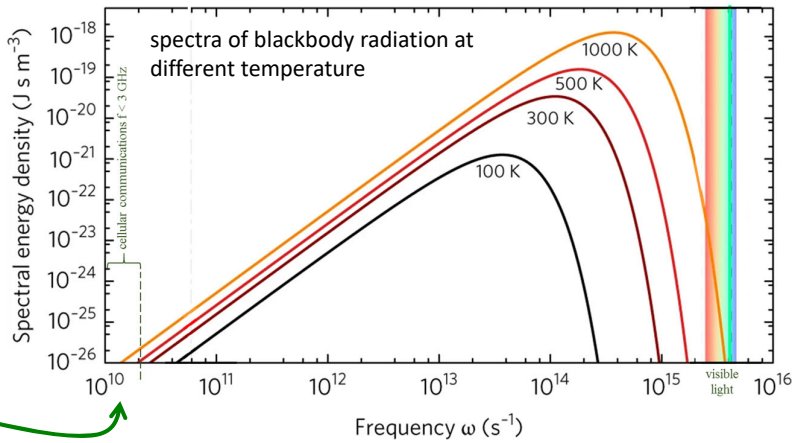
$$v_{n,rms} = \sqrt{\overline{v_n^2(t)}} = \sqrt{4kTR\Delta f} \quad , \quad i_{n,rms} = \sqrt{\overline{i_n^2(t)}} = \sqrt{4kTG\Delta f}$$

- the „noisy resistor“ modeled as voltage (or current) source with load:



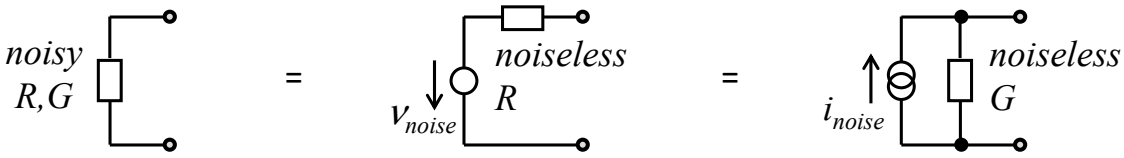
- the noise power spectral density is **approximately** constant („**white noise**“) if temperature is „high“ and frequency is „low“:

not really „constant“ over frequency, but considering rather narrowband electronic systems ...



[wikipedia]

- the „noisy resistor“ modeled as voltage (or current) source with load:



- the noise power spectral density is approximately constant („**white noise**“) if temperature is „high“ and frequency is „low“:

e.g., room temperature

e.g., RF, microwaves, millimeter waves

$$\frac{1}{G}\frac{i_{n,rms}^2}{\Delta f} = \frac{1}{R}\frac{v_{n,rms}^2}{\Delta f} = 4kT\,\frac{hf}{kT\left(\exp\left(\frac{hf}{kT}\right)-1\right)} \approx 4kT$$

$e^x \approx 1 + x$

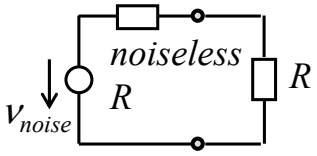
Planck: $h = 6.62 \times 10^{-34}\ Ws^2$

Boltzmann: $k = 1.38 \times 10^{-23}\ Ws/K$

- the noise power in a matched load is:

$$P_{noise} = \frac{1}{R}\left(\frac{v_n}{2}\right)^2 = G\left(\frac{i_n}{2}\right)^2 = kT\Delta f$$

:



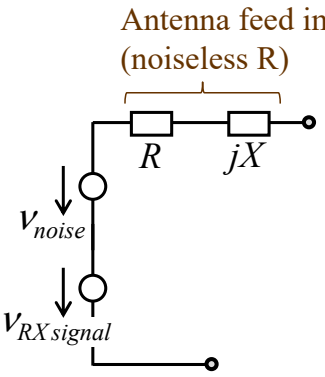
- example: „noise floor“ for engineers (T=290K, Δf=1Hz):

$$\left.\frac{P_{noise}}{\Delta f}\right|_{\Delta f=1Hz} = kT|_{T=290K} = 3.985 \times 10^{-23}\ {}^{W}/_{Hz} \cong -174\ {}^{dBm}/_{Hz}$$

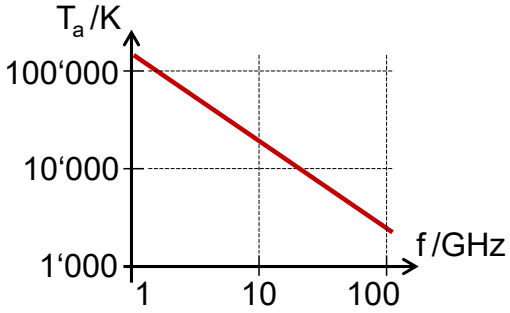
- since bandwidth (Δf) is often constant in a system with many noise sources, each thermal noise source can also be characterized (in particular, the noise source spectral power density) by an associated **noise temperature T**

Noise sources — others than thermal noise

- some sources can be approximately modeled as a resistor (a source of „thermal noise“) at a specified temperature. Example: receive antenna:



Noise temperature of the sun (note: must be weighted with antenna directivity pattern!):



[Zinke/Brunswig, Lehrbuch der Hochfrequenztechnik 2, Springer, p. 216]

where v_{noise} corresonds to a source with R at the antenna equivalent noise temperature T_a

T_a < 10 K for cosmic background and stars (exeption: sun)

T_a ~ 290 K for earth surface

- **Shot noise** is due to the particule nature of current. It can be observed in vacuum devices, at reverse biased junctions and in optical devices (e.g., in fiber amplifiers).

- **Flicker noise** (or: **1/f noise**) is particularly strong in/at low frequency, physically small, low current devices/operations.
It can be upconverted in non-linear decvoices (mixers, oscillators) to become significant at microwave frequency.

- Avalanche noise is due to impact ionization, e.g. in strongly reverse-biased junctions.

Notes: - thermal noise occurs in every resistive (not in reactive) device, including zero-/low-biased junctions

- 1/f noise occurs in biased junctions (also in a biased resistor)

Noise in microwave circuits

- since noise relates to noise power, and signal to signal power, an obviously useful measure is signal-to-noise ratio:

$$SNR = \frac{P_{signal}}{P_{noise}}$$

→ for analogue signals, examples for „good“ SNR are:
- (historic) analogue TV: SNR ~ 44…48 dB
- analogue speech: SNR ~ 0 dB

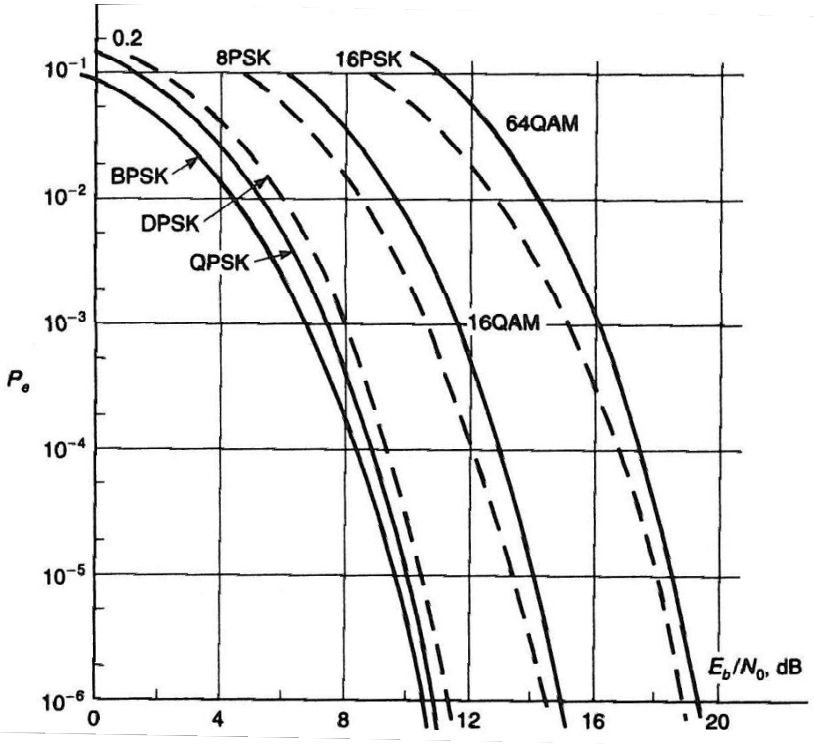
→ for digital signals, a „good“ SNR means low BER
- for spread-spectrum systems: SNR < 0 dB possible

- ways to improve SNR are mostly quite expensive:
  - increase (transmitted) signal power
  - reduce transmission distance
  - improve receiver electronics (less added noise)
  - increase antenna focusing (gain) (in wireless systems)

Parameter and measurement method needed!

Bit error rate (BER) versus symbol energy (SNR):

→ note that BER is highly non-linear and quite insensitive on varying SNR, so even for digital systems, SNR is useful



[J.B. Anderson, Digital Transmission Engineering, Wiley / IEEE Press, 2005, S. 113]

Noise in two-port networks

- assume a two-port with noise power at the input, $P_{N,source}$, and gain G

- assume matched output

$$P_{N,output} = G \cdot P_{N,source} + P_{N,two-port}$$

noise contribution of the two-port

$$= k\Delta f \cdot \left(G \cdot T_{source} + T_{two-port}\right)$$

$$P_{N,output} = Gk\Delta f \cdot (T_{source} + T_e) = k\Delta f \cdot T_{output}$$

$$T_e = \frac{T_{two-port}}{G}$$

$$T_{output} = G \cdot (T_{source} + T_e)$$

- the noise contribution of the two-port is described by a noise source at its input, or by its **equivalent input noise temperature** T_e

- the **signal-to-noise ratio** (SNR) at the output of a two-port is always worse than the SNR at its input:

$$SNR_{input} = \frac{P_{signal,input}}{P_{N,source}} = \frac{P_{signal,input}}{k\Delta f\, T_{source}}$$

$$SNR_{output} = \frac{P_{signal,output}}{P_{N,output}} = \frac{G\, P_{signal,input}}{Gk\Delta f \cdot (T_{source} + T_e)} = \frac{P_{signal,input}}{k\Delta f \cdot (T_{source} + T_e)}$$

$$\frac{SNR_{input}}{SNR_{output}} = \frac{T_{source} + T_e}{T_{source}} = 1 + \frac{T_e}{T_{source}} > 1$$

- The SNR is quite easy to measure. However, T_{source} is not a parameter of the two-port ! For a specific, defined T_{source}, the two-port parameter called **noise figure**, F, is defined as:

$$F = \frac{SNR_{input}}{SNR_{output}}\bigg|_{T_{source}\ = 290\ K} = 1 + \frac{T_e}{T_{source}}\bigg|_{T_{source}\ = 290\ K} = 1 + \frac{T_e}{290\ K}$$

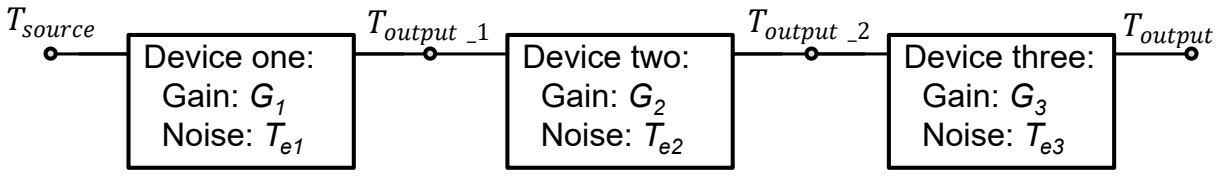
- the noise figure F can be expressed in dB: $F|_{dB} = 10 \log F$

- for a **passive two-port** at physical temperature T=290 K and with attenuation A=1/G, its equivalent input noise temperature is:

$$T_e = (A - 1) \cdot 290\ K \qquad \text{and:} \qquad F = A$$

… logical, because for A=G=1, there is no attenuation and therefore no added noise!

- consider a **cascade connection** of three two-ports:



$$T_{output\ _1} = G_1 \cdot (T_{source} + T_{e1})$$

$$T_{output\ _2} = G_2 \cdot (T_{output\ _1} + T_{e2})$$

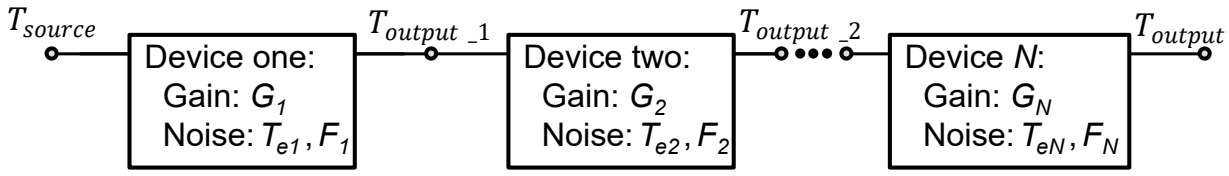
$$T_{output} = G_3 \cdot (T_{output\ _2} + T_{e3})$$

$$T_{output} = G_1 G_2 G_3 (T_{source} + T_{e1}) + G_2 G_3 (T_{e2}) + G_3 (T_{e3})$$

$$T_{e,all} = \frac{T_{output}}{G_1 G_2 G_3} - T_{source} = T_{e1} + \frac{T_{e2}}{G_1} + \frac{T_{e3}}{G_1 G_2}$$

… in a receiver, T_{e1} must be small and G_1 must be large → **LNA** !

- consider a cascade connection of *N* two-ports:



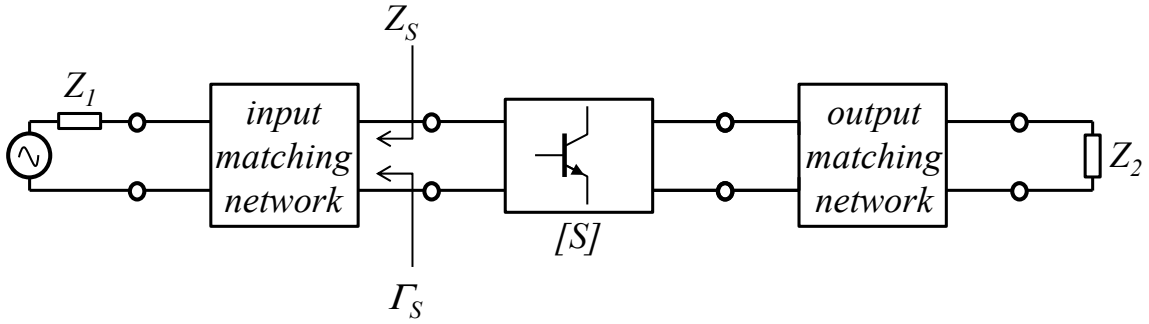
$$T_{e,all} = T_{e1} + \frac{T_{e2}}{G_1} + \frac{T_{e3}}{G_1 G_2} + \frac{T_{e4}}{G_1 G_2 G_3} + \cdots + \frac{T_{eN}}{G_1 G_2 G_3 \ldots G_{N-1}}$$

$$T_{e,i} = (F_i - 1) \cdot 290\ K$$

$$F_{all} = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2} + \frac{F_4 - 1}{G_1 G_2 G_3} + \cdots + \frac{F_N - 1}{G_1 G_2 G_3 \ldots G_{N-1}}$$

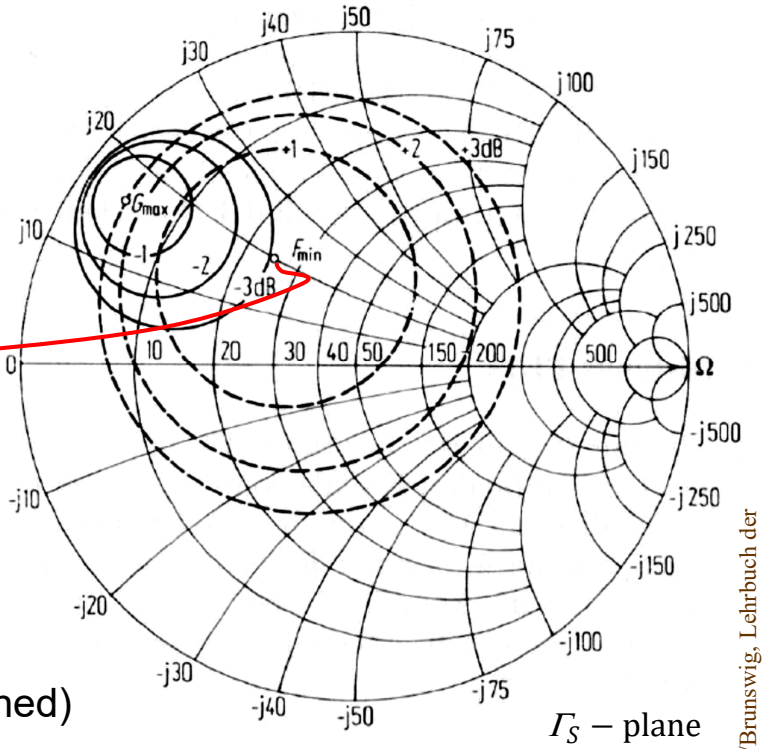
Measurement of noise figure

- for a LNA, *F* must be small and *G* must be large. Often, in a practical realization, both conditions cannot be met at the same time, therefore a trade-off must be made.
- for a LNA, minimum $F=F_{min}$ is obtained at some Z_S and will increase for other Z_S → **constant noise figure circles** are drawn in Γ_S plane



- constant noise figure circles for (historic) GaAs MESFET type CFY10 at V_{DS} = 4 V at 6 GHz :

$$\Gamma_{opt} = -0.24 + j\,0.33$$
$$F_{min} = 1.45 = 1.6\ dB$$



Γ_S − plane

[Zinke/Brunswig, Lehrbuch der Hochfrequenztechnik 2, Springer, p. 113]

- thus for measurement of noise figure, a source impedance must be specified (and output matched)

- „noise tuners“ exist for transforming (mechanically!) impedance in well-defined low-loss ways

- input noise can not be set to zero for the measurement of noise added by the device under test (DUT) …

- … but assuming the DUT is linear, measurement of two well-known input noise levels will allow to subtract and find the noise contribution of the DUT:

- a noise source delivers two different (i.e., switchable) noise power levels at well-specified source impedance (e.g., 50 Ohms)

- the diode-based noise source is a resistor at „off“ state (low-level noise power, T ≈ 290 K) and is noisy diode (operating in breakdown) plus attenuator (to achieve 50 Ohm source impedance) at „on“ state (T ≈ 10‘000 K)

- the two states of the noise source are called „hot“ and „cold“

$$P_{N,source\ ,cold} = k\ \Delta f\ T_{cold}$$

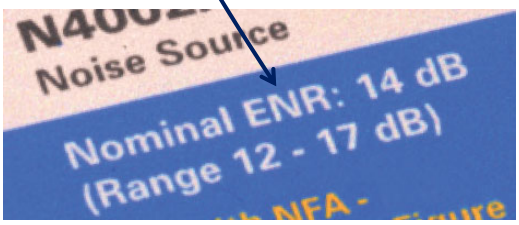
$$P_{N,source\ ,hot} = k\ \Delta f\ T_{hot}$$

- the noise source is characterized by ist Excess Noise Ratio (ENR):

$$ENR = \frac{T_{hot} - T_{cold}}{T_0}$$

which simplifies for $T_{cold} = T_0 = 290\ K$ to:

$$ENR = \frac{T_{hot}}{290\ K} - 1$$



- if the noise source is connected to the DUT input, power measured at the DUT output is:

$$P_{N,output\ ,cold} = k\ \Delta f\ G\ (T_{cold} + T_e)$$

$$P_{N,output\ ,hot} = k\ \Delta f\ G\ (T_{hot} + T_e)$$

… as defined previously, the noise contribution of the two-port DUT is described by a noise source at its input, or by its equivalent input noise temperature T_e

- in fact, only the ratio of these two powers must be measured precisely, the so-called Y-factor:

$$Y = \frac{P_{N,output\ ,hot}}{P_{N,output\ ,cold}} = \frac{T_{hot} + T_e}{T_{cold} + T_e}$$

- then:

$$\frac{ENR}{Y-1} - 1 = \frac{T_e}{290\ K} \quad \Rightarrow \quad \boxed{F = \frac{ENR}{Y-1}}$$

- this is called **Y-factor method**

- the method can include corrections for deviations of temperature (of noise source and ambient) different than 290 K

- the „hot“ state must be calibrated, which is done at metrology labs using noise surces at „real“ temperature (e.g., liquid N_2 at 77 K)

- the measured noise figure so far is the one of the cascade connection of DUT (F_1) and measurement equipment (F_2):

$$F_{sys} = \frac{ENR}{Y-1} = F_1 + \frac{F_2 - 1}{G_1} \qquad \text{or} \qquad F_1 = F_{sys} - \frac{F_2 - 1}{G_1}$$

- in a „calibration“ step, F_2 is found from measuring the noise source directly (without DUT) and G_1 from:

$$G_1 = \frac{P_{N,output,hot,with\ DUT} - P_{N,output,cold,with\ DUT}}{P_{N,output,hot,without\ DUT} - P_{N,output,cold,without\ DUT}}$$

Phase Noise

Phase noise is a measure of frequency stability over time.

Drift describes long-time frequency stability and is not considered here.

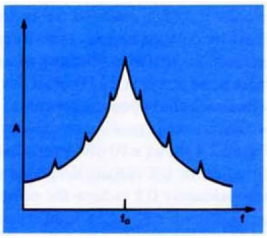
A signal with short-time fluctuations can be described as

$$V(t) = \left[V_o + \epsilon(t)\right] \sin\left[2\pi f_o t + \Delta\phi(t)\right]$$

amplitude variations / amplitude noise

phase variations / phase noise

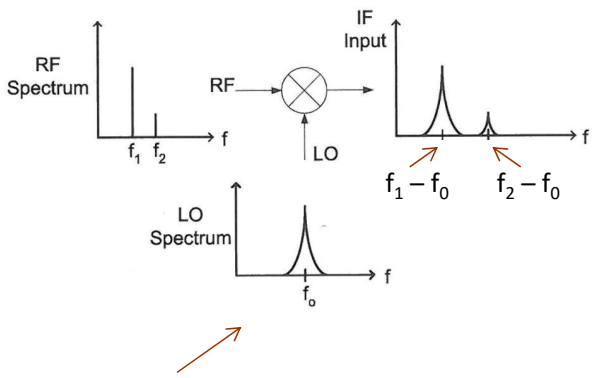
A signal with phase noise typically looks on an ideal spectrum analyzer as:



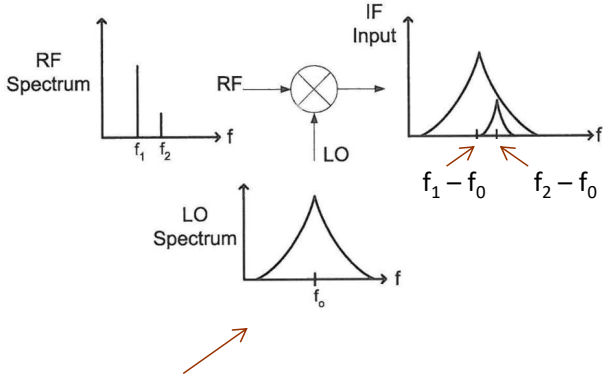
Why is low phase noise important ?

Microwave systems often use modulated signals being up-/downconverted.

LO phase noise adds at each conversion and reduces SNR:



low phase noise LO

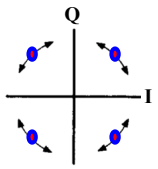


high phase noise LO

Consequences of LO phase noise:

→ in Doppler radar, a small moving target response may be masked

→ in digital comms, EVM is increased



QAM state diagram

Description of phase noise spectrum

- spectral density of phase fluctuations per unit bandwidth:

$$S_\phi(f_m) = \frac{\Delta\phi^2_{rms}(f_m)}{\text{BW used to measure } \Delta\phi_{rms}} \quad \frac{\text{rad}^2}{\text{Hz}}$$

- (single sideband) noise power density:

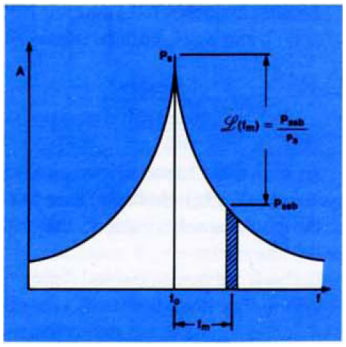
$$\mathcal{L}(f_m) = \frac{\text{power density (in one phase modulation sideband)}}{\text{total signal power}} = \frac{P_{ssb}}{P_s}$$

approximate relation (for small phase variations):

$$\mathcal{L}(f_m) \simeq \frac{1}{2} S_\phi(f_m)$$

… does not hold for small f_m
(where $\Delta\varphi_{peak}$ is not small)

can be derived from a spectrum analyzer reading:



- spectral density of frequency fluctuations per unit bandwidth:

$$S_{\Delta f}(f_m) = \frac{\Delta f^2_{rms}(f_m)}{\text{BW used to measure } \Delta f_{rms}} \quad \frac{Hz^2}{Hz}$$

… a frequency discriminator (as will be shown later) outputs a voltage proportional to $S_{\Delta f}(f_m)$

interrelations:

$$S_\phi(f_m) = \frac{S_{\Delta f}(f_m)}{f_m^2} \qquad \mathcal{L}(f_m) = \frac{S_{\Delta f}(f_m)}{2f_m^2}$$

dB scale:

$$S_{\Delta f}(f_m)\ [\text{dBHz/Hz}] = 20 \log \frac{\Delta f(\text{Hz})}{1\ (\text{Hz})} \text{ per Hz}$$

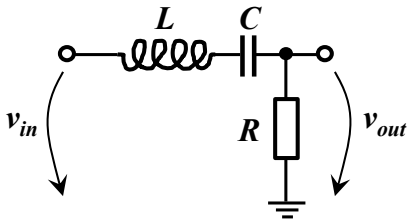
$$S_\phi(f_m)\ [\text{dBr/Hz}] = 20 \log \frac{\Delta\phi(\text{rad})}{1(\text{rad})} \text{ per Hz}$$

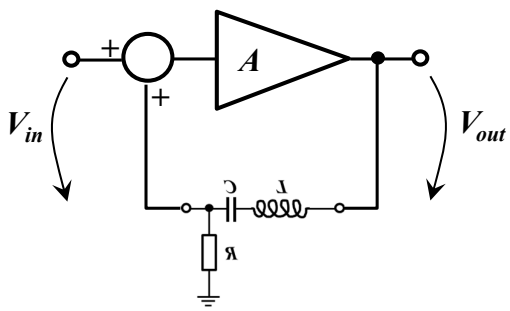
$$\mathcal{L}(f_m)\ [\text{dBc/Hz}] = 10 \log \frac{P_{noise}}{P_{carrier}} \text{ per Hz}$$

$$20 \log \frac{f_m}{1\ (\text{Hz})} \quad \text{3 dB}$$

Oscillator phase noise

consider an oscillator composed of an amplifier and a bandpass resonator in the feedback path:





$$H_{reso} = \frac{v_{out}}{v_{in}} = \frac{R}{R + j\omega L + \frac{1}{j\omega C}} = \frac{1}{1 + jQ_0\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right)}$$

$$\omega = \omega_0 + \omega_m \qquad \omega_0 \gg \omega_m \qquad \frac{\omega}{\omega_0} - \frac{\omega_0}{\omega} \approx \frac{2\omega_m}{\omega_0}$$

$$Q_0 = \frac{1}{\omega_0 RC} \qquad \omega_0 = \frac{1}{\sqrt{LC}}$$

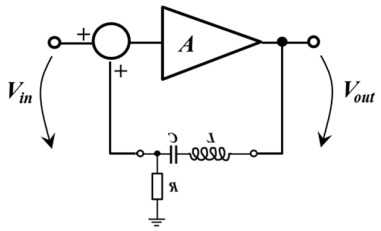
$$H_{osc} = \frac{V_{out}}{V_{in}} = \frac{A}{1 - AH_{reso}} = \frac{1}{1 - H_{reso}}$$

at resonance & compression (A=1):

$$|H_{osc}|^2 = 1 + \left(\frac{\omega_0}{2Q_0\omega_m}\right)^2 \quad (1)$$

$$H_{reso} = \frac{1}{1 + jQ_0\frac{2\omega_m}{\omega_0}}$$

spectral density of phase fluctuations at the oscillator output is given as:



$$S_\phi(f_m)\big|_{\text{out}} = S_\phi(f_m)\big|_{\text{in}} \cdot |H_{\text{osc}}|^2 \quad (2)$$

… where the „input“ of the oscillator is given by the thermal noise:

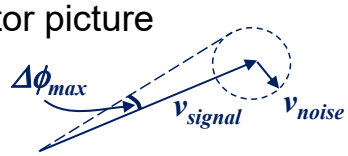
at the input of the loop amplifier (of noise figure F), there is a vectorial superposition of

the (large) signal of power P_S and voltage $v_{\text{signal}} = \sqrt{RP_S}$

and the (small) thermal noise of power FkTB and voltage $v_{\text{noise}} = \sqrt{FkTBR}$

noise at amplifier output referenced back to its input

from the vector picture

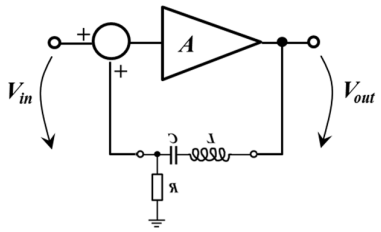


follows (at B = 1 Hz) :

$$\Delta\phi_{\max} \approx \tan\Delta\phi_{\max} = \frac{v_{\text{noise}}}{v_{\text{signal}}} = \sqrt{\frac{FkT}{P_S}}$$

therefore: $S_\phi(f_m)\big|_{\text{in}} = (\Delta\phi_{\max})^2\big|_{B=1\text{Hz}} = \frac{FkT}{P_S}$

this input thermal noise is independent of offset frequency f_m !

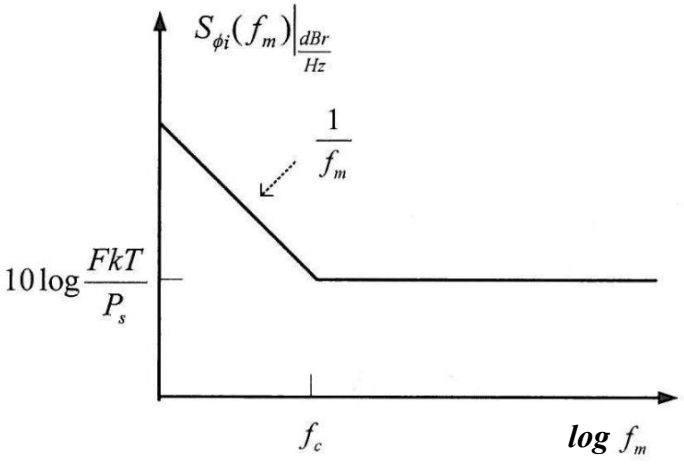


a semiconductor amplifier also generates another noise called Flicker noise at low offset frequency

in a loglog-plot (dB over log f), the Flicker noise slope is a line with a corner frequency, f_c

f_c depends on semiconductor type and operation conditions

JFETs have lowest f_c (< 1 kHz), bipolar transistors moderate fc (~ 10 kHz) and GaAs FETs have high f_c (> 10 MHz)



therefore: $S_\phi(f_m)\big|_{\text{in}} = \frac{FkT}{P_S}\left(1 + \frac{f_c}{f_m}\right)$ (3)

combining equations (1) (2) (3) gives Leeson's formula:

$$L(f_\mathrm{m}) = \frac{1}{2} S_\phi(f_\mathrm{m})\big|_\mathrm{out} = \frac{FkT}{2P_\mathrm{S}}\left[1 + \frac{f_c}{f_\mathrm{m}} + \frac{1}{f_\mathrm{m}^2}\left(\frac{f_0}{2Q_0}\right)^2 + \frac{1}{f_\mathrm{m}^3}\left(\frac{f_0^2 f_c}{4Q_0^2}\right)\right]$$

phase noise floor

flicker noise

up-converted thermal noise

up-converted flicker noise

f_0 … frequency of oscillation (center frequency of phase noise spectrum)

f_m … offset frequency in displaying phase noise spectrum

f_c … (flicker noise) corner frequency in phase noise spectrum

according to Leeson's formula, a low oscillator phase noise is achieved by:

- low phase noise floor (low temperature, low amp noise figure, large signal power)
- low flicker noise corner frequency
- low operation frequency
- high Q bandpass feedback filter

this is tricky because amplifer input transistor may work at non-standard input impedance and/or large signal conditions

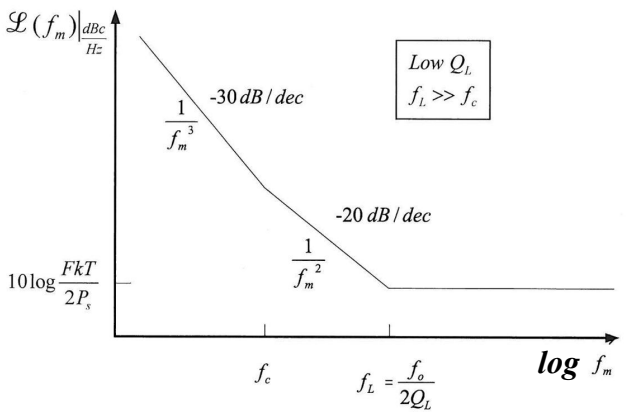
[D. B. Leeson, "A simple model of feedback oscillator noise spectrum," Proc. IEEE, vol. 54, pp. 329–330, Feb. 1966.]

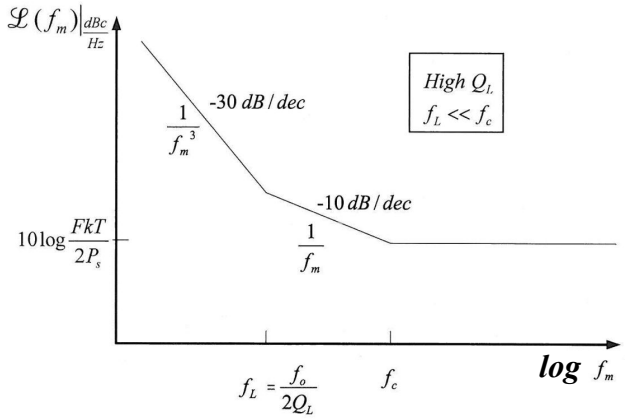
typical SSB phase noise spectrum of an oscillator:

$$L(f_\mathrm{m}) = \frac{1}{2} S_\phi(f_\mathrm{m})\big|_\mathrm{out} = \frac{FkT}{2P_\mathrm{S}}\left[1 + \frac{f_c}{f_\mathrm{m}} + \frac{1}{f_\mathrm{m}^2}\left(\frac{f_0}{2Q_0}\right)^2 + \frac{1}{f_\mathrm{m}^3}\left(\frac{f_0^2 f_c}{4Q_0^2}\right)\right]$$

20 dB / decade f_m
6 dB / octave f_m

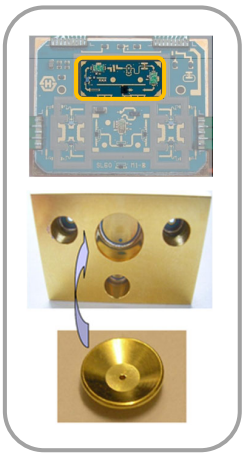
30 dB / decade f_m
9 dB / octave f_m

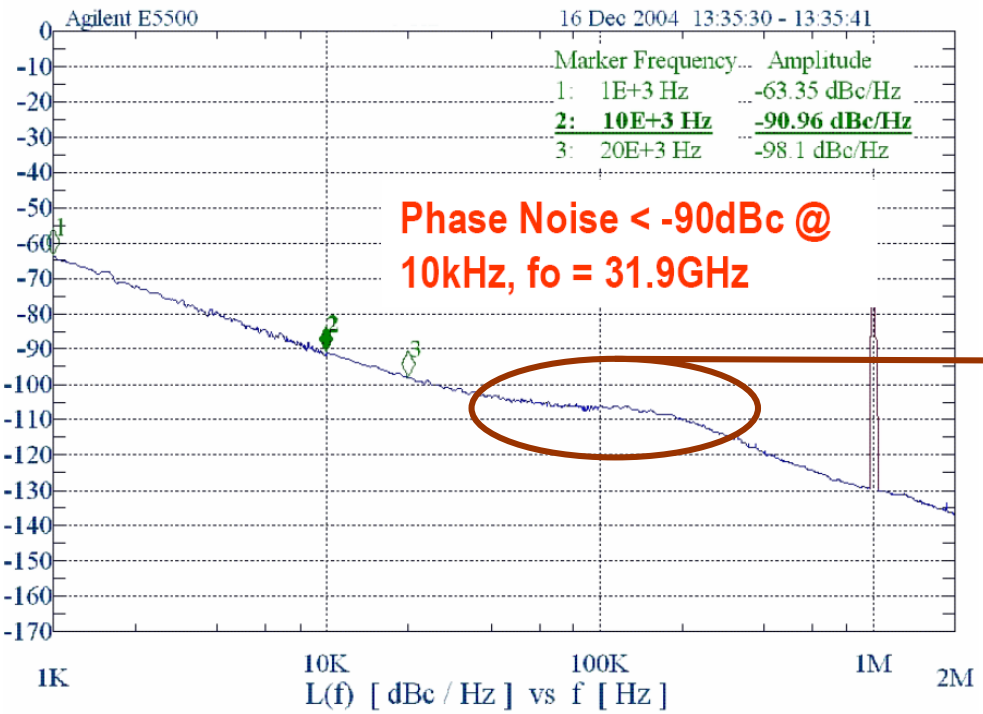




measured SSB phase noise spectrum of an oscillator

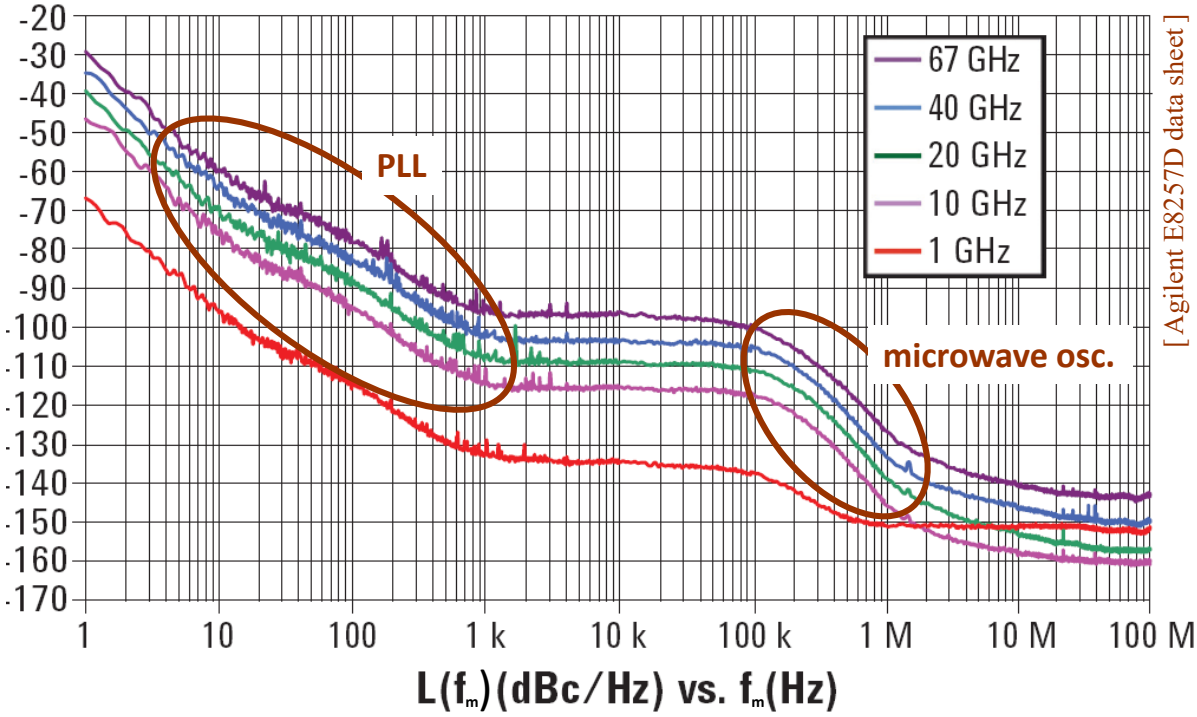
high-Q cavity (TM010, see B.65), GaAs FET amplifier





What is that ?
→ *measurement problem:*
at low offset, the reference (PLL to quartz) is better than DUT and DUT is „measured“, while at high offset, the DUT is better than the reference and thus phase noise slope of reference is shown

measured SSB phase noise spectrum of a high-quality synthesizer:



Note: at f_m = const. and constant „effort“, phase noise increases with 20 dB / decade f_0

Phase noise measurement techniques

Measurement of phase noise of a microwave oscillator is conceptually simple and practically quite difficult. As a result, there exist several techniques with advantages and drawbacks each.

Phase noise of a microwave oscillator can be measured using the following strategies:

A. Down-convert to DC with a reference LO of the same frequency.

The reference LO must have much better phase noise characteristics.

The reference LO and the DUT must have exactly the same frequency, that is, one of the two must be tuneable and frequency-locked.

Instead of better reference LO, another (identical) DUT can be used and the resulting phase noise spectrum be reduced by 3 dB.

→ Spectrum analyzer „method“

→ Phase detector method

B. Down-convert to DC with a time-delayed version of the DUT signal

The DUT signal is split in 2 paths, one path is delayed, and the 2 signals are then mixed to DC.

For high accuracy, long delays are needed. For measurements at small offset frequencies, delay lengths become prohibitive.
Long delays cause attenuation, therefore amplification is needed.

Method of choice for unstable sources, free-running sources, measurements at large offset frequency.

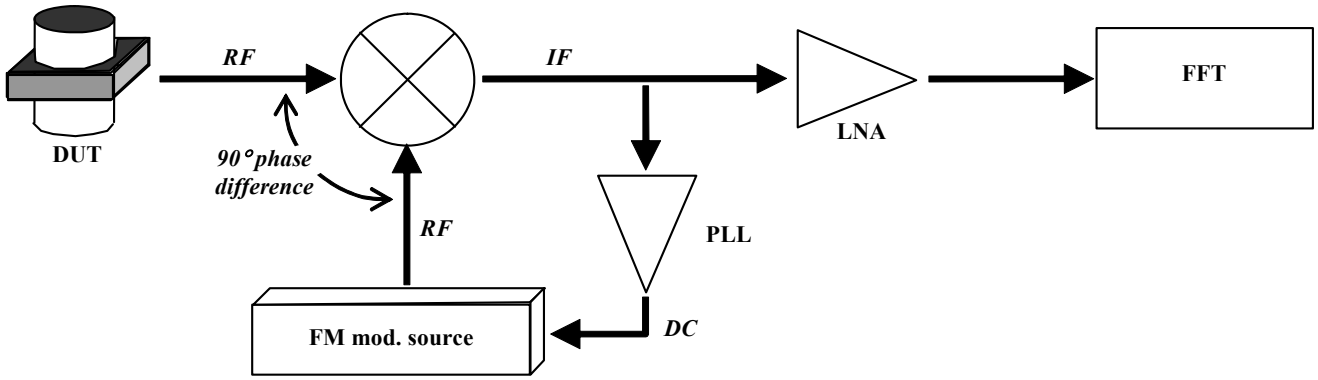
→ Frequency discriminator („delay line“) method

C. Split then down-convert to DC with 2 identical, uncorrelated reference LOs

Comparing the noise sidebands of the two downconverted signals, they contain correlated as well as uncorrelated noise. The uncorrelated noise contributions stem from the 2 reference sources and can be averaged away. In digital signal processing of the baseband data, this can be done many times. The noise contribution of the 2 reference LOs reduces. DUT with less noise than the reference LO can be measured.

→ Cross-correlation method

Phase detector method



either the DUT or the reference source must be frequency-tuned and locked.

the reference source must have lower phase noise than the DUT

E.g., if the reference noise is 10dB below DUT noise, then 110% noise power are displayed, causing 0.41 dB too high a noise level (which can be accounted for if known precisely)

the 90° phase difference must be maintained during the measurement

Why? Only at 90 deg phase difference („phase quadrature“), the mixer converts small phase variations in small output voltage variations

The role of phase quadrature in the phase detector method:

consider a diode $I = I_0\left(\exp\left(\frac{U}{U_{th}}\right) - 1\right)$ and its Taylor expansion $i = c_0 + c_1 u + c_2 u^2 + c_3 u^3 + \cdots$

and the DUT signal at the mixer input $v_0 \cos(\omega t + \varphi(t))$ and the reference LO signal $v_0 \cos(\omega t + \varphi_0)$

the quadratic term with the superposition of the two input signals reads

$c_2 v_0^2 (\cos(\omega t + \varphi(t)) + \cos(\omega t + \varphi_0))^2$ which can be written as

$c_2 v_0^2 (\cos(\omega t + \varphi(t)))^2 + c_2 v_0^2 (\cos(\omega t + \varphi_0))^2 + 2 c_2 v_0^2 \cos(\omega t + \varphi(t)) \cos(\omega t + \varphi_0)$

where the third term can be modified according to $2 \cos x \cos y = \cos(x + y) + \cos(x - y)$ as

$c_2 v_0^2 \cos(2\omega t + \varphi(t) + \varphi_0) + c_2 v_0^2 \cos(\varphi(t) - \varphi_0)$ where the first term at the doubled frequency

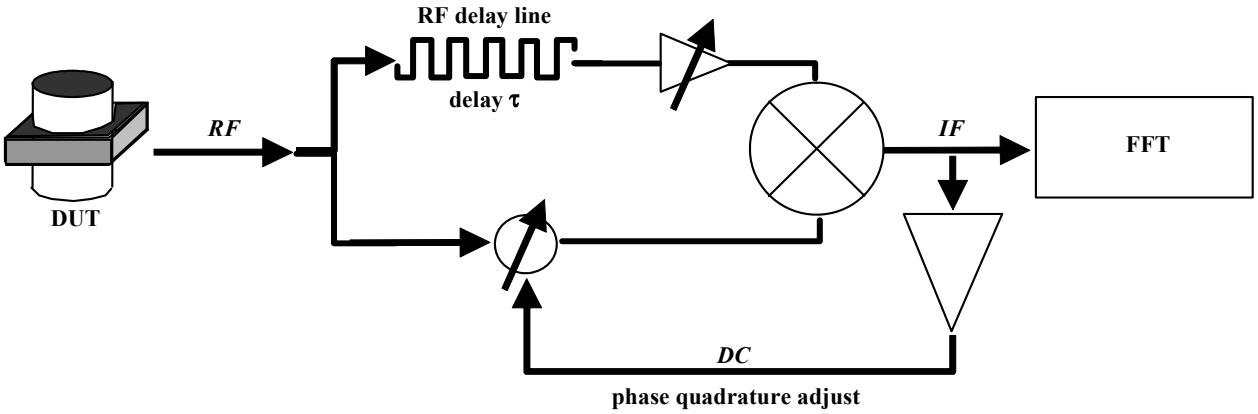
is filtered and the second term at DC is modified according to $\cos(x + y) = \cos x \cos y - \sin x \sin y$

and one obtains $c_2 v_0^2 [\cos \varphi(t) \cos \varphi_0 + \sin \varphi(t) \sin \varphi_0]$ which for $\varphi_0 = \frac{\pi}{2}$ simplifies to

$$c_2 v_0^2 \sin \varphi(t) \approx c_2 v_0^2 \cdot \varphi(t)$$

That is, the mixer current is proportional to the phase fluctuation φ(t)

Delay line method



Frequency fluctuations of the DUT are translated into phase fluctuations at the mixer. The sensitivity for $S\phi(f_m)$ is reduced for small f_m by $1/f^2$:

$$S_\phi(f_m) = \frac{S_{\Delta f}(f_m)}{f_m^2}$$

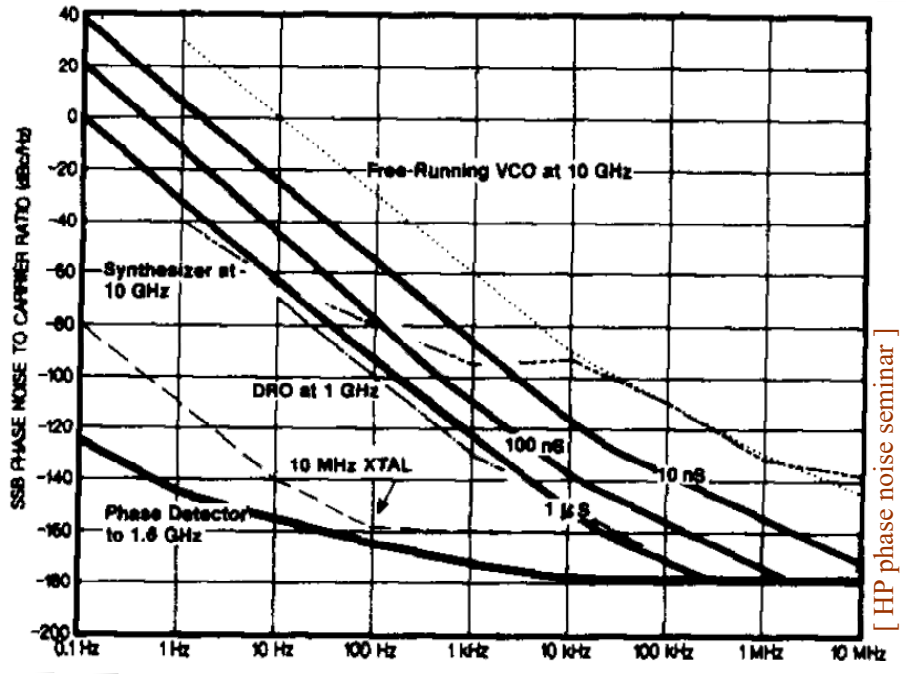
The two carrier signals at the mixer must be in phase quadrature.

The loss of the delay line can (advantageously) be compensated.

… but beware of amplifier noise !

Advantages: - frequency instability (e.g., drift) of the DUT does not matter
- no need for a „better“ reference source

Disadvantages: - insensitive for „close-in“ phase noise (small f_m)
- maximum f_m approx. $1/(2\tau\ldots 4\tau)$

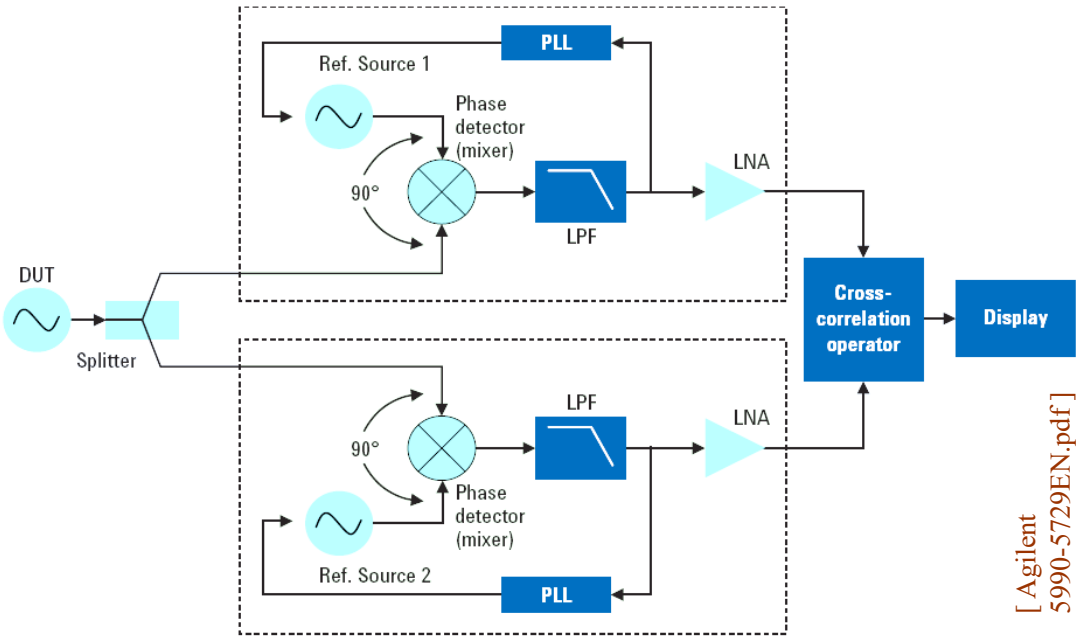


[HP phase noise seminar]

typical system sensitivity using delay line method and delay lines of diffrent length; also shown some typical DUT phase noise characteristics

Note: a 100ns delay corresponds to a length of 30m!

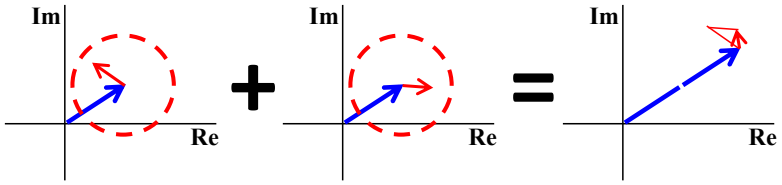
Cross-correlation method



The only method where the reference source phase noise can be worse than the DUT phase noise (by up to about 15…20 dB*))

*) W.F. Walls, „Practical problems involving phase noise measurements," Proc. 33rd Annual Precise Time and Time Interval (PTTI) Meeting, Long Beach, Nov. 2001, pp. 407-416

Principle of operation is „complex averaging":



blue … DUT phase noise phasor

red … reference source noise phasor

the un-correlated reference source noise (red phasors) will statistically cancel

one correlation reduces the reference source phase noise in relation to the DUT phase noise

reference noise floor reduction increases with the number of cross-correlations *N* (that is, also with the measurement time) as

$$5 \log N$$

for example, N=1000 results in an improvement of 15 dB

Signal Analyzers

RF signals are most completely described in time domain. A time-domain analyzer must sample the signal voltage frequently (at least twice as fast as the highest frequency content of the signal) and with high amplitude precision. Very fast and high-resolution analogue-to-digital converters (ADC) are limiting this approach.

Real-time **oscilloscopes** do the fast sampling in the RF domain.

Sampling oscilloscopes assume periodicity of the signal and sample the periodic RF signal at equidistant points in time.

Time-domain reflectometers (**TDR**) and transmitters (TDT) generate a step excitation to the DUT and watch for reflected/transmitted signal shapes.

Spectrum analyzers use a frequency-tuned local oscillator (and some filters) to down-convert a single RF frequency, followed by an amplitude detector.

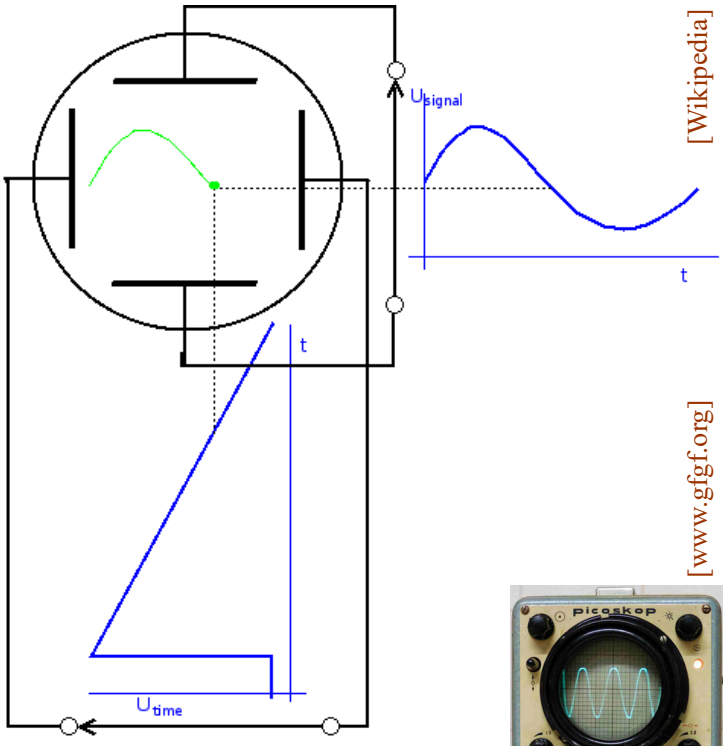
Modulation analyzers (or **vector signal analyzers**) use a frequency-tuned local oscillator to down-convert a narrow RF frequency band, followed by digital signal processing to analyze phase and amplitude states of the modulated signal.

Oscilloscope

In an analogue oscilloscope, using a **cathode ray tube (CRT)** with vertical und horizontal ray deflection, an internal saw-tooth (ramp) generator deflects the ray horizontally, whereas the (amplified) signal deflects the ray vertically.

In the digital equivalent (**digital storage oscilloscope** – DSO), a fast analog-to-digital converter (ADC) samples the input signal.

Note that Nyquist's sampling theorem must be complied, meaning that *very fast* ADC is needed for microwave signals — today's DSOs are limited to about 2 GHz max frequency.



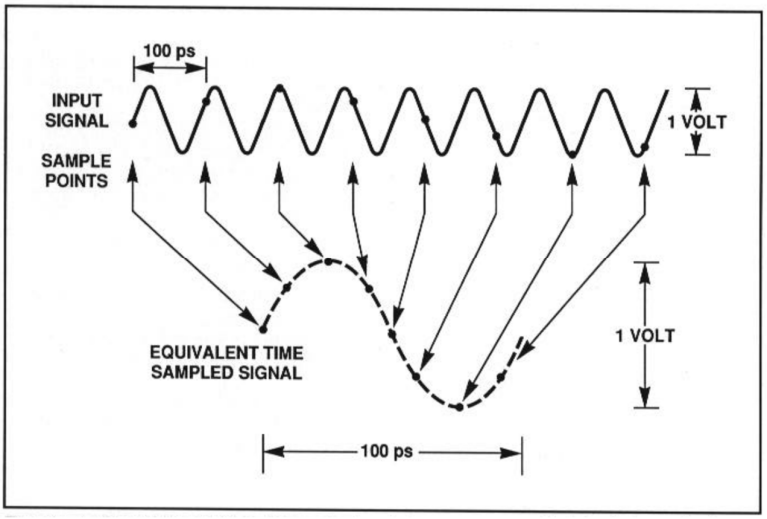
[Wikipedia]

[www.gfgf.org]

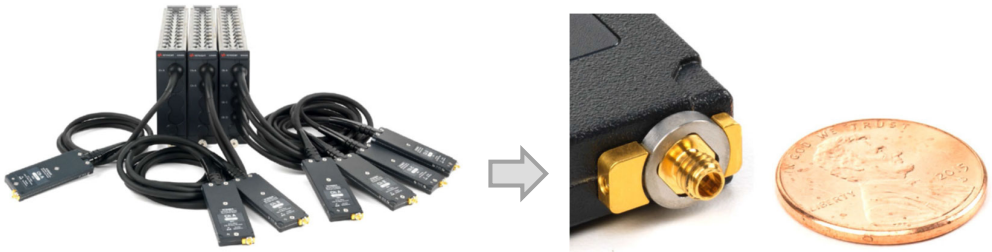
[Tektronix]

Sampling oscilloscope

A (digital) **sampling oscilloscope**, takes samples of the (assumed periodic) signal at (much) lower repetition rate than the frequency of the signal. Thus, a slower ADC (with higher accuracy) can be used. Sampling scope heads with up to 100 GHz bandwidth do exist.



[Tektronix]



[Keysight N1046A]

TDR — time domain reflectometer (1/6)

Reflection coefficient (or impedance) over time (and therefore, space) can be measured using a step generator (or, impulse generator) and a wideband receiver followed by a data logger (or, by an oscilloscope).

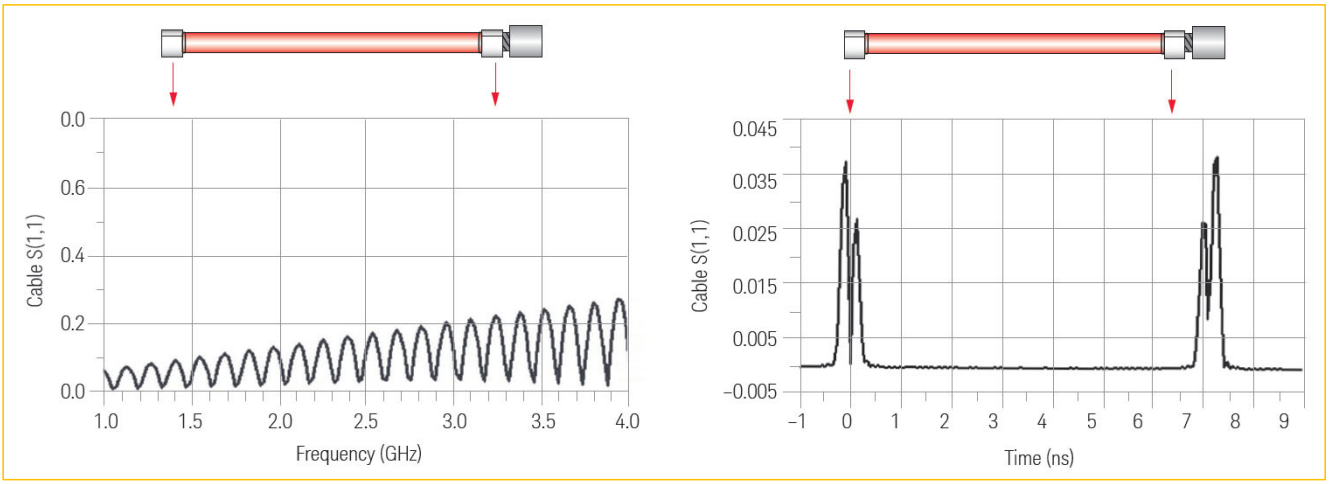
Applying Fourier transform, time-domain data can also be obtained from (wideband) frequency-domain data as measured by a VNA (vector network analyzer).

The two approaches are not exactly identical:

- calibration of impedance and test-set directivity is much easier (and more accurate) for VNA than for an oscilloscope.
- noise level and dynamic range are an issue for the wideband receiver of an oscilloscope.
- the VNA does not operate below a certain minimum frequency, therefore, its signals show bandpass behavior (though low-pass signals can be approximated by extrapolation to DC).
- a time-domain converted VNA measurement assumes a highly linear DUT.

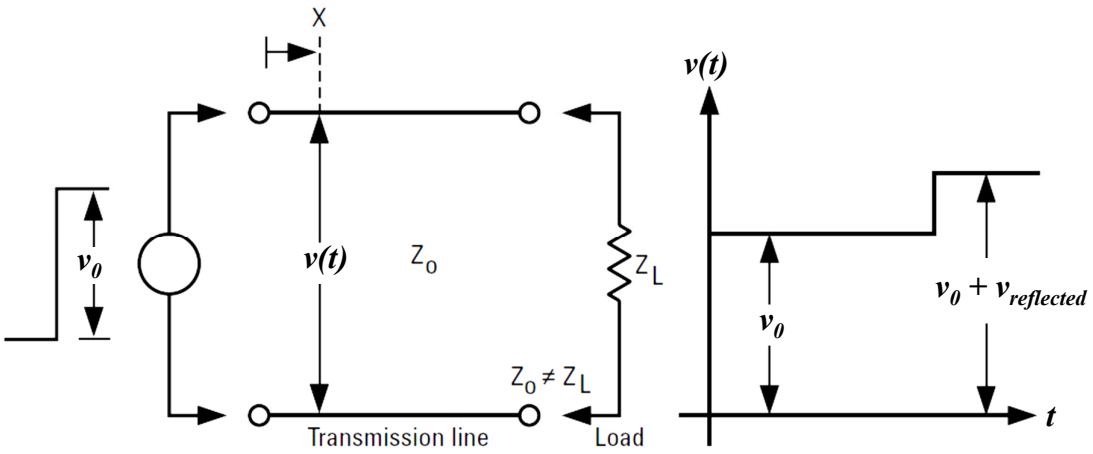
TDR — time domain reflectometer (2/6)

A typical application of TDR is the identification of faults (or, discontinuities) along a cable: A localized discontinuity (local and small variation of impedance) leads to small ripples in the magnitude of the input reflection coefficient over frequency. Transformed to time domain (by inverse Fourier transform), the discontinuity is clearly identified over time, and by considering propagation velocity of the wave, over space.



[Keysight 5989-5723]

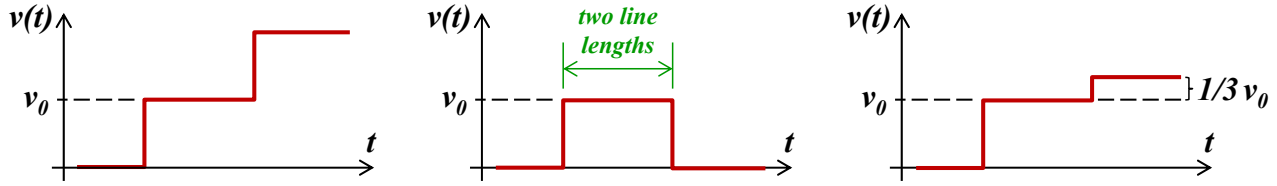
TDR — time domain reflectometer (3/6)



[Keysight 5966-4855E]

Different “types” of discontinuities are readily identified:

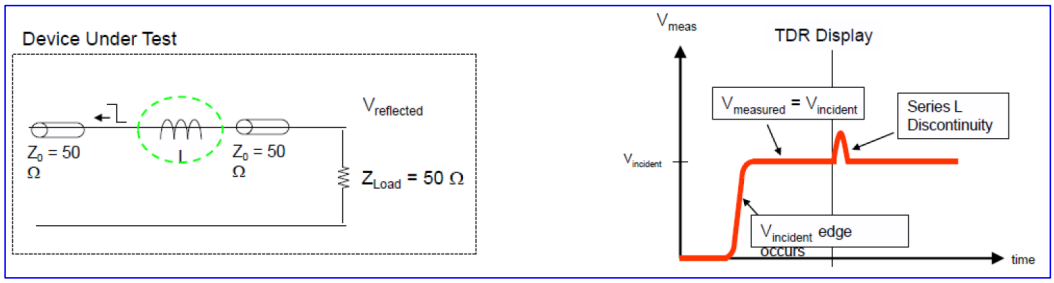
open circuit, $\Gamma = +1$: short circuit, $\Gamma = -1$: load $Z_L = 2\,Z_0$, $\Gamma = +1/3$:



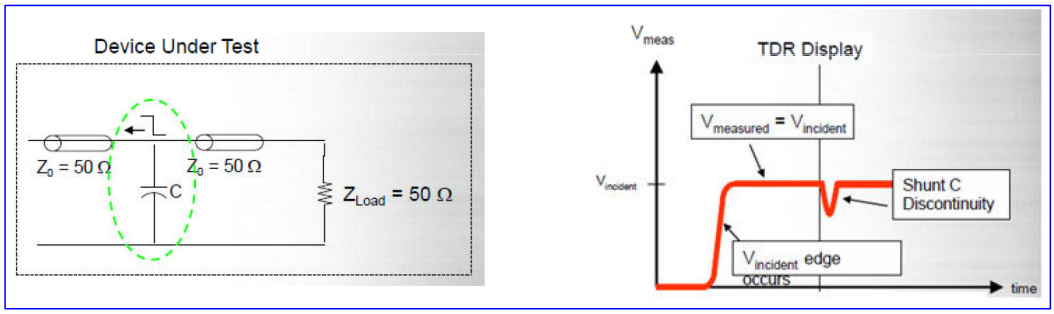
TDR — time domain reflectometer (4/6)

Different “types” of discontinuities are readily identified:

- series inductance (or a section of high-impedance line):



- shunt capacitance (or a section of low-impedance line):

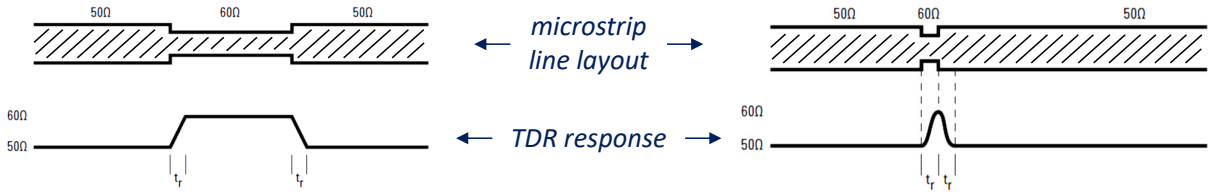


[Anritsu]

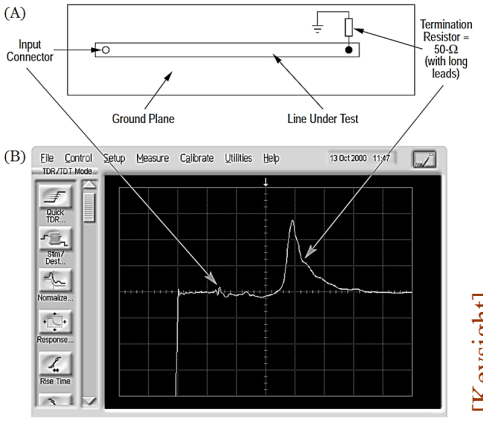
TDR — time domain reflectometer (5/6)

Accurate localization requires smallest TDR step risetime (or broadest VNA band-width). TDR modules use step generators with as small as 6 ps risetime (10%-90%).

Example: effect of TDR step risetime t_R :



Example: 50 Ohm trace ending with 50 Ohm resistor to ground – resistor with long leads causing series inductance:



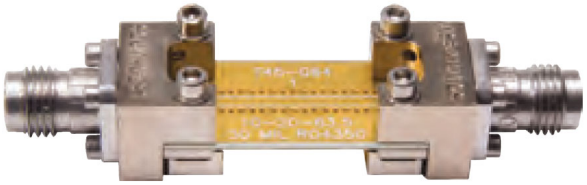
[Keysight]

TDR — time domain reflectometer (6/6)

Limitation of TDR: Accuracy is difficult (impossible?) to quantify.

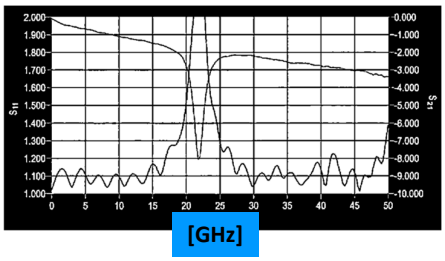
„Looking behind“ a narrowband discontinuity can produce rather arbitrary results:

Example of end-launch coax connectors on a 50 Ohm trough line:

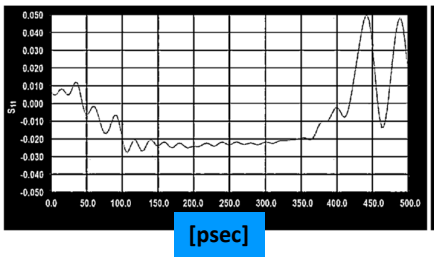


Failure of one connector causing a narrowband dip – frequency domain measurement:

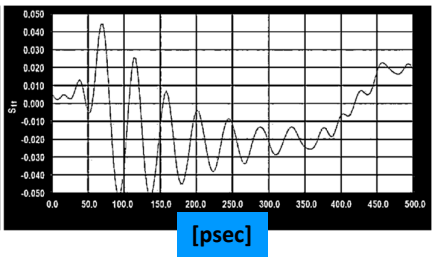
TDR measurement: failed connector at the „far end“ (left) and at the „near end“ (right) – „*masking*“ – :



strange dip at some frequency



reflection at far end

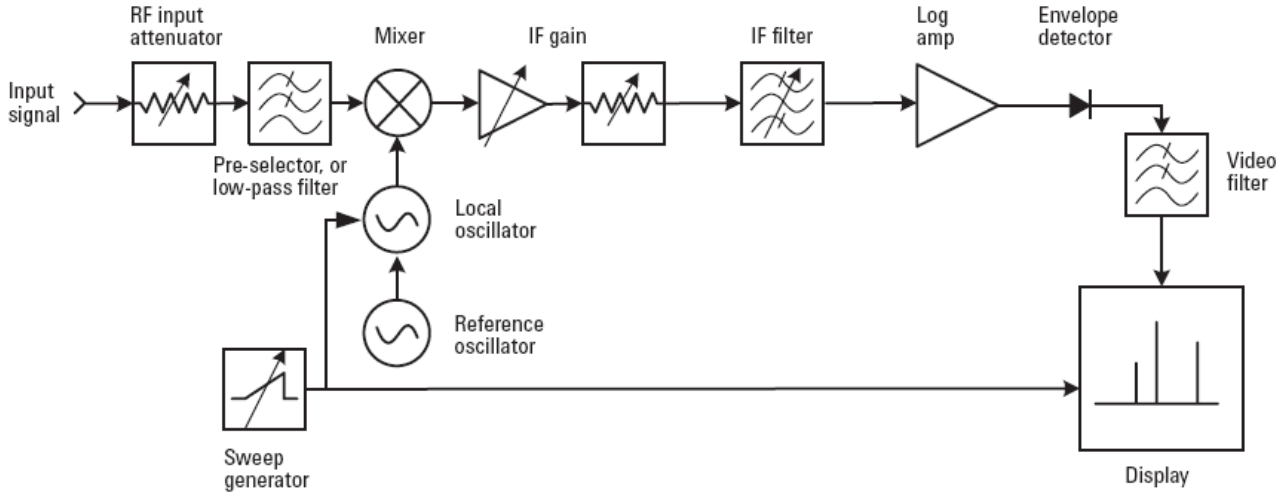


near-end reflect masks everything behind it

[Southwest Microwave]

Spectrum analyzer

The spectrum analyzer shows signal power (or magnitude) over frequency for a continuous signal. Obviously, a tuneable bandpass filter in combination with a power meter would do the job. Narrowband analogue tuneable bandpass filters are rather inexistent. Therefore, heterodyne system approach allows to use a fixed-frequency bandpass filter:

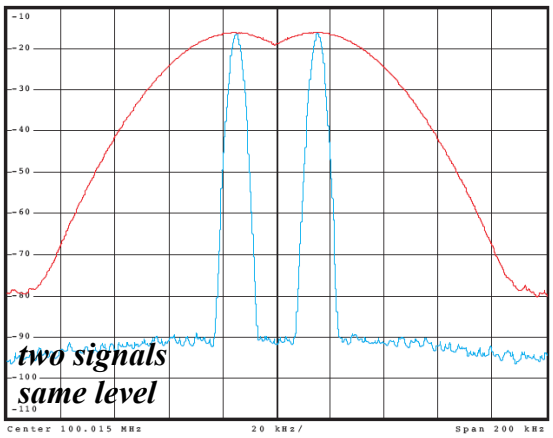


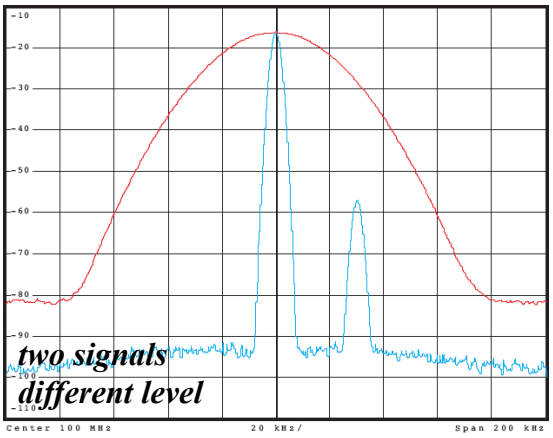
Notes: up to 4 mixer stages; pre-selector filter very sophisticated (e.g., YIG-filter)

Spectrum analyzer

Measurements with the spectrum analyzer must consider many parameters:
- noise floor (depends on RF input attenuation and IF filter bandwidth (RBW))
- dynamic range & linearity (depends on RF input attenuation)
- frequency resolution (depends on IF filter bandwidth (RBW))
- sweep time (depends on freq span and on IF filter bandwidth (RBW))
- …

Example: two signals separated by 30 kHz: **RBW = 30 kHz** , **RBW = 3 kHz**



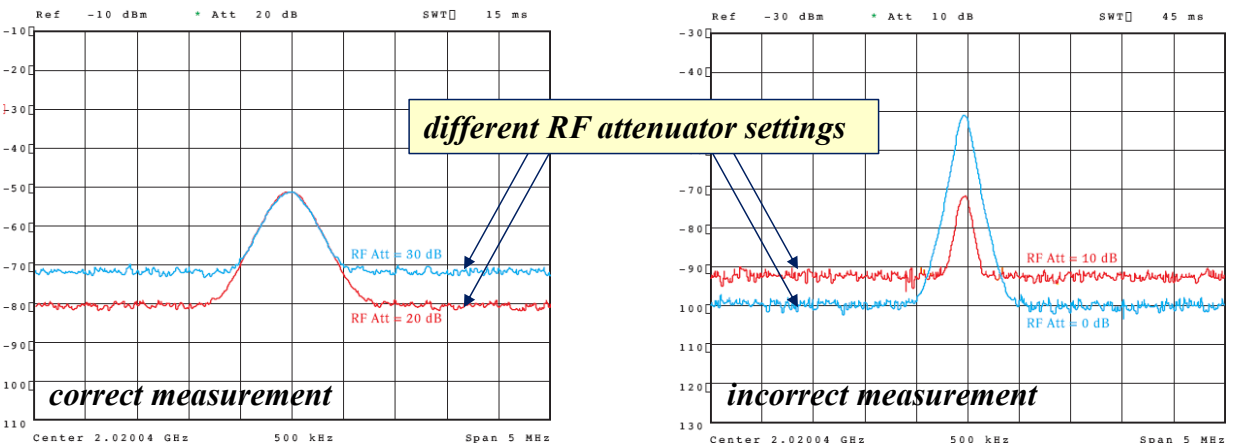


[C. Rauscher, „Fundamentals of Spectrum Analysis" Rohde & Schwarz GmbH & Co. KG, 2001]

Spectrum analyzer

Example:
is a nonlinearity disturbing the measurement caused by the spectrum anlyzer itself?

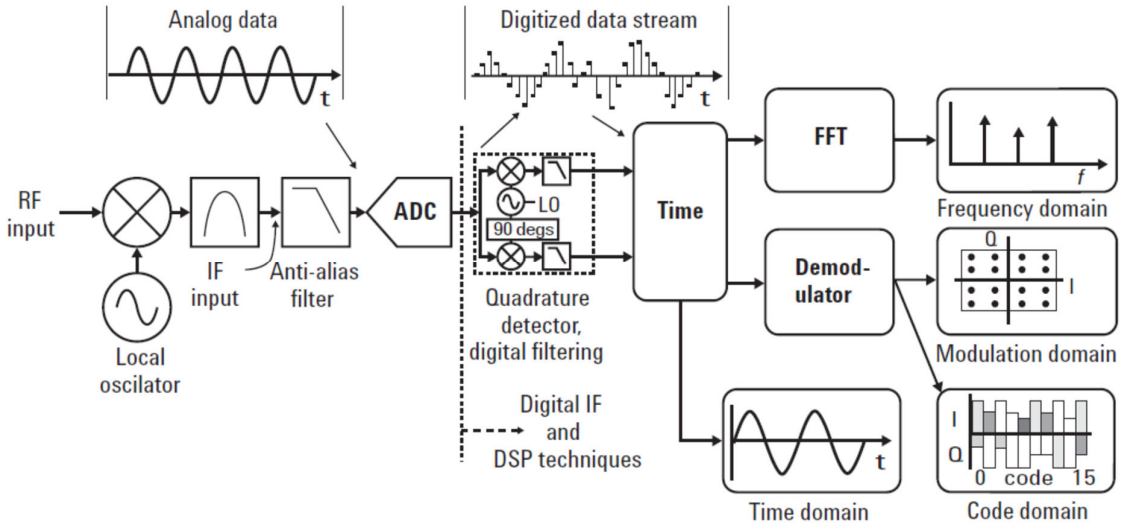
Variation of RF input attenuator should not change the signal. If it does, then because of the nonlinearities which are caused be the spectrum analyzer.



[C. Rauscher, „Fundamentals of Spectrum Analysis" Rohde & Schwarz GmbH & Co. KG, 2001]

Vector signal analyzer

A vector signal analyzer downconverts the analog microwave signal, subsequently digitizes the IF signal and, by digital processing, analyzes phase, magnitude and frequency content of the signal. Thereby, all sorts of advanced modulation schemes, including digital modulations, can be investigated. Important for the performance are the digital part as well as bandwidth (today: 250 MHz), dynamic range and linearity of the analogue part (and of the ADC).



[Keysight]